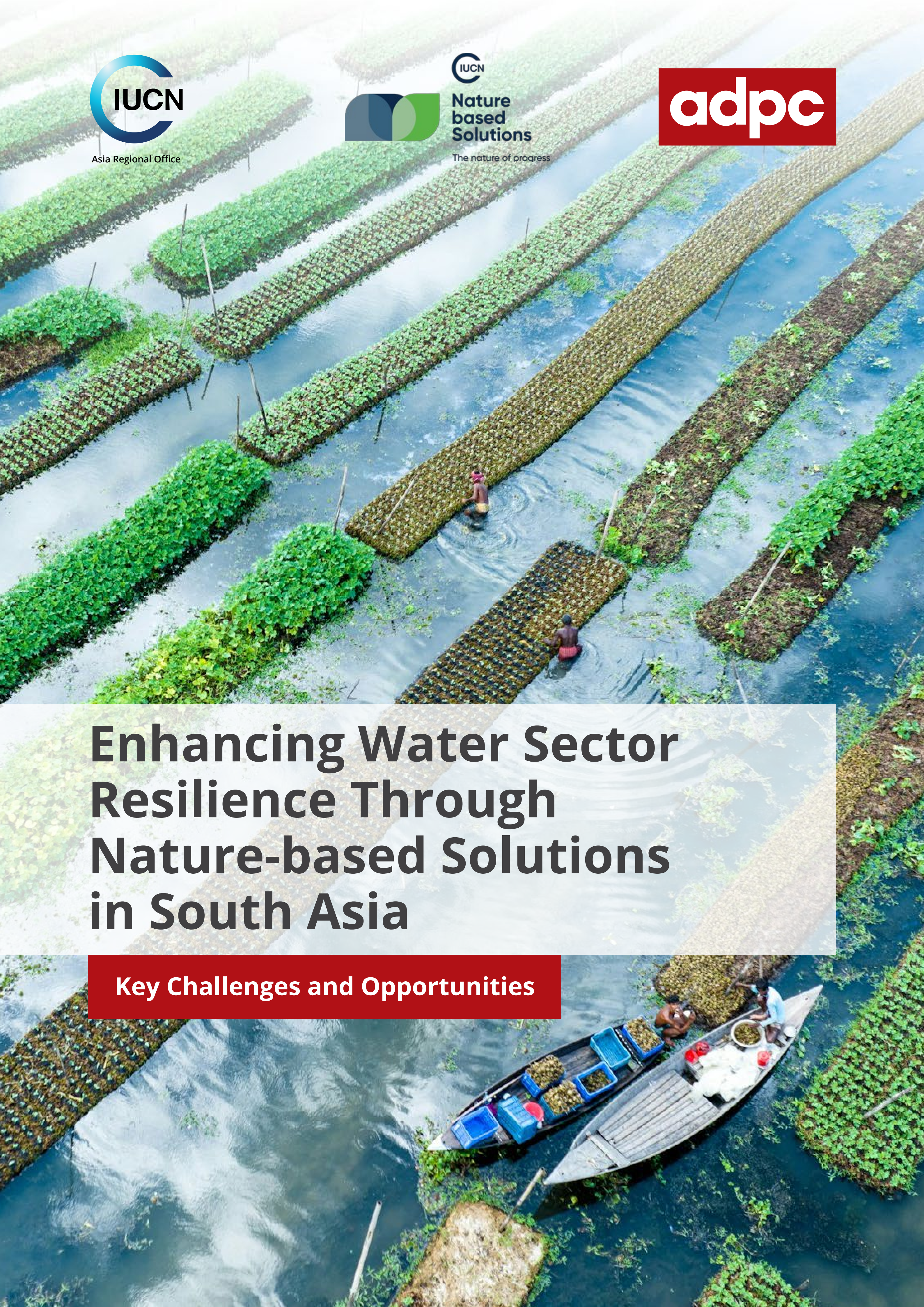

Enhancing Water Sector Resilience Through Nature-based Solutions in South Asia

Key Challenges and Opportunities

Enhancing Water Sector Resilience Through Nature-based Solutions in South Asia:

Key Challenges and Opportunities

Asia Regional Office

This publication is produced jointly by Asian Disaster Preparedness Center (ADPC) and IUCN (International Union for Conservation of Nature), Asia Regional Office as part of the project "Climate Adaptation and Resilience (CARE) for South Asia," funded by the World Bank (WB Reference: P171054). The findings, interpretations, and conclusions expressed in this work do not necessarily reflect the views of IUCN and ADPC, its Board members, or the governments they represent. The boundaries, colors, denominations, and other information shown on any map in this work do not imply any judgment on the part of IUCN and ADPC concerning the legal status of any territory or the endorsement or acceptance of such boundaries.

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Cover Photo: Floating farms in Bangladesh, Mostafijur Rahman Nasim

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Preface

It gives me great pleasure to introduce this guidance document, *"Enhancing Water Sector Resilience Through Nature-based Solutions in South Asia: Key Challenges and Opportunities,"* developed in partnership between the Asian Disaster Preparedness Center (ADPC) and IUCN (International Union for Conservation of Nature). South Asia's growing population and complex water challenges demand solutions that harness natural processes to bolster water security, protect ecosystems, and sustain livelihoods. By leveraging ecosystem functions, we can create adaptive landscapes that support communities and wildlife alike, forging resilience that endures through change.

South Asia, home to over one-fifth of the world's population, consistently ranks among the most climate-impacted regions. It also faces intensifying heat waves, erratic monsoon patterns, glacial melt, and sea level rise. These phenomena exacerbate floods, droughts, and saltwater intrusion, undermining agricultural productivity, human health, and economic stability.

Nature-based Solutions (NbS) offer an integrated pathway to addressing these interconnected challenges. By protecting, restoring, and managing ecosystems—such as wetlands, forests, and floodplains—NbS deliver multiple benefits: regulating water flows, sequestering carbon, enhancing biodiversity, and strengthening community resilience. Across South Asia, pioneering projects demonstrate how NbS can mitigate floods, alleviate drought impacts, and improve water quality, all while supporting local economies and cultural traditions.

This document synthesizes regional learning and policy analysis to identify five strategic priorities, providing actionable recommendations. It also features insightful case studies from the region, such as India's forest carbon targets, Nepal's community forestry models, Pakistan's ecosystems restoration fund, and Bhutan for Life's payments for ecosystem services schemes, demonstrating how these can be easily translated into action

I sincerely thank the World Bank, IUCN, authors, and all government officials for their valuable resources, time, and effort in creating this report. I hope this guidance encourages policymakers and practitioners to embrace Nature-based Solutions in water and disaster risk management, fostering a South Asia where water is safe, ecosystems flourish, and communities prosper.

Aslam Perwaiz
Executive Director
Asian Disaster Preparedness Center (ADPC)

Foreword

IUCN Asia is proud to present this timely and critical publication, *Enhancing Water Sector Resilience Through Nature-based Solutions in South Asia: Key Challenges and Opportunities.* Developed jointly with the Asian Disaster Preparedness Center (ADPC) under the World Bank-funded Climate Adaptation and Resilience (CARE) for South Asia project, this report reflects our shared commitment to advancing sustainable and inclusive solutions to climate and water challenges in one of the world's most vulnerable regions.

South Asia faces profound threats from climate change — from intensified floods and prolonged droughts to saltwater intrusion and ecosystem degradation. These climate-induced pressures severely impact water security, agriculture, and human well-being across the region. As the world strives to move towards the UN Sustainable Development Goals, we must ensure that investments in nature contribute to the health and well-being of people and the planet. In this context, Nature-based Solutions (NbS) offer a transformative pathway by harnessing the power of ecosystems approach to build climate resilience while enhancing biodiversity and providing co-benefits for communities, especially the most vulnerable.

The IUCN Global Standard provides practical guidance for high-integrity and impactful NbS by designing and implementing NbS which harness nature to deliver valuable ecosystem services. In addition, the Global Standard sets clear benchmarks – with composite criteria and indicators -- to measure NbS progress and outcomes. This report provides a comprehensive review of NbS integration within national policies in Bangladesh, India, Nepal, and Pakistan, and outlines clear, actionable recommendations to advance NbS implementation at scale. These recommendations are grounded in extensive consultations with government officials and a robust review of existing frameworks and IUCN's experiences across the region.

At IUCN, we believe that a resilient and water-secure future must be built upon the foundation of healthy ecosystems, human well-being, and inclusive governance. The findings and recommendations from the study – which served as the basis for this publication – are intended to support policymakers, practitioners, and partners in making informed, coordinated decisions that place nature at the heart of climate and water strategies. We are grateful to our

colleagues at ADPC and to all contributors from governments and institutions across South Asia who enriched this process with their insights and commitment.

I encourage readers to share this publication more widely through adapting the content in preparing derived knowledge products, communication materials, and learning events. Let this report serve as a catalyst – not only for mainstreaming NbS in policy and practice – but for transforming the way we approach water resilience and regional cooperation in South Asia.

Dr Dindo Campilan
Regional Director for Asia and Hub Director for Oceania

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This publication, *Enhancing Water Sector Resilience Through Nature-based Solutions in South Asia: Key Challenges and Opportunities,* is a product jointly developed by Asian Disaster Preparedness Center *(ADPC)* and IUCN (International Union for Conservation of Nature), under the project titled "Climate Adaptation and Resilience (CARE) for South Asia," funded by the World Bank (WB Reference: P171054). ADPC and IUCN extend their deepest gratitude to all those who contributed to the development of this report, particularly the government officials across various ministries and departments in Bangladesh, India, Nepal, and Pakistan, for participating in the regional NbS survey and providing invaluable insights on the current application of NbS in the water sector and candidly sharing challenges and opportunities.

ADPC expresses gratitude to Anu Adhikari, Fauzia Bilqis Malik, Rohit Kumar Singh, Tareq Aziz, Yan Yang and Ariela Mcdonald from IUCN's national teams in Bangladesh, India, Nepal, Pakistan, and the Asia Regional Office in Bangkok, for their efforts in conducting surveys, developing national reports, supporting drafting, and copy editing this document.

ADPC also wishes to acknowledge the tireless efforts of its team at the country office and the headquarters in Bangkok for their support to IUCN during the development of this document. We extend our sincere thanks to all stakeholders, partners, and contributors whose valuable input and feedback were instrumental in shaping this report.

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List of Acronyms and Abbreviations

ADB	Asian Development Bank
ADPC	Asian Disaster Preparedness Center
BCCSAP	Bangladesh Climate Change Strategy and Action Plan
BWDB	Bangladesh Water Development Board
CCKP	Climate Change Knowledge Portal
CCMD	Climate Change Management Division
CGWB	Central Ground Water Board
CPCB	Central Pollution Control Board
GPFLR	Global Partnership on Forest Landscape Restoration
CRAP	Climate Resilient Agriculture Program
CRC	Climate Resilient Cities
CRCP	Climate Risk Country Profile
CRWMP	Climate Resilient Water Management Practices
CSO	Civil Society Organizations
DANIDA	Danish International Development Agency
DAC&FW	Department of Agriculture, Cooperation & Farmers Welfare
DBT	Department of Biotechnology
DDMA	District Disaster Management Authority
DHM	Department of Hydrology and Meteorology
DoA	Department of Agriculture
DoFSC	Department of Forest and Soil Conservation
DWSSM	Department of Water and Sewerage Management
ECCCP	Extended Community Climate Change Project-Drought
EPF	Environment Protection Fund
ESG	Environmental, Social, and Governance
FAO	Food and Agriculture Organization
FFC	Federal Flood Commission
GCF	Green Climate Fund
GCISC	Global Change Impact Study Centre
GDP	Gross Domestic Product
GEF	Global Environment Facility
GIZ	German Corporation for International Cooperation
ICIMOD	International Centre for Integrated Mountain Development
IIT	Indian Institute of Technology
IITM	Indian Institute of Tropical Meteorology
IPLCs	Indigenous People and Local Communities
IRSReM	Indus River System Rehabilitation and Modernization
IUCN	International Union for Conservation of Nature
IWMI	International Water Management Institute
IWRM	Integrated Water Resource Management

JICA	Japan International Cooperation Agency
LiFE	Mission Lifestyle for Environment
MoA	Ministry of Agriculture
MoALD	Ministry of Agriculture and Livestock Development
MoCC&EC	Ministry of Climate Change and Environmental Coordination
MoDMR	Ministry of Disaster Management and Relief
MoEFCC	Ministry of Environment, Forest and Climate Change
MoEWR	Ministry of Energy, Water Resources and Irrigation
MoF	Ministry of Finance
MoLGRDC	Ministry of Local Government, Rural Development & Co-operatives
MoWR	Ministry of Water Resources
MoWS	Ministry of Water Supply
NAP	National Adaptation Plan
NARC	National Agriculture Research Council
NAST	Nepal Academy of Science and Technology
NbS	Nature-based Solutions
NBSSLUP	National Bureau of Soil Survey and Land Use Planning
NDC	Nationally Determined Contributions
NDMA	National Disaster Management Authority
NDRMF	National Disaster Risk Management Fund
NIH	National Institute of Hydrology
NIUA	National Institute of Urban Affairs
NPC	National Planning Commission
NTF	NbS Task Force
PARC	Pakistan Agriculture Research Council
PCRWR	Pakistan Council of Research in Water Resources
PMD	Pakistan Meteorological Department
PES	Payments for Ecosystem Services
RCA	River-Cities Alliance
REDD	Reducing Emissions from Deforestation and Forest Degradation in Developing Countries
UNDP	United Nations Development Programme
UNFCCC	United Nations Framework Convention on Climate Change
UN OCHA	United Nations Office for the Coordination of Humanitarian Affairs (OCHA)
WCC	World Conservation Congress
WECS	Water and Energy Commission Secretariat
WEFE	Water–Energy–Food-Ecosystem
WGIN	World Green Infrastructure Network
WRDC	Water Research and Development Center
WRI	World Resources Institute
WRIS	Water Resources Information System
WARPO	Water Resources Planning Organization

Executive Summary

South Asia remains one of the world's most climate-vulnerable regions, with countries such as Bangladesh, India, Nepal, and Pakistan frequently ranking among the ten most affected nations according to the Global Climate Risk Index. These four countries together represent more than 95% of the South Asian land mass and population and is representative of the region. Therefore, this regional guidance was developed based on the learning from these four countries. The impacts of climate change in this region manifest through severe heatwaves, drastic rainfall fluctuations, and accelerated glacial melt. These trends are predicted to worsen, posing severe economic and social challenges. Projections for India and Pakistan indicate that by the end of this century, temperature increases will exceed the global average, potentially reaching approximately five degrees Celsius above levels observed in 1990-2000. Such changes could lead to profound, widespread impacts on South Asian ecosystems, human health, and agricultural productivity.

As temperatures rise and rainfall patterns become less predictable, water-related disasters such as floods, droughts, and water scarcity have escalated, directly affecting food and cash crop productivity across the region. Government surveys in Bangladesh, India, Nepal, and Pakistan have identified three urgent climate-related issues: (a) the escalating frequency and intensity of flash floods, (b) increasing water scarcity and drought vulnerability, and (c) rising sea levels causing saltwater intrusion. Addressing these interconnected challenges is vital for sustaining regional resilience, water and food security.

In this context, Nature-based Solutions (NbS) emerge as a promising, integrative approach to mitigating the impacts of climate change on water resources. NbS focuses on conservation, restoration, and sustainable management of ecosystems to address societal challenges. By harnessing natural processes, NbS contributes to climate mitigation and enhances water and food security, while simultaneously benefiting biodiversity. Many South Asian initiatives already employ NbS, aiming to build resilience against floods and droughts or enhance water security.

A review of climate and water policies in Bangladesh, India, Nepal, and Pakistan reveals that NbS principles are increasingly integrated within national frameworks. For instance, India's Nationally Determined Contributions (NDC) set an ambitious goal to mitigate 2.5 to 3 billion tonnes of carbon through additional forest and tree cover by 2030. Nepal's NDC underscores the importance of community-based forest management, targeting at least 60% of forest area under such systems, with women comprising half of the management committees. These policies offer strategic entry points for designing and upscaling locally led NbS efforts that can bolster resilience across South Asia.

Despite these positive steps, significant challenges impede the effective implementation of NbS. Barriers include limited understanding of NbS and its co-benefits, gaps in institutional capacity, insufficient data on NbS cost-effectiveness, a lack of reliable financing mechanisms, and poorly developed or missing inter-agency coordination mechanisms. Additionally, regional governments often approach NbS implementation in a project-focused manner, which is frequently reactive and uncoordinated, and lacking a broader strategic vision.

Based on the analysis of the context, the guidance suggests five strategic directions that national governments can prioritize to accelerate NbS mainstreaming and upscaling for enhancing water resilience for communities and different economic sectors.

1. Establish a National NbS Task Force to guide the development of a comprehensive NbS strategy for water management: Currently, South Asian countries do not have national-level NbS strategies for water management. This guidance proposes that countries may start with the establishment of an inter-ministerial 'National NbS Task Force for water management (NTF)' to coordinate the development of a national NbS strategy for accelerating the adoption of NbS in water management across sectoral policies, program, as well as national accounting methods and standards. The NTF can be tasked with developing a strategy, highlighting priorities and key water-related issues the country is facing, identify examples and successful initiatives, and link to several national policies and plans. Based on the analysis, the NTF can define country-level objectives, targets, and indicators, along with the preferred strategies for the country, which should be aligned and harmonized with other sectors and approaches. The strategy could highlight the benefits provided by NbS implementation at scale and its linkage to the commitments made by the governments under several Multilateral Environmental Agreements (MEAs), demonstrating the multiple benefits provided by NbS.

The NTF would ideally be hosted by a national institution, coordinating efforts among environmental, agricultural, fisheries, planning, and finance agencies. Annex II of this document provides a comprehensive list of agencies mentioned by participants during regional government survey as important when it comes to building water resilience. To support the activities of NTF, the guidance proposes creation of the following three technical groups, a) Scientific Committee, with representation from academia and governments, to support the documentation of best practices, undertake needs-based research and development of policy recommendations and action plans for community participation; b) Thematic User Groups, with representation from state and provincial level disaster management and climate change authorities, indigenous groups, and women leaders. The user group will bring on the ground perspectives for the design of guidelines and capacity building program; and c) NbS Investment Committee, with representation from relevant disaster and environment management funds and private sector lenders. The technical user group will support the development and financing of proposals for large-scale NbS projects for water resilience.

2. Enhance research and knowledge on NbS for water management: An effective NbS strategy requires a systems-based approach that recognizes the interdependencies among ecosystems, communities, and the economy and builds them into the decision-making process. Considering that the impacts of climate change are manifested through water, a river basin or watershed provides the right scale to understand the ecosystem linkages and their roles in water security and disaster risk reduction. Furthermore, NbS initiatives can build on ongoing research, such as the global Natural Capital Accounting and Ecosystem Services project in India, and the results of the ecosystem services assessment provides input for the development of integrated river basin management strategies. The results of these type of studies provides input to the development of a structured approach for engaging Indigenous peoples and local communities (IPLCs) in the co-management of ecosystems and the design of national and local guidelines and frameworks.

3. Design and implement national capacity-building programs on NbS, tailored to the needs of diverse stakeholder groups: Capacity gaps hinder the effective design and implementation of NbS across South Asia and the lack of comprehensive understanding of the technical aspects of NbS is one of the barriers to the design and implementation of NbS for water resilience. It is being proposed that national governments can start by mapping relevant government agencies, initiatives and academic institutions for the development of sector and context specific training. The capacity building program will build on the comprehensive data collection and assessment of the status of ecosystems, biodiversity, and their links to climate and water resources coordinated by the NbS Scientific Committees. The priority themes for capacity building includes a better understanding of the values and trends in the ecosystem services, enabling provisions of NbS in current climate, water, and disaster management policies, role of traditional knowledge and indigenous governance systems in the design of NbS at the landscape level, and role of wetlands and forest-based adaptation in floods and drought resilience. The national capacity-building initiatives could be funded through existing climate and disaster management funds, promoting comprehensive ecosystem data collection and assessments managed by NbS Scientific Committees.

4. Develop national financing strategies for NbS: Financing remains a major barrier, as current funding is primarily project-based. The government survey identified a diversity of financing schemes for NbS applied by countries, but also highlighted an overall gap in financing NbS. Countries are making efforts to develop national financial mechanisms for ecosystem restoration and NbS financing, such as Pakistan's Ecosystem Restoration Fund and the Bhutan For Life (BFL) initiative. Financing of NbS is an important catalysing factor for developing a national NbS strategy and needs to be developed at the country level. Several options for financing NbS exist. Grants and public funding such as the Green Climate Fund (GCF), Adaptation Fund, International Climate Initiative (IKI) of the German Government, and Global Environment Facility (GEF) remain key sources of funding for medium to large NbS initiatives. Accredited agencies including the World Bank and IUCN can help countries to develop large-scale NbS programs.

Furthermore, for water related NbS, blended finance can be particularly effective for larger projects or those with the potential for revenue generation. For climate mitigation and water related NbS projects, blended finance can be particularly effective for national or basin scale projects which combine climate and conservation outcomes with a long-term vision. The example being Bhutan for Life (BFL), the BFL fund till date has mobilised USD 7,270,034 of GCF funds and USD 9,996,940 of private donor funding (GCF 2022. FP050). Additionally, through the BFL project government is facilitating signing of local level PES schemes where upstream communities (service providers) commit to manage the watersheds and water resources and downstream water users such as municipality, hotels and institutions (service users) pay a certain fee to the community groups. Through the PES scheme, the BFL in incentivising the service providers carry out activities such as cleaning springs and rivers, control overgrazing and overharvesting of forest resources. In 2015, a PES agreement was formalised in Namey Nichu watershed, located in Paro district. Currently watershed assessment is ongoing in several locations, including Gangzur Gewog in Lhuntse District, and Langthel and Korphu villages in Trongsa Dzongkhag district.

5. Promote regional exchange and cooperation on NbS: Regional cooperation is limited, with few NbS-related dialogues within the South Asian Association for Regional Cooperation (SAARC). The government survey identified limited regional level interactions between South Asian countries on topics linked to NbS and water. The SAARC supported dialogues currently do not include NbS-related topics. Governments and development partners must continue advocating that SAARC includes NbS and consider it as a key strategy for regional development. The implementation of regional, transboundary NbS projects are often required in the water sector as several river basins in South Asia are transboundary. This is sometimes difficult to implement as countries tend to prefer national-level initiatives, but potentially piloting NbS projects in small, shared watersheds are recommended to demonstrate the benefits of regional cooperation.

The guidance is designed for national, subnational, and local policymakers and government agencies responsible for disaster management, environmental protection, water, land, climate change, and rural development. The guidance has been developed through extensive interaction with government officials in Bangladesh, India, Nepal, and Pakistan, and backed by a review of academic and policy literature published by reputed institutions, academic journals and news media.

The guideline aims to foster improved inter-agency coordination and policy alignment. Implementing the recommendations provided in the guideline can aid in designing and executing river basin or landscape-level water management plans that address the pressing climate, water, and disaster challenges facing South Asia.

Photo Credit: Asikul Islam Himel

CHAPTER 1

Introduction

South Asia is one of the regions most severely affected by climate change. This chapter outlines the background, objectives, and methodology used in developing this guidance, and provides a summary of its key contents.

South Asia is one of the regions most impacted by climate change in the world. As per World Bank (WB) estimates, more than half of all South Asians, or 750 million people, were affected by at least one natural hazard in the last two decades. If the trend continues, it is estimated that annual economic losses in the region will average US$160 billion by 2030. (UN News, April 2018).

The projected shifts in temperature and precipitation patterns, increased glacier melting, and fluctuations in river flow will impact the availability of water resources for drinking and irrigation.

This, coupled with the increasing population pressure and the haphazard land-use changes, will drive the loss of forests and wetlands, which serve as natural water infrastructure, further enhancing the vulnerability of communities and economic sectors to climate change.

Globally, and in South Asia, Nature-based Solutions (NbS) are gaining recognition among governments, the private sector, and researchers as a proven approach for addressing societal challenges, including climate change impacts and food and water insecurity. NbS interventions conserve or restore ecosystems to address societal challenges, while also providing biodiversity benefits. NbS encompasses various strategies, including Ecosystem-based Adaptation (EbA), Ecosystem-based Disaster Risk Reduction (Eco-DRR), and Forest Landscape Restoration (FLR), which are well recognized in environment and climate change policies in South Asia.

1.1 NbS in policies

The Nationally Determined Contributions (NDCs) of Bangladesh, India, Nepal, and Pakistan include NbS measures such as ecosystem restoration, afforestation, and tree cover enhancement to create additional carbon sinks. For example, the NDC of Bangladesh sets a restoration target of 150,000 ha of forest in the coastal region and islands by 2030. Similarly, the India NDC sets a target of 2.5 to 3 billion tonnes of carbon mitigation through additional forest and tree cover by 2030. Beyond the inclusion of specific land-use targets, the NDC of India also emphasises creating a mass movement for behavioural changes at the individual level as key to combating climate change. The Indian government launched the Lifestyle for Environment - LiFE initiative, to promote a sustainable way of living that is based on traditional values and the protection of nature. Such initiatives provide entry points for the national government to design large-scale capacity-building programmes on sustainable behaviour, value chains, and markets to incentivise investments in NbS. Similarly, by linking the NDC targets to rural and urban development projects, local governments and Indigenous People Organisations (IPOs) can leverage funding for addressing locally relevant societal challenges.

The NDC of Nepal targets the development of a nature-based ecotourism plan for at least five tourist destinations by 2025 and achieve carbon-neutral status for at least five tourist destinations by 2030. The recognition of the role of women in forest and natural resource governance is strongly integrated into indicators such as, "at least 50% participation of women in the forest management committees". The Pakistan NDC identifies several large-scale national-level NbS flagship projects for achieving the NDC targets, including the Ten Billion Tree Tsunami (TBTT) and Recharge Pakistan. The NDC estimates the TBTT programme will help sequester more than 148 mega tonnes of carbon dioxide in ten years, whereas the Recharge Pakistan will reduce flood risk and enhance groundwater recharge across the Indus river basin. This approach has helped the country mobilize financing from the Green Climate Fund (GCF) for large-scale initiatives.

It is not only the NDCs and National Adaptation Plans (NAPs) of South Asian countries that are integrating NbS as an approach to support climate change adaptation and mitigation. National water and disaster management policies are also acknowledging the value of nature and ecosystem services in achieving national targets. In India, the National Water Policy (NWP) 2012, explicitly recognizes the need for the conservation of rivers, river corridors, water bodies, and water infrastructure in a scientifically planned manner, and through community participation. Chapter Four of this Guidance presents a more detailed discussion of the specific NbS enabling provisions in the NDCs, NAPs, and national water and disaster management policies of Bangladesh, India, Nepal, and Pakistan. At the national level, the policies will benefit from the review and adoption of NbS enabling provisions and good practices discussed under different policies in this guidance.

1.2 Need for NbS guidance

The design of NbS requires a full understanding of the interaction between society, economy, and ecology at the landscape or river basin scale, as well as the engagement of different sectors and stakeholders in the design and implementation of solutions. Although NbS or ecosystem-based approaches are included in policies across South Asia, the absence of an institutional coordination mechanism for the design and implementation of large-scale NbS initiatives is preventing integrated application of NbS to address water resilience issues, such as floods and droughts.

To strengthen regional understanding of gaps and opportunities in enhancing NbS application in water sector in South Asia, IUCN collaborated with the ADPC under the regional Climate Adaptation and Resilience (CARE) for South Asia project, funded by the World Bank, to develop this:

Enhancing Water Sector Resilience Through Nature-based Solutions in South Asia.

The guidance aims to support increased application of NbS in South Asia by:

i. Providing an overview of how climate change is impacting water resources in Bangladesh, Bhutan, India, Nepal, and Pakistan;
ii. Defining ecosystem services and NbS and analysing the extent that NbS principles are integrated into policies in the region;
iii. Documenting the policy and planning strategies currently promoted by governments to mitigate the impacts of climate change, including opportunities and barriers in the application of NbS for building water resilience;
iv. Suggesting a pathway for achieving better inter-agency coordination for filling gaps in capacity, knowledge, and policies for the design of inclusive NbS at scale.

1.3 Methodology

This regional guidance analyses inputs from a government survey and is supported by a review of policies and online literature available from national ministries, academic institutions, peer-reviewed articles, and intergovernmental organizations such as the IPCC, the UN agencies, and the World Bank.

IUCN conducted a regional survey through face-to-face interviews with government officials, from May to August 2023, interviewing more than 30 government officials from Bangladesh, India, Nepal, and Pakistan. The survey targeted government officials in central ministries responsible for water, environment, climate change, and agriculture sectors. Survey participants were mid- to senior-level government officials, working in different capacities as project directors and joint secretaries, as well as technocrats and experts working on irrigation, hydrology, engineering, geology, and conservation. The guidance also incorporates inputs provided by government officials during the "Regional Online Training and Dialogue on NbS for the Resilient Water Sector in South Asia" facilitated jointly by IUCN and ADPC from 16 - 18 January 2023. The training was attended by more than 30 policymakers from South Asia.

To develop the report, a team of legal experts from the Indian Environment Law Organisation (IELO) in New Delhi supported the assessment of the NbS enabling provisions in climate, water, and disaster management policies. For the policy analysis, the following criteria were developed to assess NbS integration in the selected national policies in Bangladesh, India, Nepal, and Pakistan:

1. Recognition of NbS and/or ecosystem management in building water resilience
2. If NbS is recognized, does the policy promote the inclusion of communities in the decision-making process?
3. Does policy recognize the role of biodiversity, ecosystem integrity and connectivity in achieving policy objectives?
4. Whether policy promote cross-sectoral collaboration?
5. Does the policy include mechanisms for regular monitoring and upscaling of best practices?
6. Does the policy recognize the need for financial innovation and the development of market-based mechanisms?

Based on the analysis of gaps in current policies, and the challenges in translating enabling policy provisions on the ground identified by the regional government survey, the guidance suggests five actions to promote and accelerate the mainstreaming of NbS for water resilience at national and local levels. This includes the creation of NTF to facilitate the inter-ministerial and sectoral coordination related to the development of nationally relevant NbS guidelines, as well as research and capacity-building initiatives that will support the development of large-scale and landscape-level NbS initiatives required to tackle water resilience issues in South Asia.

1.4 Target audience

The intended audience for this guidance is officials from the central ministries and their agencies working on water, forest and climate change, disaster management and rural development in the South Asia region. The officials from these agencies will benefit from a clear understanding of the NbS concept and what is needed for its application at national or river basin scale. The officials managing local governance and rural development, particularly those working on enhancing rural communities' resilience to climate change impacts and achievement of sustainable goals will benefit from better insights on the role of NbS in providing multiple benefits and help them build arguments for its integration into their work and government policies.

The officials engaged in or responsible for national and local level development planning and coordinating of development projects, will benefit from a better understanding of the role of ecosystem services and NbS, and current government strategies and enabling policy provisions they can leverage to facilitate it's on ground implementation. Considering the critical role of ecosystem management in disaster risk reduction, those working with disaster management issues will get better insights for prioritising their long-term actions for mainstreaming of NbS in cross-sectoral dialogue and planning. The officials and experts working on road and infrastructure development will gain a better understanding of what NbS means within a landscape and why it is important to consider ecosystem connectivity in infrastructure planning.

1.5 Guidance structure

The guidance is structured into an introductory chapter and an eight additional chapters.

Chapter 1: Introduction

The introductory chapter provides the background, objective, and methodology for the development of this guidance and a summary of the content discussed in each chapter.

Chapter 2: Climate Change Drivers and Risks to the Water Sector

Provides an overview of trends in temperature and rainfall in Bangladesh, Bhutan, India, Nepal, and Pakistan, and the role of anthropogenic factors in magnifying climate vulnerabilities. Based on the review of inputs from a regional government survey, the chapter presents three pressing threats to water resilience in South Asia, a) increasing frequency and intensity of flash floods; b) water scarcity and droughts; and c) sea level rise and salt intrusion. The chapter also highlights the potential of NbS in addressing climate change-related water risks, and shares examples of NbS frameworks and case studies from Asia and globally.

Chapter 3: Understanding Ecosystem Services and NbS

Introduces the reader to the concept of ecosystem services and NbS definition, demonstrating the various types of NbS and examples of NbS in a river basin context.

Chapter 4: Review of NbS in Government Policies

Provides an assessment of the degree of integration of NbS in the NDCs, NAPs, National Water Policies, and National Disaster Management Policies in Bangladesh, India, Nepal, and Pakistan. The chapter highlights relevant provisions and targets that could be leveraged by government officials working at different levels during the design and approval of large-scale projects on NbS water resilience.

Chapter 5: Financing for Ecosystem Protection and Water Resilience in South Asia

Provides examples of financing strategies adopted by South Asian countries. These include the development of national-level environment and disaster management funds, testing of Payment for Ecosystem Services (PES), and policy measures for enhancing private sector engagement and funding for ecosystem-based approaches to water resilience. This chapter also includes examples of bilateral and multilateral funding mechanisms leveraged by Bangladesh, Bhutan, India, Nepal, and Pakistan.

Chapter 6: Analysis of Barriers to NbS Implementation at Scale in South Asia

Presents a summary of key challenges to implementing NbS at scale, and the strategies that governments are applying to address them. This chapter is based on the analysis of inputs received from the regional government survey, and groups the barriers into the following categories: a) Governance; b) Capacity; c) Research and Knowledge; and d) Financing.

Chapter 7: Recommendation for Mainstreaming NbS for Water Resilience

Provides recommendations for promoting the mainstreaming of NbS in water resilience initiatives, starting from the constitution of the National-level NbS Task Force (NTF) multi-agency and multi-disciplinary platform of different government agencies, including those who need to work together to design a national NbS strategy for water management and address the current knowledge, capacity, and funding gaps. The chapter ends with a short conclusion section, providing a summary of the main findings and recommendations to address these barriers.

Chapter 8: Conclusion

Provides a summary of South Asian climate context, and references the barriers, and enabling policy provisions discussed in the guidance that could be leveraged by the target audience to support the implementation of recommendations.

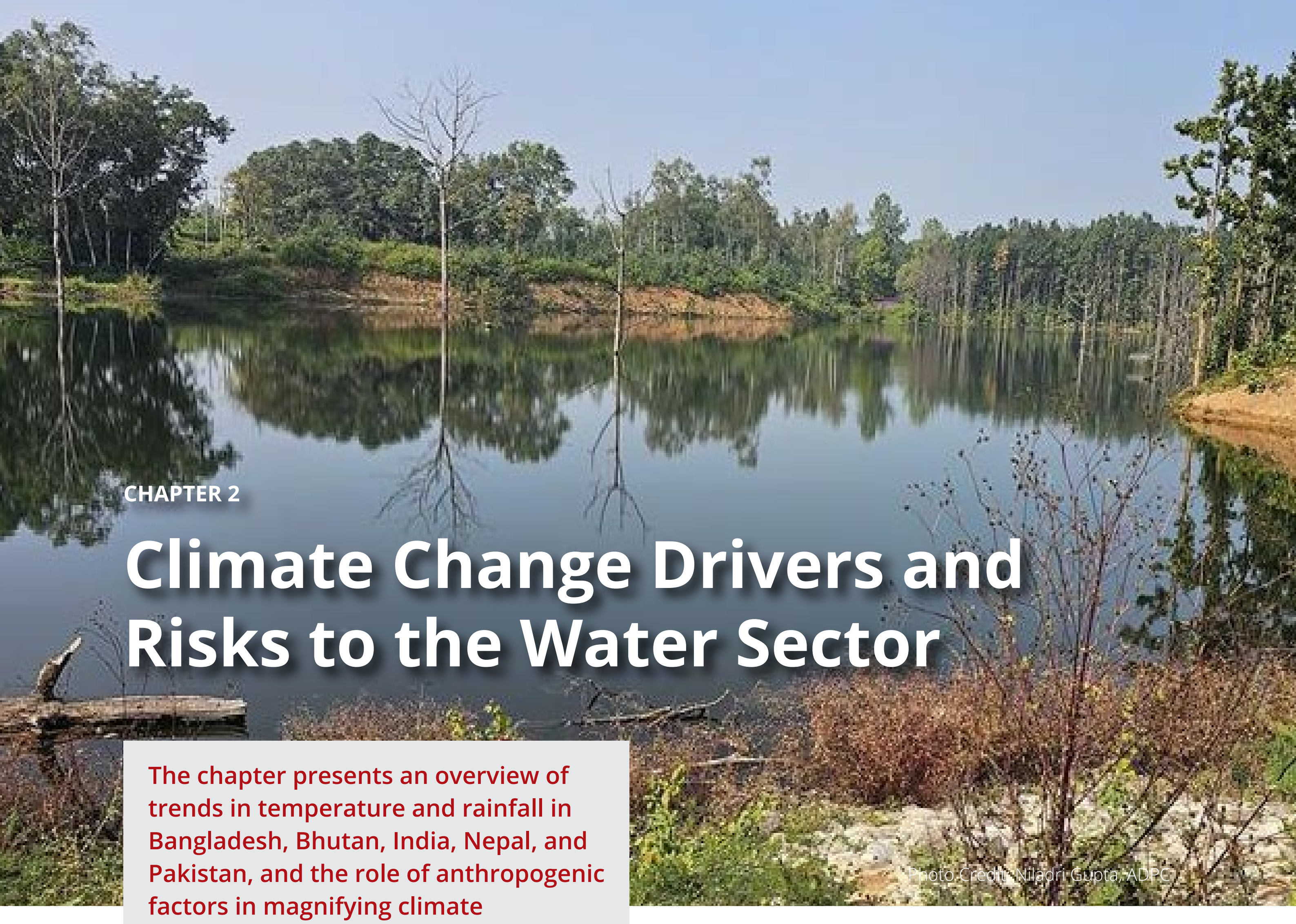


Photo Credit: Niladri Gupta, ADPC

CHAPTER 2

Climate Change Drivers and Risks to the Water Sector

The chapter presents an overview of trends in temperature and rainfall in Bangladesh, Bhutan, India, Nepal, and Pakistan, and the role of anthropogenic factors in magnifying climate vulnerabilities. Based on the review of inputs from a regional government survey, the chapter presents three pressing threats to water resilience in South Asia. The chapter also highlights the potential of NbS in addressing climate-related water risks, and presents relevant frameworks and case studies from both Asia and around the world.

2.1 Climate as a driver of change in South Asia

Bangladesh, Nepal and Pakistan were in the top ten list of most affected countries by climate change from 2009 to 2019 (Global Climate Risk Index 2021). The Intergovermental Panel on Climate Change (IPCC) Six Assessment Report (ARC6 2021) warns of escalating extreme weather events in the future, adversely affecting up to 800 million people in the South Asia region.

The region is also witnessing heightened heatwaves, intense monsoons, and glacial melting, contributing to rising sea levels. The IPCC's "Special Report on the Ocean and Cryosphere in a Changing Climate" projects that sea level rise in South Asia will increase from 3–4 mm/year currently, to 10–20 mm/year at the end of the century (RCP8.5). This will lead to salinization, frequent and intense coastal flooding, enhanced coastal erosion, and the loss of coastal ecosystems.

Below is a summary of the temperature and rainfall projections in selected South Asian countries.

Bangladesh

The temperature is projected to increase between 0.44°C to 0.69°C in the near term (the 2030s), and 1.3°C to 2°C in the mid-term (2050s) under various scenarios (IPCC ARC6 2021). This will contribute to moisture stress, particularly in the Barind region, one of the most drought-prone areas in the country.

By the 2030s, annual rainfall will be slightly higher or similar across the country. The rainfall variations for the country will range from 0.1% to 1.4% by the year 2030 and 2.4% to 3.5% by 2050, resulting in varying impacts in different regions of the country. The northeastern and eastern hill regions will receive higher rainfall, while the western region will receive lower-than- average rainfall. By the 2050s, rainfall is projected to increase all over the country (NAP 2022) leading to a higher incidence of floods.

Bhutan

An increase from 0.8°C to 3.2°C between 2021-2100 is projected for the country, based on various scenarios (RCP 4.5 and RCP 8.5, IPCC AR6). Warming will be more significant during the spring and winter seasons in Bhutan. An increase in temperature will be seen across the country, but the greatest increase is projected in the highlands.

The mean annual rainfall in Bhutan is projected to increase by 10-20% between 2021 and 2050 (RCP 8.5). Despite the overall increase in rainfall, the north and north-western areas are expected to experience decreases in rainfall and droughts.

India

In a high-emission scenario, the temperature in India could rise by 5.1 degrees Celsius by the end of this century (Kumar et al. 2023). Due to increasing temperature, the length and intensity of heat waves are projected to increase by 2,515%, leading to a 25-fold increase in heat-related deaths in the country compared to the 1990 level (G20 Climate Risk Atlas, India). The analysis of temperature data collected from 125 stations across India, by the National Institute of Hydrology (NIH) India indicates an increase of 0.42°C, 0.92°C, and 0.09°C in annual mean temperature, mean maximum temperature, and mean minimum temperature, respectively, over the past century.

The review of rainfall patterns by the Indian Institute of Tropical Metrology (IITM) indicates a substantial spatial difference in the projected rainfall changes. Central India will experience an increase of 10-30% in rainfall, the maximum expected increase in the country. The World Bank Commissioned Potsdam Institute for Climate Impact Research and Climate Analytics for India indicates a 2°C rise in the world's average temperatures will make India's summer monsoon highly unpredictable, whereas, at 4°C warming by the end of this century, the frequency of extreme rainfall will increase tenfold. India's northwestern and southeastern coastal regions could see higher than average rainfall.

Nepal

In Nepal, the mean temperatures are expected to rise by 0.92-1.07°C in the near term (the 2030s) and by 1.3-1.8°C in the long term (the 2050s) (MoFE 2021). Higher mountain elevations and the Terai region are anticipated to experience the greatest changes.

The post-monsoon rainfall is projected to increase by 20% in the long term (by 2050s), with greater uncertainty compared to temperature forecasts (MoFE, Nepal 2021). An overall increase in the number of extremely wet days is projected, increasing the risk of water-related hazards. The north-western and northern districts are projected to be impacted most.

Pakistan

Pakistan's Climate Risk Country Profile documents that the rate of temperature increase will be higher than the global average with a potential rise of 1.3°C to 4.9°C by the 2090s against the 1986–2005 baseline (WB and ADB 2021). Projections highlight higher variability in the annual maximum and minimum temperature, with impacts on human health, livelihoods, and ecosystems. The northern regions of the country will be most impacted by extreme heat waves.

Regarding rainfall trends, the country's climate profile suggests an overall increase in precipitation in the Upper Indus Basin and a decrease in the Lower Indus Basin. The number of people affected by flooding will increase to 5 million people between 2035-2044, and around 1 million people annually are expected to be exposed to extreme river bank erosion and coastal flooding by 2100 (WB and ADB 2021).

2.2 Anthropogenic factors magnifying climate vulnerability

South Asia is one of the most populated and fastest-developing economic regions in the world. However, high rates of development are also putting increasing pressure on land due to urbanisation, as well as agriculture and industrial expansion. Some of these anthropogenic factors that contribute to increased vulnerability to climate change are discussed below:

Urban expansion

The World Bank report, Leveraging Urbanization in South Asia, estimates that 250 million people will be living in urban areas by 2030. With an increasing number of people in cities, water security is becoming a challenge, particularly in megacities with a very high population growth rate such as Delhi, Dhaka, and Kathmandu.

Rapid urban expansion along the steep slopes in the Himalayan region, particularly in the flood plains of the rivers and the coastal areas, has increased vulnerability to landslides, erosion, and flash floods.

The recent floods in Sylhet, Bangladesh demonstrate how the loss of wetlands and a lack of landscape or basin planning contributes to increased urban flood risks. The city is located in the northwestern part of Bangladesh in the Haor region, known for its numerous wetlands. In 2022, Sylhet experienced the worst flood in recorded history, affecting 70% of the city and impacting both urban and rural areas, with more than 10,000 houses destroyed (TBS News 2023). Although climate change certainly had an impact, experts warned against attributing the current floods in the city solely to climate change, emphasizing the loss of natural wetlands, locally referred to as 'haors,' and their degradation as a significant factor contributing to flooding.

Across South Asia, the number of coastal cities and their populations have been increasing rapidly. India's coastal region is home to about 170 million people (UN OCHA 2020), and approximately 6% of this population live in low-lying coastal areas (IITM 2022). Furthermore, 26% of the Indian coastline is prone to erosion, and 450 hectares of coastal land is lost every year (IITM 2022), leading to impacts on infrastructure and putting individuals at risk. Urban expansion in coastal areas has led to the loss of mangrove forest cover, making coastal communities highly vulnerable to sea-level rise, erosion, and natural hazards such as tropical storms and cyclones.

Agriculture expansion and increasing water demand

The agricultural sector is the main source of livelihood for more than 60% of the population of South Asia (FAO and IFAD 2019). It is the biggest consumer of surface water resources, and therefore also the sector most impacted by droughts and water scarcity. More than 90% of surface water is consumed by the agriculture sector in the South Asia region (Golam Rasul 2014). Increasing water demand from the agricultural sector impacts the availability of water for domestic water supply, industrial activities, and hydropower generation, leading to increased competition for water resources among sectors. There is a need to move away from water-intensive crops and apply landscape and basin-level planning approaches to increase water use efficiency in the agricultural sector.

To meet the ever-increasing demand for water in agriculture, countries are investing in inter-basin water transfer canals and dams to meet agricultural demands for water. During the regional government survey, participants from India, Pakistan, and Nepal suggested inter-basin water transfers as a viable management option to meet the demand for water for irrigation and manage climate-induced droughts and floods. In the Indus basin, Pakistan has developed several river link canals, linking the Chashma River with the Jhelum River and the Ravi River with the Sutlej River, allowing for the transfer of significant amounts of water between these river basins for irrigation.

In India, there has been an ongoing discussion on the National River Linking Project (NRLP) which will connect 44 rivers via 9,600 km of canals (Higgins et al. 2018). In 2023, forest clearance was granted for the construction of 5.372 billion USD (44,605-crore) Ken-Betwa River Link Project (KBLP) (Sirur 2023). While clearance has been provided for KBLP, the NRLP project remains under consideration for its impact in India and downstream Bangladesh. Studies have highlighted that the changes to mean annual water discharge will impact 29 rivers in the case of full implementation of the NRLP, leading to significant impacts (Higgins et al. 2018). The study forecasts reductions in annual suspended sediment transport to deltas by 40–85% in the Mahanadi river basin, 71–99% in the Godavari river basin, and 60–97% in the Krishna river basin due to reservoir trapping and peak streamflow reductions. Before its confluence with the Brahmaputra, the Ganges is projected to experience a 39–75% reduction in annual suspended load (Higgins et al. 2018).

Pollution impacts on water quality and availability

Uncontrolled industrial discharge and untreated sewage pose a threat to water quality and aquatic biodiversity throughout South Asia. The contamination of rivers and water bodies changes the local species composition and contributes to the loss of unique biodiversity within the river basins. The loss of biodiversity affects the functioning of the aquatic ecosystem and compromises its ability to provide services, such as the provision of clean water for consumption and agriculture. Addressing these challenges requires the integration of sustainable land use planning, enforcement, and protection of natural water infrastructure in climate adaptation and disaster management strategies.

2.3 Priority water resilience issues in South Asia

During the regional survey, government officials highlighted several priority climate change issues impacting the water sector, including the increasing frequency and intensity of flash floods, water scarcity, and increasing vulnerability to droughts, sea level rise, and salt intrusion.

Increasing frequency and intensity of flash floods

The altered rainfall and glacial dynamics in the Himalayan region have contributed to a significant increase in flood events. Mega-floods and landslides in Himachal Pradesh and Uttarakhand, India in August 2023 impacted over 150,000 people, resulting in the death of 91 people in the nearly 50 incidents of landslides and dozens of flash floods in the hilly region, with more than 1,000 roads remaining unusable (UN OCHA Services).

Changes in rainfall patterns, leading to heavier and more intense precipitation events are disproportionately impacting low-lying countries like Bangladesh. This is particularly true in the Haor region in Bangladesh, which is characterized by a high density of wetlands. In 2022, flash floods affected 7.2 million people in the Sylhet region, causing damage to croplands, water storage ponds, and sanitation facilities (UN OCHA Services). Similar challenges were observed in Pakistan, where the 2022 mega floods impacted 33 million people, resulting in an estimated economic loss of US$ 30 billion (WB 2022).

In Nepal, government survey participants shared concerns regarding the impacts of landslides on the agricultural sector, due to increasing precipitation and floods. Flood frequency is projected to increase in the future, amplifying threats to agriculture and the livelihoods of vulnerable farming communities (MoALD 2019). The Post Flood Recovery Need Assessment Report of the Nepal Planning Commission (NPC) highlighted that the 2017 floods affected more than half (58%) of Nepal's agriculture and livestock-based livelihoods. The report also estimated an economic loss of about US$ 69.5 million in agriculture (11.9%), US$ 102.7 million in livestock (17.6%), and US$ 168.1 million in irrigation infrastructure (28.8%).

Flooding in the Himalayan and Terai regions is also caused by Glacial Lake Outburst Floods (GLOFs), which occur when there is a sudden release of water from a glacier-fed lake in high altitudes of the Himalayas. As global temperatures continue to rise, glaciers in the Himalayas are melting rapidly, resulting in new glacier lakes and expanding those that already exist. In 2023, a GLOF event in South Lohanak Lake in Sikkim, India, caused flash floods in the Teesta River downstream, claiming the lives of more than 100 people draining more than 105 hectares of land, and causing extensive damage to infrastructure (Livemint News 2023).

Water scarcity and increasing vulnerability to droughts

Changing rainfall patterns, increasing temperatures, and enhanced evaporation are contributing to water security and droughts in many parts of South Asia. In Bangladesh, Bhutan, India, Nepal, and Pakistan this is particularly a concern because of the resulting impacts on agricultural productivity and national food security. Increasing temperatures and water loss will contribute to a decline in the productivity of food and cash crops, including cotton, wheat, sugarcane, maize, and rice.

The Barind region in northwestern Bangladesh is one of the most drought-prone areas of the country. The region is impacted by a longer spell of dry days, of 32-48 days annually, and temperatures above 40°C. This has affected the availability of water for agriculture, sanitation, and health services. Droughts are also projected to have large-scale impacts on rice production, with an overall decline of 17 percent by 2050 (NAP Bangladesh 2022).

Across the Himalayan region, the drying of springs, the main source of water for household use and irrigation, is becoming a challenge. Bhutan's Watershed Management Division reports that approximately 25% of the country's spring-shed is running dry. In Bhutan, India and Nepal the revival of springs is recognized as an important strategy for achieving household water security, and investing in research and implementation of spring revival plans by rejuvenating their watersheds. In Bhutan, organizations like the Tarayana Foundation and the Royal University of Bhutan College of Natural Resources are actively involved in investigating springshed revival through the mapping of springshed catchment areas.

In Nepal, more than 150 drought events from 1971 to 2007 affected about 330 thousand hectares of agricultural land (UNDP 2009). The drought in the eastern region decreased rice production by 30% (Gahatraj et al. 2018), whereas heavy flooding in the western regions during 2006 and 2008 destroyed millions of dollars of crops (Practical Action 2008).

In India, the number of districts impacted by droughts is increasing. In 1972, the National Irrigation Commission identified nearly 67 Indian districts that were prone to droughts distributed in eight Indian states. However, by 2023, the situation was worse than projected, with the Indian Meteorological Department (IMD) declaring over 500 districts (78% of total districts) impacted by meteorological drought conditions, ranging from mildly dry to extremely dry.

In Pakistan, droughts are becoming more frequent and severe due to climate change. The country has experienced at least five major drought episodes between the year 2000 to 2018. The most intense and prolonged episode of drought was in 2000-2002, which affected over 60% of Pakistan (Latif, M et al. 2024).

Countries are applying several strategies to address drought issues. In Bangladesh, the government is actively advocating for interagency partnerships and collaboration with the academic sector and civil society organisations in climate resilience projects. Noteworthy examples include initiatives promoting agricultural resilience, such as the Climate-Smart Agricultural Water Management Project funded by the World Bank. This project is jointly implemented by the Bangladesh Water Development Board (BWBD), the Department of Agricultural Extension (DAE), and the Department of Fisheries (DoF). The primary objective of the project is to enhance flood management, improve drainage, and irrigation infrastructure systems, and foster climate-smart agricultural production and marketing.

Sea level rise and salt intrusion

Changes in precipitation patterns, coupled with sea level rise, are leading to increased frequency and severity of coastal flooding events. This is particularly a concern for coastal areas of Bangladesh, India, and Pakistan. Across South Asia, annual coastal flooding damages infrastructure, disrupts water supply systems, contaminates surface water sources, and increases the risk of waterborne diseases. Along the Indian coast, the sea level has risen by 8.5 centimeters in the past 50 years, and predictions indicate that 36 million Indians are likely to be living in areas experiencing chronic flooding, cyclones, and coastal erosions by 2100 (UNDRR 2020).

In the regional survey, government officials shared concerns regarding the impact of floods and sea-level rise on water, sanitation, and hygiene (WASH) services due to disruptions in water supply systems and source contamination. In 2020, tropical Cyclone Amphan displaced nearly five million people across India and Bangladesh. In countries like Bangladesh, such events lead to the shortage of safe drinking water due to the mixing of seawater with natural freshwater resources. At the household level, this will disproportionately impact women, who are responsible for collecting drinking water for their families. Increasingly, saline drinking water may also result in health hazards, especially for pregnant women.

CHAPTER 3

Understanding Ecosystem Services and NbS

This chapter introduces the concept of ecosystem services and NbS definition, demonstrating the various types and examples of NbS in a river basin context are discussed.

Photo Credit: Vishwa Sinha, IUCN

NbS offer a holistic approach to addressing the water-related impacts of climate change. By definition, NbS involve the conservation, restoration, and protection of natural processes and ecosystem services to address societal challenges such as climate change. The section below explores the potential of NbS in climate mitigation and adaptation, and the need to understand and document ecosystem services and the criteria for designing and implementing NbS for the water sector based on IUCN Global Standard for NbS.

3.1 The potential of NbS for climate mitigation and adaptation

By leveraging the power of natural ecosystems, NbS can enhance resilience to climate extremes, mitigate greenhouse gas emissions, and provide multiple co-benefits for communities and biodiversity. Globally and in South Asia, there are several examples of successful NbS implementations that demonstrate their potential in mitigating and adapting to climate change. There is a need to document the cost-benefit analysis of these success stories and use them for the development of Standard Operating Procedures (SOP) to facilitate their replication regionally.

Research highlights that NbS could provide around 30% of the cost-effective mitigation needed by 2030 to stabilise warming to below 2°C and thus can be a powerful defence against the impacts and long-term hazards of climate change. Therefore, finding ways to work with ecosystems, rather than relying solely on conventional engineered solutions, can help communities adapt to climate change impacts. Using nature to green cities can also result in significant energy savings and health benefits.

In the United States, forests play a crucial role in climate mitigation by absorbing carbon dioxide at a rate equivalent to 10% of the country's annual emissions (The White House 2022). This not only helps in reducing greenhouse gases but also supports over 2.5 million jobs and contributes approximately $282 billion annually to the national economy through the forest products industry (USDA Forest Service 2021). The restoration and conservation of forests have proven to be effective NbS, enhancing carbon sequestration and providing flood control, water purification, and habitat for biodiversity.

The European Union (EU) has been a pioneer in promoting NbS through various initiatives, including the EU Adaptation Strategy and the Nature Restoration Law. The Tullstorp Stream Project in Sweden is a notable example, where a holistic approach to river basin management has led to significant improvements in water quality and agricultural resilience. Since its inception, the project has restored 39 wetlands and 10 km of the stream, reducing nitrogen content by 30% and phosphorus content by 50%, thereby enhancing flood resilience and agricultural productivity (European Climate Adaptation Platform Climate-ADAPT, n.d.).

Form the South Asia region, Bangladesh which is facing significant climate risks, has developed a Bangladesh Delta Plan 2100 (BDP 2100) that integrates NbS to enhance flood control, improve water quality, and support agricultural productivity. The plan includes strategies for the restoration of natural reservoirs and water bodies, along with biodiversity conservation. These NbS approaches are integrated into the broader framework of the BDP 2100, highlighting the importance of leveraging natural processes to enhance water resilience and sustainability.

India is promoting NbS through National Water Policy and various national missions, such as the Clean Ganga Mission and the Green India Mission. By integrating green-grey infrastructure solutions, these initiatives aim to improve water quality and availability for household and agriculture and reduce the risk of floods and droughts. Additionally, the restoration of mangroves along the coast has provided natural barriers against storm surges and coastal erosion, protecting communities and infrastructure.

In Nepal and Bhutan, the revival of springs and the development of multi-functional water reservoirs are the strategies being used to address water scarcity and support agricultural productivity. The revival of springs in the Himalayan region has improved water availability for household use and irrigation, benefiting local communities and enhancing food security. These efforts have also reduced the risk of landslides and soil erosion, contributing to overall landscape stability.

In Pakistan, Recharge Pakistan project is working towards building an ecosystem approaches to minimize the impacts of droughts and floods. This project has restored wetlands and floodplains, reducing flood risks and improving water storage capacity. Additionally, the Federal Flood Protection Plan IV (NFPPIV) includes NbS to enhance flood protection and water management, benefiting both urban and rural communities. The restoration of mangroves in coastal areas has also provided natural protection against sea-level rise and storm surges.

3.2 Ecosystem services, types, and examples

Ecosystem services are the benefits that people receive from ecosystems (MEA 2005). They encompass a wide array of benefits derived from nature, ranging from material resources like food and water to intangible values such as cultural, aesthetic, and recreational benefits. They are commonly grouped into four categories: provisioning services, regulating services, cultural services, and supporting services.

Provisioning services: Provisioning services include the tangible products obtained from ecosystems, including food, water, and raw materials. Examples from South Asia include rice and fish products that rely on the fertile flood plains of the Ganges, Indus, and other rivers. In Bangladesh, the Hilsha fishery (Tenualosa ilisha) represents the largest single-species estuarine fishery in the world and contributes 1% to the country's GDP (Mahmud 2020). Indigenous People living across the Himalayas depend on non-timber forest products (NTFPs), such as bamboo, honey, and mushrooms. Similarly, the high-altitude rangelands in Nepal and Bhutan support pastoralism and agriculture.

Regulating services: Regulating services include the services that ecosystems provide to moderate natural processes, including water and climate regulation, water purification, and disease control. The mangrove ecosystems of Sundarbans in India and Bangladesh act as a natural barrier to cyclones and help regulate floods and protect coastal areas. The Western Ghats in India, also known as the Sahyadri mountain range, stretch over 1,600 km across the western coast of India and play an important role in regulating the Indian monsoon system. The forested areas across the Himalayas play an important role in regulating water flow, preventing downstream flooding, and ensuring sustained water supply. The floodplains across the Ganges and Indus river basins provide regulatory services by filtering water and preventing soil erosion.

Cultural services: Cultural services are non-material benefits from the ecosystem, such as spiritual, cultural and recreational values. Many communities hold natural sites as sacred. For example, the Ganges River is revered as holy, providing spiritual services to Hindu communities.

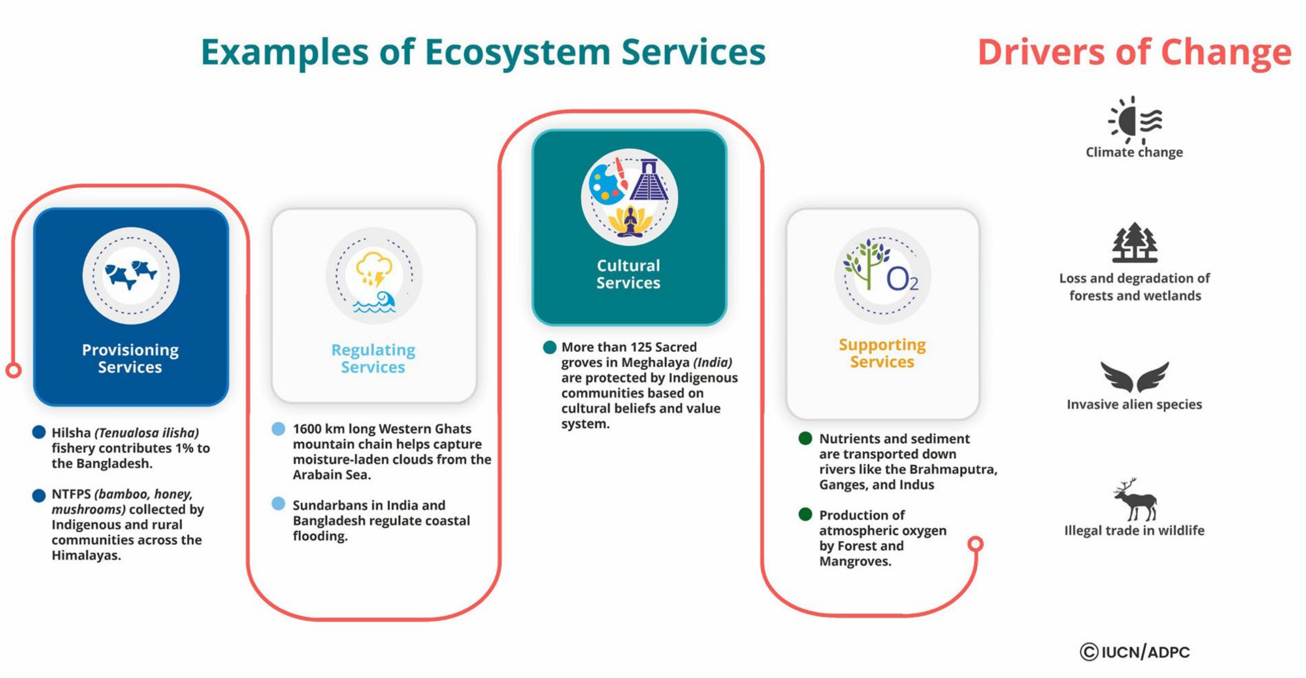
The spiritual and cultural values promote local stewardship for the ecosystem protection and thus contributes to biodiversity conservation. One example is the sacred groves in the state of Meghalaya, India. These are forests preserved by local Indigenous communities through centauries due to religious and cultural significance linked to these forests. More than 125 sacred groves have been recorded in the state of Meghalaya, including the Mawphlang sacred grove, spread across 78 hectares and now internationally known tourist destination. The Mawphlang sacred grove is protected by Khasi communities, due to the belief that the forest is home of the local deity Labasa worshiped by Khasi Indigenous community.

Supporting services: Supporting services are all ecosystem services that are necessary for the production of all other ecosystem services, including nutrient cycling and soil formation, which are essential to maintain the conditions for life on earth. This includes nutrient and sediment transportation down the rivers like the Ganges, and the production of atmospheric oxygen by coastal mangrove ecosystems along the coast in Bangladesh, India, and Pakistan.

Ecosystem services and indigenous people

IPLCs play an important role in ecosystem management, particularly in the Himalayas. According to the 2011 census of Nepal, the Indigenous Nationalities of Nepal comprised 36% of the total population of 30.2 million. As per the National Statistical Bureau of Nepal, in 2022, there were 22,380 Community Forest User Groups (CFUGs) and 3.34 million households involved in community forests, many of these user groups being led by Indigenous People including representatives from the Sherpa, Tamang, Kiranti, and Dolpali communities.

Infographic 1 | **Ecosystem services types, examples and drivers of change**



In the state of Meghalaya, Indigenous communities, including the Khasi, Garo, and Jaintia are responsible for the management of more than 90% of the natural forest and thus have an important role to play in ensuring the sustainable supply of forest-based ecosystem services. Under the cultural services, there are more than 100 documented sacred groves protected by Indigenous People in the state of Meghalaya. These provide refuge for endangered species and supporting climate and water regulation for nearby communities. IUCN film, *Living in Harmony with Nature - Application of Nature-based Solutions in the Meghna Basin* highlights locally led examples of ecosystem management and NbS initiatives led by Indigenous People to address water and food security issues (IUCN 2023a).

3.3 NbS definition, types, and criteria

NbS are emerging as an important tool for achieving water sector resilience in South Asia. NbS is an umbrella term for several strategies, including Ecosystem-based Adaptation (EbA), Ecosystem-based Disaster Risk Reduction (Eco-DRR), and the Green Infrastructure approach.

In 2022, the Fifth Session of the United Nations Environment Assembly (UNEA-5) adopted a multilaterally agreed definition of Nature-based Solutions: “actions to protect, conserve, restore, sustainably use and manage natural or modified terrestrial, freshwater, coastal, and marine ecosystems, which address social, economic, and environmental challenges effectively and adaptively, while simultaneously providing human well-being, ecosystem services and resilience and biodiversity benefits” (UNEA-5, 2022). This demonstrates the increasing recognition of the role of NbS in addressing global challenges while promoting the conservation of ecosystems and the services they provide.

Types of NbS

NbS is an umbrella term for a diversity of ecosystem or sector-specific approaches for restoring, protecting, and managing ecosystems to address societal challenges and provide biodiversity.

Since the late 1990s, the Dutch government has promoted the idea of ‘Making Space for Rivers’, allowing rivers to occupy their natural floodplains. Urban development and construction on riverbanks constrict spaces for rivers, increasing vulnerability to riverbank erosion and flood risk, and decreasing the riverscape or landscape water recharging potential. Therefore, protecting the riverbank through an area-based conservation strategy (biodiversity park, river-bank forest) and zoning are increasingly promoted by national governments in South Asia. Recently, India launched the River Rejuvenation program across the country, aimed at restoring major rivers and their surrounding basin, as is thus promoting the idea of making spaces for the river. One key component of this initiative is the Namami Gange Mission, which focuses on the Ganga River. This mission has successfully rejuvenated 1,500 km of the river and restored 30,000 hectares of forests (United Nations 2022). The initiative prioritizes several NbS approaches, including afforestation, soil conservation, and the construction of groundwater recharge structures for improving water quality, enhancing biodiversity and mitigating climate change impacts.

Some of the commonly used terminologies in policy and literature that align with the NbS definition are discussed below. These are categorized for better understanding: a) Ecosystem restoration approaches; b) Issue-specific ecosystem-related; and c) Infrastructure-related approaches relevant to the water-related climate risk reduction. These approaches are inspired by nature and aim to restore natural ecosystem processes and connectivity within the context of a river basin or landscape. The individual measures can be combined to achieve specific objectives within a target river basin or landscape.

a) Ecosystem restoration approaches:

Ecosystem Restoration (ER) is defined as actions leading to the recovery of the structure, function, and processes of the original ecosystem that has been either degraded, damaged, or destroyed (Society for Ecological Restoration International and IUCN Commission on Ecosystem Management 2004). In a river basin context, restoration of a river stream or wetland will include a range of measures, including regulatory measures to make the space available for the natural meandering of the mainstream of river, connecting it to former oxbow lakes, changing agricultural practices to make it ecosystem friendly and restoring the forest in the watersheds. These measures can slow water velocity, and recreate habitat for both aquatic and terrestrial species, reduce erosion and increase soil water retention. Wetlands buffer against floods and droughts and reduce coastal erosion. In the headwaters of a river, reforestation and afforestation enhance carbon sequestration, reduce soil erosion, and improve water infiltration. Similarly, reconnecting floodplains in urban areas provides natural floodwater retention and habitats for fisheries and waterbirds. These measures reduce flow discharge extremes, benefiting agriculture, tourism, and ecology. Local communities need to be engaged in the consultation process and implementation of measures for long-term sustainability and to also ensure that the proposed measure does not impact water availability and storage in the dry season for the local communities.

Forest Landscape Restoration (FLR) is one of the tools for implementing ES. It is aimed at regaining ecological functionality and improving human well-being in deforested or degraded forest landscapes. The FLR tools, such as a guide to Restoration Opportunities Assessment Methodology (ROAM) published by IUCN and WRI, provide a stepwise approach to analyse FLR potential and identify context-specific opportunities at a national or sub-national level (Laestadius, L. 2014). The guide offers practical advice and options to bare in mind when considering or conducting an FLR assessment, by providing real-life examples of the types of outputs one can expect from a particular intervention.

One example of an ongoing project applying FLR in South Asia is the "Scaling Up Forest Landscape Restoration in Chittagong Hill Tracts (CHT)" in Bangladesh. The project started in early 2024 and aims to significantly enhance FLR efforts in the CHT district, which hosts almost one-third of Bangladesh's state-owned forests and is home to 12 forest-dependent ethnic communities. The project is implemented by Arannayk Foundation, Tahzingdong, and Bolipara Nari Kalyan Somity (BNKS). The project is developing and testing cost-effective agroforestry models for jhum fields and strengthening local stakeholders' capacities to reach a restoration target of at least 1,000 hectares in degraded forest lands. This initiative is supported by the "Reversing Environmental Degradation in Africa and Asia (REDAA)" program, funded by UK International Development from the Foreign, Commonwealth and Development Office (FCDO). Additionally, the project is promoting the creation of groundwater recharge structures and other sustainable agricultural practices to deal with water security stress. The initiative also aligns with national and international environmental goals, such as the Bonn Challenge, which aims to restore 0.75 million hectares of forests in Bangladesh (Sarkar et al. 2024).

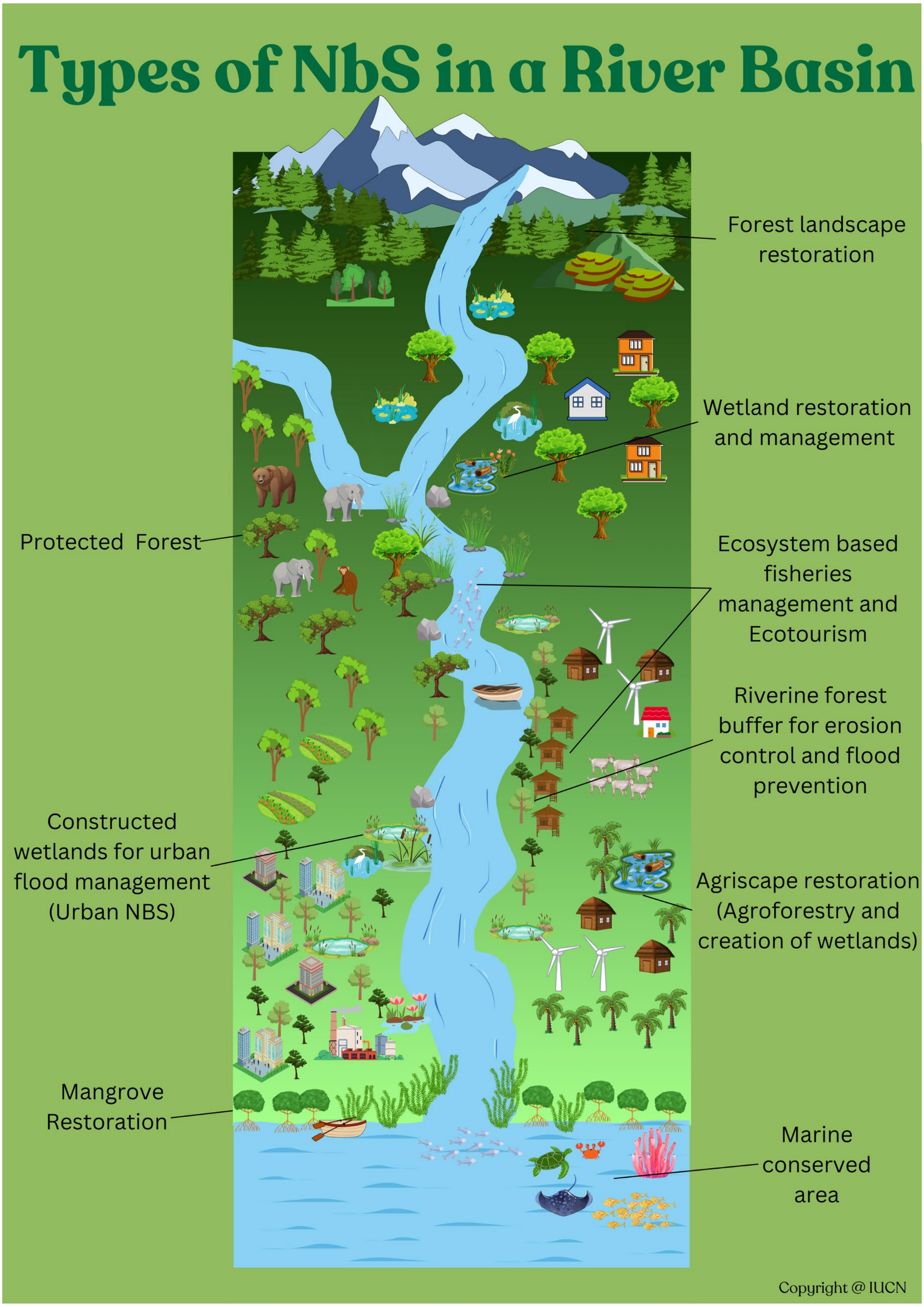
b) Issue-specific ecosystem-related:

Ecosystem-based Adaptation (EbA) and Ecosystem-based Disaster Risk Reduction (Eco-DRR) are examples of issue-specific NbS. The Convention on Biodiversity (CBD) defines Ecosystem-based Adaptation (EbA) as the use of biodiversity and ecosystem services as part of an overall strategy to help people adapt to the adverse effects of climate change (CBD, 2009). EbA applications may include both the protection and restoration of rivers or canals to moderate or eliminate the risks of flooding for a particular population group or city. Considering climate adaptation focus, the restoration of forests in case of EbA project will consider more future climate-tolerant species to adapt to climate change (Doswald & Osti, 2011). Similarly, Eco-DRR is NbS designed to reduce overall disaster risk within a specific river basin or landscape by mitigating hazards and increasing livelihood resilience through sustainable management, conservation, and restoration of ecosystems (PEDRR n.d.).

The World Bank-supported Forest-based Landslide Risk Management Programme in Sri Lanka is an example of Eco-DRR project designed to reduce the landslide risk on local communities due to development and climate risks. Implemented by the National Building Research Organization (NBRO) of Sri Lanka with support from the Asian Disaster Preparedness Center (ADPC), the programme aims to enhance the capacity of the National Building Research Organization (NBRO) to understand and apply forest-based solutions for landslide risk management, the programme promotes training, study tours, and knowledge exchanges between different government agencies. Specific NbS interventions under the program includes measures like afforestation, reforestation, and the development of detailed forest-based landslide risk management plans for selected areas to stabilize slopes and manage water flow (World Bank 2019). Similarly, countries like Bangladesh are promoting Managed Aquifer Recharge (MAR) techniques to deal with water stress issues. The MAR includes a range of interventions such as recharge ponds, injection wells, and restoring watersheds to enhance the recharging of underground wells increasing drought resilience. Approximately 100 MAR systems have been constructed in coastal Bangladesh to provide safe drinking water to rural communities. MAR captures excess rainwater during the monsoon season and infiltrates it through wells into underground aquifers.

c) Infrastructure-related approaches:

Green Infrastructure (GI) and Ecological Engineering (EE) are commonly used in published literature for the infrastructure-related NbS approach. The GI and EE refer to a "strategically planned network of natural and semi-natural areas, with other environmental features designed and managed to deliver a wide range of ecosystem services, including


Types of NbS in a River Basin
Forest landscape restoration
Wetland restoration and management
Protected Forest
Ecosystem based fisheries management and Ecotourism
Riverine forest buffer for erosion control and flood prevention
Constructed wetlands for urban flood management (Urban NBS)
Agriscape restoration (Agroforestry and creation of wetlands)
Mangrove Restoration
Marine conserved area
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green spaces (or blue if aquatic ecosystems are concerned) and other physical features in terrestrial (including coastal) and marine areas" (WGIN n.d.).

The World Bank publication "Integrating Green and Gray – Creating Next Generation Infrastructure" provides insights on how to create synergy between natural (green) and engineered (gray) infrastructure and/or solutions to enhance water-related ecosystem services. The report defines green infrastructure as natural systems, such as forests, wetlands, and floodplains that provide essential services like water filtration, flood control, and habitat for biodiversity. These natural systems can absorb and filter water, reducing the need for extensive water treatment facilities. For example, restoring wetlands can improve water quality and provide natural flood protection by absorbing excess water during heavy rains.

Gray infrastructure, on the other hand, consists of engineered solutions like dams, levees, reservoirs, and water treatment plants designed to manage water resources and mitigate risks. The publication highlights that integrating these gray components with green infrastructure can create more resilient and cost-effective water management systems. For instance, combining levees with restored wetlands can enhance flood protection while also supporting biodiversity and improve water quality. This integrated approach not only addresses immediate water management needs but also provides long-term benefits such as climate resilience and poverty reduction.

Within an urban context, creating green spaces (forests and wetlands) can help in reducing the impact of sudden heavy downpours. In urban areas, GI involves innovation in gray infrastructure design, such as the creation of additional green roofs and walls, and permeable pavements for reducing the natural flow and impact of stormwater, reducing urban heat island effects, and improving air quality. The World Bank is supporting urban NbS initiative in Sri Lanka's capital city of Colombo, which aims to restore and protect 20 square kilometres of freshwater lakes and wetlands, and the creation of additional grasslands and swamps in the city as a buffer for flash floods. A review of project documents suggest that the wetlands and forests of the city absorb up to 90 percent of Colombo's greenhouse gas emissions and also provide other benefits like food and supplemental income for the urban poor. The project integrates natural solutions with engineered infrastructure, such as pumping stations and tunnels, to manage floodwaters effectively. This hybrid approach has benefited 2.5 million residents by enhancing flood resilience and improving urban living conditions. The success of this initiative has inspired the Sri Lankan government to develop 50 additional plans for wetland conservation across the country, demonstrating the project's scalability and potential for broader impact. Overall, the Colombo wetlands project highlights how NbS can address urban flood risks while providing multiple co-benefits, such as climate mitigation, conservation of biodiversity, and socio-economic benefits for vulnerable communities.

For drought management, gray infrastructure like treatment plants and sewer systems can be integrated with Managed Aquifer Recharge (MAR) techniques to enhance natural infiltration processes and replenish groundwater supplies. This includes creating rainwater harvesting structures, including constructed wetlands, and riparian buffer plantations.

Similarly, constructing wetlands can help treat wastewater and stormwater and improve water availability in the dry season. One example of this is the East Kolkata Wetlands (EKW) in India. The EKW spans over 12,500 hectares and acts as a natural buffer against storm surges and flooding, significantly contributing to Kolkata's water resilience (The Climate Group 2024). The wetlands naturally recycle approximately 910 million litres of the city's untreated sewage daily. This process not only improves water quality but also helps in recharging groundwater and regulating hydrological flow.

By absorbing excess rainwater, the wetlands mitigate the risk of urban flooding, which is particularly vital during the monsoon season. Recognized as a Ramsar site, the EKW is a prime example of how integrating natural processes into urban planning can yield sustainable and cost-effective solutions for environmental and socio-economic challenges (Ramsar Sites Information Service 2002). Table 1 presents gray and green components in water-related services, such as water supply, flood and drought management and urban climate resilience.

Table 1 | **Green and gray component in water-related ecosystem services**

Services	Gray infrastructure	Green infrastructure component
Water supply and treatment	Dams, reservoirs, pipelines	Restoring forests and wetlands in watersheds (FLR). Hybrid system combining pipelines and treatment plants with constructed wetlands, such as East Kolkata wetlands.
Flood management	Levees, floodwalls, embankments	Combining levees with restored wetlands and planting vegetation to absorb floodwaters. Planting trees and grasses along riverbanks to slow down water flow, reduce erosion, and improve water quality. Reconnecting rivers to their floodplains allows for natural water storage during high-flow events.
Drought management	Treatment plants, sewer systems	Managed Aquifer Recharge (MAR) to enhance natural infiltration processes to replenish groundwater supplies. It may include a mix of approaches, including creation of rainwater harvesting structures, such as constructed wetlands, dams and reservoirs, creation of riparian buffer plantation for enhancing the groundwater recharge potential to mitigate drought effects. Soil moisture retention techniques in the agriculture sector, such as mulching and cover cropping can improve soil moisture retention, reducing the impact of droughts.
Urban drainage	Green roofs, permeable pavements	Green roofs and permeable pavements integrated with traditional drainage. Construct wetlands to provide buffering against flash floods and heat island effects.

Criteria for defining NbS

Following the adoption of a definition of NbS and the increasing use of the term in literature and projects on the ground, there was a pressing need for greater clarity on how NbS could be developed to ensure benefits to both people and biodiversity. To address this, IUCN launched the Global Standard for Nature-based Solutions in 2020 (IUCN 2020). The design of the standard was based on consultations with more than 10,000 stakeholders globally. To date, IUCN Global Standard for NbS provides the most comprehensive guidance on NbS criteria.

The standard is designed around eight criteria and 28 indicators, which are linked to the ecological, social, and economic dimensions of NbS (Box 1). These criteria and indicators aim to ensure that the application of NbS is credible and that negative impacts are mitigated by thorough design and monitoring processes.

The criteria of IUCN Global Standard for NbS provide a systematic learning framework for NbS and acts as a tool to help design, implement, monitor, and evaluate NbS inclusively and sustainably. Criterion 1 provide guidance on identifying societal challenges, that the NbS intervention can address, such as water and food security, climate change mitigation, and adaptation, among others. The identification of societal challenges should be done jointly with the rights holders and affected communities.

Criterion 2 promotes a 'systems approach' to the design and implementation of NbS that responds at the appropriate scale. In NbS interventions, scale refers to both geographic scale across spatial landscapes, as well as understanding and integrating economic, ecological, and societal aspects in landscape management.

Criteria 3, 4, and 5 are linked to the three pillars of sustainable development and aim to promote the design of environmentally sustainable, economically viable, and socially equitable NbS where IPLCs are at the centre of decision-making processes. Criterion 6 is linked to balancing trade-offs to achieve short and long-term gains in a landscape or a river basin, while also ensuring that there is a transparent, equitable, and inclusive process to determine who benefits and who pays the cost, and the associated trade-offs or compensation required to ensure sustainable and successful NbS.

Criterion 7 responds to the need for adaptive management through continuous monitoring and evaluation and adjusting the NbS intervention as needed to ensure the original goals are achieved. Criterion 8 promotes documenting and sharing lessons learned from the intervention to inform policy at all levels and ensure upscaling of the intervention in future work.

Infographic 3 | Criteria - IUCN global standard for Nature-based Solutions

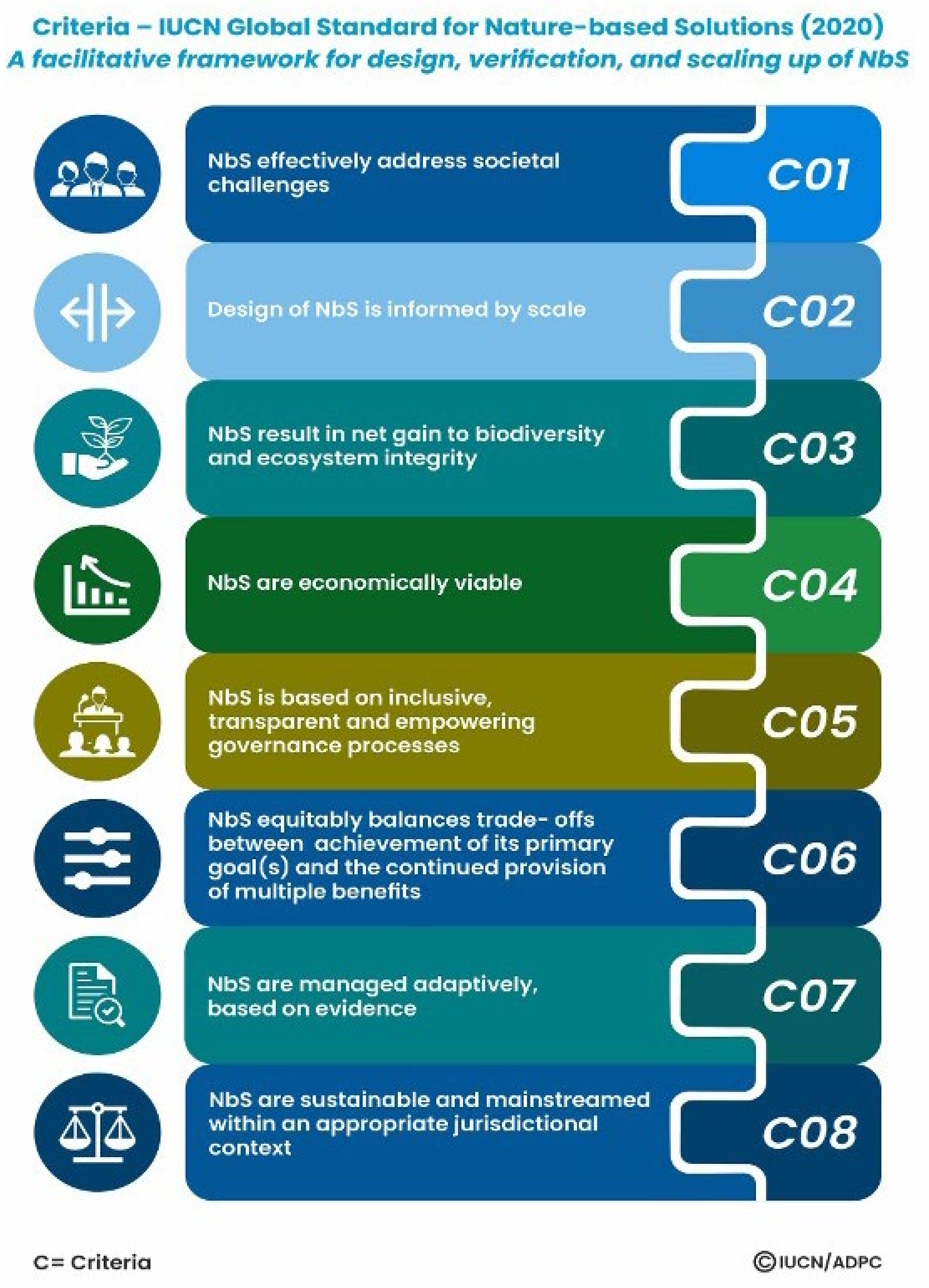

Photo Credit: IUCN Pakistan

CHAPTER 4

Review of NbS in Government Policies

This chapter provides an assessment of the degree of integration of NbS in the NDCs, NAPs, National Water Policies, and National Disaster Management Policies in selected countries South Asia. The chapter also highlights relevant provisions and targets that could be leveraged by government officials working at different levels during the design and approval of large-scale projects on NbS water resilience.

The section below provides an overview of the enabling provisions for NbS in the NDCs, climate action plans, and water and disaster management policies of Bangladesh, India, Nepal, and Pakistan. These countries were selected for detailed policy analysis, as these countries together represents more than 95% of people and land, and regularly feature among the top ten most impacted by climate change globally. For the identification of the enabling provisions, six criteria were developed for this report and detailed in the section 4.1 below. The six criteria were applied for analysing the following three policy types, a) Climate change: Nationally Determined Contribution (NDC) and National Action Plan on Climate Change; b) National Water Policy; and c) National Disaster Management Policies. These policies were targeted due to the overall alignment with the objective of this report and considering the climate change and water related disasters were among the priority water resilience issue for government as per the regional survey. Considering the main objective of this report is to provide guidance on the opportunities and strategies for NbS mainstreaming in water sector for resilience, the mandate and role of the climate change, water and disaster management institutions are best aligned with the guidance objectives.

4.1 Criteria for identification of NbS enabling policy provisions

The policy analysis applies six criteria to assess NbS integration in the selected national policies. These criteria are built on the NbS definition adopted at the Fifth Session of the United Nations Environment Assembly in 2022, and criteria proposed under IUCN Global Standard for NbS (2020). These criteria together consider important aspects that enable NbS in policy, starting with the recognition of NbS, inclusion of specific mechanisms for community engagement and cross-sectoral collaboration, and promotion of innovative mechanisms to mobilize resources, such as payment of ecosystem services (PES) and carbon markets.

The six criteria are discussed below:

1. **Recognition of NbS and/or ecosystem management in building water resilience:** Recognizing NbS is important as it sets the foundation for integrating ecosystem considerations and processes into water management and sectoral policies and strategies. This criterion evaluates whether the policy acknowledges the role of NbS and/or ecosystem management, including practices like reforestation, and wetland restoration to address climate resilience issues.
2. **If NbS is recognized, does the policy promote the inclusion of communities in the decision-making process?** Community inclusion ensures that NbS are tailored to local needs and fosters local ownership of proposed solutions. Therefore, this criterion aims to assess whether the policy encourages the involvement of local communities in the decision-making process and what mechanisms are available for their engagement in planning, implementing, and monitoring NbS or ecosystem restoration initiatives.
3. **Does policy recognize the role of biodiversity, ecosystem integrity and connectivity in achieving policy objectives?** Healthy, connected ecosystems provide essential services such as water purification, flood regulation, and climate resilience. This criterion therefore examines whether the policy recognizes the role of biodiversity, ecosystem integrity, and connectivity in achieving its objectives.
4. **Whether policies promote cross-sectoral collaboration?** For the success of NbS, different sectors need to work together, leveraging synergies and working towards a joint vision. This criterion, therefore, aims to evaluate whether the policy recognize the value of the cross-sectoral approach and provides guidance for joint decision-making among different sectors (e.g., agriculture, water, forestry).
5. **Does the policy include mechanisms for regular monitoring and upscaling of best practices?** Monitoring and upscaling are needed for continuous improvement and wider adoption of NbS. This criterion therefore aims to assess whether the policy includes mechanisms for regular monitoring of the achievements.
6. **Whether policy recognizes the need for financial innovation and the development of market-based mechanisms?** Globally, market-based mechanisms such as carbon trading, green bonds, and public-private partnerships are being utilized to mobilize resources for NbS implementation.

4.2 NbS enabling provisions and gaps in selected policies in South Asia

The review of policy provisions applying the six criteria above suggests that NbS are integrated as approaches in addressing environmental challenges, particularly in enhancing water resilience. For instance, Bangladesh's National Adaptation Plan (NAP), India's National Water Policy (NWP), and Nepal's National Climate Change Policy (NCCP) all recognize the critical role of NbS in sustainable development and climate resilience. Table 2 to 5 below presents the NbS enabling policy in Bangladesh, India, Nepal and Pakistan. The provisions are presented by each of the six criteria discussed in section 4.1, C1 to C6.

4.2.1 Bangladesh

Table 2 NbS enabling provisions in selected policies in Bangladesh

Policy	NbS recognition and enabling provisions
Nationally Determined Contributions, Bangladesh 2021	C1: The NDC does not mention NbS, however, the role of ecosystems in adaptation and mitigation is recognized through several provisions. Under adaptation measures, the NDC mentions nature-based agriculture and fisheries management. As a part of proposed mitigation actions to deliver on the unconditional contribution from Bangladesh by 2030, the policy sets a target of increasing tree cover from 22.37% (2014) to 24% by 2030, including – the restoration of 150,000 ha of forest in the coastal areas and islands, through measures like afforestation and forest restoration. The policy includes a target for the creation of 137,800 hectares of forest in deforested areas (clear forest in hill and Sal forest in the foothills to be restored) and 200,000 hectares of degraded forests in these areas.) C2: Promotes community inclusion in the maintenance of forest and tree cover through collaborative forest management, social forestry, and other programmes. This includes implementing co-management practices in 72,000 hectares of protected areas and scaling up alternative income-generating activities for 55,000 forest-dependent families. C3: No recognition of the role of ecosystem connectivity in achieving policy objectives. To fill this gaps, future revisions of NDC shall include documentation and mapping of ecological corridors and provide mechanism to ensure their integration into land-use planning and development projects. By maintaining and enhancing ecological connectivity, forest and wetlands ecosystems can function more effectively as carbon sinks, helping to mitigate climate change. C4: Policy promotes cross-sectoral collaboration, and MoEFCC Bangladesh is identified as implementing agency, however, the mechanism for cross-sectoral collaboration and powers of ministry in the implementation of the NDC are not clear. The implementation of the NDC is envisaged through 12 initiatives, including Mujib Climate Prosperity Plan up to 2030, Bangladesh Climate Change Trust Fund, Bangladesh Delta Plan 2100.

Policy	NbS recognition and enabling provisions
	C5: There is no monitoring system proposed although the NDC mentions that a Monitoring, Reporting, and Verification (MRV) will be established and includes the need for data collection, analysis, and sharing across sectors. This is particularly for developing and maintaining GHG inventories, designing appropriate MRV systems to be able to monetise the forest and tree cover through REDD+ or other carbon market schemes. C6: NDC clearly identifies the funding gaps and the need for market-based mechanism and role of REDD+ as critical for filling this funding gaps. Among the 12 initiatives recognized for NDC implementation. The Mujib Climate Prosperity Plan 2030, includes aspiration to develop carbon market regime, and whereas Bangladesh Delta Plan (BDP) 2100 proposes the establishment of BDP 2100 resilience bonds. Under the climate financing section, NDC mentions the role of banks and financial institutions in Bangladesh will be vital, however, NbS is not recognized as environmental-friendly technology.
National Adaptation Plan of Bangladesh (NAP), 2023-2050	C1: The NAP explicitly recognizes NbS (Goal 4) and the role of ecosystem restoration in climate change adaptation and mitigation. The conservation of ecosystems, wetlands, and biodiversity is prioritised in the design of adaptation measures. C2: Promotes locally led adaptation (LLA) and prioritisation of adaptation measures based on the local community preferences ensuring that their preferences and knowledge are central to the adaptation strategies (Chapter 4, Section 4.3). The NAP also recognizes the need for socially inclusive adaptation including for women, youth and people with disabilities (Clause 4.8 Guiding Principles). Community-based coastal afforestation or reforestation (Clause 3.3.7) were identified as a preferred adaptation options. C3: Restore and conserve ecosystems and biodiversity are strategic actions (Outcome 3, Goal 4) identified under NAP, and it is supported by indicators, such as critical and fragile ecosystems restored and conserved, abundance of flora and fauna, number of migratory birds and dolphins. The ecological connectivity aspects and its monitoring can be strengthened in NAP by inclusion of indicators that capture landscape and basin scale ecosystem connectivity, such as, availability of ecological c*orridor maps, as well as number of sectoral guidance developed to protect and enhance the ecological connectivity at the national scale.* C4: The NAP promotes cross sectoral collaboration, with 23 strategies and 28 stated outcomes, integration of diverse aspects. The NAP seeks collaboration between the infrastructure, climate resilient agriculture and socio-economic sectors through inclusive and ecosystem-based adaptation, improved governance, transformative capacity building and climate finance. The existing Interministerial Steering Committee on Climate Change (ISCCC) headed by the Minister of MoEFCC was identified to steer the NAP implementation. However, the NAP also acknowledges that Institutional arrangements need to be further strengthened to facilitate the outcome-based M&E.

Policy	NbS recognition and enabling provisions
	C5: The NAP provides a detailed guidance on the M&E framework, and proposes a tool to monitor (using a traffic light system), evaluate and report adaptation progress, including development of web-based monitoring. The NAP also recognizes the need for sharing best practices and lessons learned, and identifies media and tools for communication, such as websites, apps, national workshops, press conferences, online training, webinars and talk shows to support policy change and upscaling of best practices. C6: The NAP highlights the role of public-private partnerships in enhancing climate finance, and the need for private sector incentives through innovative financing instruments, de-risking capital & blended finance, and tax and value added tax exemptions, tax holidays, index-based insurance, development of a green bond market, enhanced corporate social and environmental responsibilities for adaptation through discussion with relevant key stakeholders like the Finance Division, Bangladesh Bank, development partners and NGOs.
National Water Policy, 1999	C1: The NWP does not mention NbS, but it does recognize the need to maintain ecosystem health for long-term water security. It also emphasises that water resource management actions should avoid or minimize environmental damage. One of the objectives of the policy is to address water-related challenges such as floods and droughts through better ecosystems management. (Clause 3: Objectives of National Water Policy) C2: Aims to ensure the availability of water to all elements of society including the poor and underprivileged, and to take into account the particular needs of women and children (Objective 3(b)). The policy also mandates regular consultation with local communities to gather their input and feedback on water management issues, and as a mechanism for inclusion of the voices of community members in decision-making processes. However, mechanisms and indicators to monitor long-term engagement of community in decision making process and the community rights within the water and disaster management context needs elaboration. C3: Recognize the role of biodiversity, as it suggests the development and management of the nation's water resources should include the protection, restoration, and preservation of the environment and its biodiversity including wetlands, mangrove and other national forests, endangered species, and the water quality (Clause 4.12). The policy includes basin planning as a core strategy for development of water resources, however, there is a need to include indicators for track river and wetlands connectivity at the landscape or basin scale for its role in enhancing water security and climate resilience. C4: Seeks cross sector collaboration in economic and fiscal management (Clause 4.14) and recognizes the need for inter-sectoral coordination and the alignment of environmental protection measures with the National Environmental Management Action Plan (NEMAP) and the National Water Management Plan (NWMP).

Policy	NbS recognition and enabling provisions
	C5: Promotes investigation/evidence-based decision making for understanding management issues in flood control and for guiding other structural interventions (Clause 4.15(c)). Upscaling of practices is also promoted through awareness and participation. C6: Recognizes that the development of water resources often requires large capital investment (Introduction). There is also a detailed guidance of economic and financial management generally for the water sector development. The policy calls for financial autonomy of water sector institutions to be able to charge and collect fees (Clause 4.14).
National Plan for Disaster Management (NPDM), 2021-2025	C1: Does not recognizes the role of the ecosystems and biodiversity in achieving policy, however, the policy talks about a paradigm shift in disaster management to be achieved by moving from conventional relief response practice to a more comprehensive risk reduction culture and by improving the response recovery management at all levels. (Clause 1.2 (a), (b)). C2: Community inclusion is promoted based on equity and participation for managing disasters. However, since the role of NbS or ecosystem management in disaster preparedness is not recognized, community rights and roles cannot be established. The NPDM empahisizes on the importance of community-based approaches to disaster risk reduction and engaging local communities in risk identification, planning, and implementation of disaster risk reduction measure. The plan also recognizes the need for integration of traditional knowledge in disaster management and enhancing capacity of local communities through training and awareness programs for better preparedness and response capabilities. C3: NPDM does not have a specific provision on the recognition and role of biodiversity and ecosystem integrity; however, it does recommend the periodic assessment of disaster risks and documentation of risks and vulnerabilities at the relevant social and ecological spatial scales, providing entry points for integrating ecosystem integrity and connectivity aspects. C4: The Plan encourages strengthening of baselines by understanding risks and their sequential effects at the relevant social and spatial scale on ecosystems (Priority 1) with particular focus on disaster and climate change adaptation (CCA). The policy proposes that a system be established to ensure a close working relationship between the policy formulating body of different ministries and set up a cell to deal with the issues of DRM and CCA within the sectoral laws, policies, plans, projects and programs. Arrangement to be made so that the operational agency must implement the decisions and prepare reports and suggests that it be hosted at Prime Minister's office. C5: Contains an indicative plan for 2020-2030 with timelines, actions and targets for regular monitoring. Many of the core targets will continue to be implemented over the long term until 2030. After each period, review and updating processes will be undertaken with stakeholders' feedback, making NPDM 2021-2025 an adaptive document (Timelines and Targets). The Plan also mandates upgrade and strengthening policy on regular basis (Priority 1).

Policy	NbS recognition and enabling provisions
	C6: Highlights the requirement of resource mobilization as a challenge and calls upon local governments to earmark appropriate resources within their budget for DRM (Clause 106). However, there is no recognition of role of market-based mechanism and private sector engagement in disaster resilience planning.

4.2.2 India

Table 3 | NbS enabling provisions in selected policies in India

Policy	NbS recognition and enabling provisions
India's Updated First Nationally Determined Contribution Under Paris Agreement (2021-2030)	C1: The role of NbS is recognized in climate mitigation. India's NDC includes several ecosystem-based mitigation measures to achieve carbon sink of 2.5 to 3 billion tonnes of CO_2 equivalent through additional forest and tree cover by 2030. The recognition can be further strengthened through the inclusion of wetlands specific targets, considering its role as carbon sink. C2: No provisions related to community inclusion. NDC recognize the need for sustainable living based on traditions and values of conservation and aims to promote it through mass movement on 'Lifestyle for Environment - LiFE' as key to combating climate change. This recognition provides an entry points to community inclusion and capacity building in the implementation of the NDC. C3: No provision or explicit recognition of ecosystem connectivity. However, the policy recognizes the role of forest and tree cover in achieving mitigation targets, and these can be linked to enhancing ecological connectivity in a particular landscape and river basin context. C4: The NDC highlights the need for international collaboration, learning exchange, and diffusion of cutting-edge climate technology in India, and suggests collaborative research and development for future technologies. C5: The need for monitoring and evaluation is recognized only with respect to national communication on NDC every five years in accordance with decision 1/ CP.21 Paris Agreement. The NDC can be strengthened by including the role of monitoring needs as a strategy to promote adaptive learning and promoting documentation and upscaling of best practices in projects designed to create additional carbon sinks through tree and forest cover. C6: Does not speak about the role of financial innovation or market-based mechanism. For the implementation of NAP, resources mobilized from domestic and new & additional funds from developed countries to implement the mitigation and adaptation actions. *Note: NDC India is a brief document providing information on the country targets and does not provide details of community engagement or cross-sectoral collaboration needs. National Action Plan on Climate Change (NAPCC) of India provides the details of implementation framework for achieving the targets set in the NDC.*

Policy	NbS recognition and enabling provisions
National Water Mission (NWM), 2008 (Revised 2021)	C1: The NWM* does not explicitly recognize NbS but does include strong ecosystem considerations and recognized the role of traditional management systems for water conservation, through initiatives such as the "Repair, Renovation, and Restoration of Water Bodies" (Chapter 7). C2: Promote active community participation in ground water recharging including water harvesting (Strategy 3.11). The policy can be strengthened by clarifying the role and responsibilities of the Panchayats (village institutions) in the management of watersheds and providing this institution autonomy to design and manage watershed projects, including design of locally-led and agreed strategies for payment of ecosystem services. C3: No explicit recognition of the role of biodiversity or ecosystem integrity, however, the need to work at scale is recognized through promotion of basin level Integrated Water Resources Management (IWRM) as a key strategy for achieving NWP objectives (Goal 3 and 5). This can be further strengthened by inclusion of indicators for monitoring ecosystem connectivity for long-term water resilience in a particular river basin context. C4: Promotes cross-sectoral collaboration through the recognition of basin level integrated water resources management (Goal 5). For the implementation of each goal, MWM proposes establishing an Inter-Sectoral Advisory Group involving the Ministry of Agriculture, Rural Development, and Water Resources (Chapter 3) for joint monitoring, policy alignment, and capacity building of stakeholders involved in water management. The policy promotes convergence and pooling of resources from national flagship programs such as Mahatma Gandhi National Rural Employment Guarantee Act (MNREGA), Watershed Development Program and Jal Jeevan Program. This integrated approach is intended to optimize resource utilization and achieve better outcomes in water conservation and management. The NWM also promotes collaboration between central and state ministries through provisions to undertake pilot projects for improvement in water use efficiency in collaboration with States. However, there is room for greater integration of NbS into these collaborative efforts. Explicitly incorporating NbS criteria into joint decision-making processes can enhance the effectiveness of proposed solutions. C5: The NWM includes several provisions for monitoring and upscaling, such as establishment of the Bureau of Water Use Efficiency (BWUE), Jal Sanchay Jan Bhagidari Dashboard, which serves as a platform for monitoring water conservation efforts and sharing best practices. This dashboard facilitates real-time tracking and reporting of water management activities. However, by incorporating specific NbS criteria and indicators into the monitoring frameworks and best practice compilations will support its mainstreaming. Also, community involvement in monitoring and upscaling efforts can be strengthen by establishing local monitoring committees and encouraging community-led initiatives. This can ensure that proposed NbS are tailored to local contexts and widely adopted.

Policy	NbS recognition and enabling provisions
	C6: Although the policy provides convergence and pooling of resources from the national flagship program, it is silent on the role of market-based mechanisms and financial innovations like PES. There is a general acknowledgement of innovative approach to tackle the problems related to water resources and their management. **NWM is one of the eight missions under the National Action Plan on Climate Change (NAPCC) of India.*
National Water Policy (NWP), 2012	C1: NbS not mentioned, its principles recognized through various provisions. The policy acknowledges that natural water bodies and drainage channels are being encroached upon, water is diverted of other uses, and groundwater recharge zones are often blocked. To address this, the policy suggests conservation of rivers, river corridors, water bodies in a scientifically planned manner, and through community participation. The policy requires that the environmental needs of the Himalayan region, aquatic ecosystems, wetlands, and embanked flood plains are considered in water management. C2: The policy prioritizes the engagement of rights holders in planning and suggests that water resource projects consider social, environmental, and techno- economic considerations in consultation with communities that are beneficiaries of the project. Some specific provisions include, a) Community based water management should be institutionalized and strengthened. (Clause 3.6); b) Water, particularly groundwater is to be managed as a community resource (Clause 2.2); c) Community based aquifer management is encouraged (Clause 5.4); d) Conservation of water resources through community participation (Clause 8.1). C3: NWP explicitly recognize the role of ecological connectivity in achieving water security and include a dedicated section on conservation of river corridors, water bodies and infrastructure (Section 8). The NAP highlights the need for determining ecological needs of the river through scientific study, and conservation of rivers, river corridors, water bodies and infrastructure should be undertaken in a scientifically planned manner through community participation. Furthermore, any encroachments and diversion of water bodies (like rivers, lakes, tanks, ponds, etc.) and drainage channels (irrigated area as well as urban area drainage) must not be allowed, and it should be restored to the extent feasible. C4: Cross-sectoral collaboration need is included through the recognition of the interconnectedness of water management with other sectors such as agriculture, forestry, and industry, and need to involve local governing bodies like Panchayats, Municipalities, Corporations, etc., and Water Users Associations, wherever applicable, should be involved in planning of the projects (Clause 9.2 and 9.6). However, these are not supported by clear monitoring indicators and needs to be further elaborated to highlight the role of landscape and basin scale collaboration through the establishment of inter-agency data sharing and joint planning platforms.

Policy	NbS recognition and enabling provisions
	C5: Promotes regular training and academic courses in water management, which would help to improve the current procedures of analysis and informed decision making in the line departments and by the community (Clause 15.5). A national campaign for water literacy needs to be started for capacity building of different stakeholders in the water sector (Clause 15.5). Development of district level water resource inventories including the information of reservoirs (small and large), streams, dams, canals, rivers, water used for various purposes. C6: Does recognize the need for financial innovation and the development of market-based mechanisms. The policy emphasizes the need for water pricing to reflect its economic value. It suggests that water charges should be determined on a volumetric basis and should cover at least the operation and maintenance costs of providing the services. Suggests project financing should be structured to incentivize efficient & economic use of Water and facilitate early completion of on-going projects. (Clause 6.4). However, it lacks specific, actionable frameworks for implementing these mechanisms, and these provisions need to be revised to allow the development of locally led payment of ecosystem services (PES) for improving watershed management and river connectivity, and deployment of new age fiscal instruments. For instance, policy mentions public-private partnerships and water pricing, these can be linked to detailed guidelines and incentives for the adoption of green bonds, carbon trading, or other innovative financial tools.
National Policy on Disaster Management, 2009	C1: The National Policy on Disaster Management makes references to the underlying benefits of biodiversity conservation for India but does not mention the potential of ecosystem-based approaches as a cost-effective approach to support disaster management planning. The Vision for disaster resilient India includes development of a holistic, proactive, multi-disaster-oriented technology driven strategy (Vision 2.2.1) and the role of traditional wisdom and environmental sustainability is encouraged (2.4.1) providing entry points for NbS integration. C2: Promotes the involvement of Panchayati Raj Institutions (PRIs), local urban bodies, and civil society organizations in disaster management (Chapter 5). Emphasizes the importance of community-based disaster management, including the integration of community plans and execution at the local level (Chapter 3). However, it could be strengthened by providing more specific mechanisms and frameworks for community engagement, particularly in the context of NbS and ecosystem restoration initiatives. This would ensure that NbS is tailored to local needs and fosters greater local ownership of the proposed solutions. C3: Policy makes a reference to the role of ecosystem in achieving its objectives, however, does not explicitly recognize the role of ecosystem connectivity. Therefore, policy could be further strengthened by explicitly addressing the role of biodiversity, ecosystem integrity, and connectivity in achieving its objectives, and by providing specific guidelines for incorporating these elements into disaster management practices. Several provisions under the policy provides entry point for strengthening the connectivity aspects. Under the topic of disaster prevention, mitigation and preparedness (Chapter 5), the need to balance between environmental

Policy	NbS recognition and enabling provisions
	considerations and developmental efforts is highlighted. In this regard, it mentions restoration of ecological balance in Himalayan regions and raising coastal shelter belt plantations wherever necessary in disaster management plans. Ecosystems of forests, islands, coastal areas, rivers, and the agricultural, urban and industrial environment are also to be considered for restoration of ecological balances and sustainable development. Zonal regulations must ensure the preservation of natural habitats (Environmentally Sustainable Development, 5.1.6).
	C4: Policy promotes cross-sectoral collaboration and inclusive institutional arrangement is suggested flood management systems and allocation of resources for efficient prevention and handling of disasters (Clause 10.1.1). It encourages the establishment of institutional frameworks such as the National Disaster Management Authority (NDMA), State Disaster Management Authorities (SDMAs), and District Disaster Management Authorities (DDMAs), which facilitate joint decision-making and collaboration among different sectors. However, the policy could be further enhanced by providing more specific guidelines and mechanisms for cross-sectoral collaboration, particularly in the context of NbS. This may include policy and financial incentives for sectors and institutions that actively engage in collaborative efforts to promote landscape or river scale management of water resources, enhances connectivity of river systems and promotes private sector water stewardship for long-term durability of the solutions. This could include financial rewards, recognition programs, or additional funding for joint projects aligned with the NbS principles.
	C5: Policy acknowledges the importance of continuous improvement through documentation and dissemination of best practices. It highlights the role of the National Institute of Disaster Management (NIDM) in conducting research, training, and capacity-building activities, which include the identification and promotion of best practices. Additionally, the policy advocates for the establishment of a comprehensive disaster management information system to facilitate the sharing of knowledge and experiences. However, the policy could be further strengthened by providing more specific guidelines for the systematic monitoring and evaluation of NbS initiatives and by creating dedicated frameworks for the upscaling of successful practices that strongly include ecosystem conservation and management as a part of disaster preparedness.
	C6: No mention of innovative funding or market-based mechanism for resourcing these funds. A National Disaster Response and Mitigation Funds established for mitigation and preparedness (Clause 4.3.1) established for immediate relief after disasters. However, in July 2021, the 15th Finance Commission and Insurance meeting on the Financial Arrangement in Disaster Risk Reduction, recommended establishment of dedicated mitigation funds at both national and state levels. These funds are intended to support proactive measures, including NbS, to reduce disaster risks.

4.2.3 Nepal

Table 4 | NbS enabling provisions in selected policies in Nepal

Policy	NbS recognition and enabling provisions
Second Nationally Determined Contribution (NDC), Nepal, 2020	C1: Nepal's second NDC recognizes NbS, through targets including increasing forest cover of the country to 45% of the total area by 2030, and community-based forest management in at least 60% of Nepal's forest area. By 2025, Nepal aims to formulate and implement nature-based tourism plans in at least five main tourist destinations. Therefore, Nepal's NDC shows a positive recognition of NbS and ecosystem management, but there are opportunities to enhance the integration and implementation of these strategies in water resilience efforts. C2: NDC outlines several mechanisms to ensure community participation: Local Adaptation Plans of Action (LAPAs), community forestry program. Promote the leadership, participation and negotiation capacity of women, Indigenous Peoples, and youth in climate change forums. Strong emphasis on domestic institutional arrangements, public participation and engagement with local communities and Indigenous People, in a gender responsive manner (4.1 Planning Processes). The NDC puts special emphasis on women empowerment, requiring at least 50% participation of women in the forest management committees, with representatives from Indigenous and poor communities, and the development of an Action Plan for integrating Gender Equality and Social Inclusion targets by 2030. However, these provisions could benefit from more detailed strategies on how to operationalize this inclusion, and development of specific guidelines for relevant sectors, such as hydropower and tourism, on engaging marginalized and vulnerable groups and inclusion of indicators to track the engagement. C3: The role of biodiversity is recognized but there is no mention of ecosystem or river and wetlands integrity and connectivity in the achievement of policy objectives. Forest, biodiversity and watershed conservation are one of themes under the Adaptation component of the NDC (Section 4. Adaptation component). Under the NDC, Nepal is committed to ensure fair and equitable benefits (carbon and non-carbon) from sustainable forest management, watershed management, and biodiversity conservation among local communities, women and Indigenous People. To enhance the adaptation capacity of the local community, the policy suggests incorporating best practices of watershed and landscape management into the national adaptation measures. C4: The need for cross-collaboration is recognized and includes formation of inter-ministerial committees and working groups focused on climate action by 2025. The formation of this interministerial committee should include the key ministries and their role in promoting application of NbS and maintaining ecological connectivity. Some of the cross-cutting areas identified by policy are Gender Equality and Social Inclusion (GESI), Livelihoods and Governance, Awareness Raising and Capacity Building, Research, Technology Development and Extension, Climate Finance Management.

Policy	NbS recognition and enabling provisions
	C5: Monitoring, Reporting and Verification (MRV) is one of the means of implementation under the NDCs. The policy also promotes data-driven tracking of NDC targets, including a mechanism for strengthening the data generation and validation framework, that leads to identification of best practices and inclusion of lessons learnt. The policy emphasis is on maintaining transparency by widely disseminating the methodology and results of the MRV (Point 5). The policy includes specific targets and indicators, and needs to be integrated in the MRV plans with specific timelines, responsible agencies, and resource allocations for monitoring. C6: NDC proposes formulation of Climate Finance Strategy and increasing National Capacity on Climate Finance Management. NDCs also target the development of nature-based tourism plans in at least five main tourist destinations by 2025 and aim to achieve carbon-neutral status for at least five tourist destinations by 2030. Further, it makes a mention to Article 6 of the Paris Agreement, with reference to the use of voluntary carbon cooperation, and suggests that Nepal may explore potential markets that allow higher mitigation ambition while promoting sustainable development and environmental integrity. However, there is need for further detailing these opportunities and how it will be operationalized.
National Climate Change Policy (NCCP), 2019	C1: The NCCP does not explicitly mention NbS, there is only a recognition of ecosystem-based approaches and their role in building community climate resilience. One of the objectives of the policy is to build resilience of ecosystems that are at risk of adverse impacts of climate change (Clause 7(b)). This provides an policy entry point for explicit recognition of NbS as a foundational approach to water resilience. C2: The policy aims at mainstreaming gender equality and social inclusion (GESI) into climate issues, strategies and plans at all levels (Clause 7.g), including training and awareness programs to empower communities with the knowledge and skills needed to adapt to the climate change impacts (Chapter 4). However, policy lacks detailed mechanism for ensuring active community participation in decision-making processes for management of ecosystems and development of NbS initiatives. Strengthening these mechanisms could enhance local ownership and effectiveness of the proposed climate solution. C3: Provides for the formulation of Action Plans on forest, biodiversity and watershed conservation and wetlands (Clause 8.2). The policy outlines the importance of integrated watershed management under the section on Forest, Biodiversity, and Watershed Conservation (Chapter 4) emphaisizing the need for coordinated efforts to manage land, water, and forest resources within watersheds to maintain ecosystem services and enhance resilience against climate change impacts. The policy could further emphasize the connectivity of ecosystems, and role of protected areas and Other Effective Area-based Conservation Measures (OECMs) as tools to ensure landscape and basin level river and wildlife connectivity.

Policy	NbS recognition and enabling provisions
	C4: Nepal's Climate Change Policy 2019 promotes cross-sectoral collaboration through several provisions (Chapter 4 - Sectoral and Inter-Sectoral Policies), such as the establishment of a multi-stakeholder coordination mechanism involving various government agencies, private sector, civil society, and local communities. It encourages collaboration between sectors such as agriculture, water resources, energy, and infrastructure to address climate impacts comprehensively. It promotes joint training programs, workshops, and awareness campaigns to enhance understanding and cooperation on climate issues. However, it lacks specific guidelines for joint decision-making among different sectors. C5: Regular monitoring is to be ensured through awareness and capacity development (Clause 8.10). The policy also promotes the implementation of Local Adaptation Plans of Action (LAPAs), which are designed to be flexible and adaptive, allowing for continuous monitoring and adjustment based on local needs and conditions. This approach helps in identifying successful practices that can be scaled up. C6: The policy mentions mobilizing financial resources and promotes innovative financial mechanisms. Under the means of verification (Section 3.2), the policy mentions that carbon market will also play a key role for Nepal to support and implement various GHG mitigation measures across the sectors. Currently given the potential energy generation forecast, cross-border energy trade provides Nepal the opportunity to engage in market mechanisms under Internationally Transferred Mitigation Outcomes (ITMOs) as referred on Article 6 of the Paris Agreement (PA). It also recognizes the need to develop non-market mechanism under Article 6.8 of PA for potentially the AFOLU. However, there is a need to include the role of NbS, and detailed guidelines and actionable strategy to operationalize the innovative finance mechanism. The current emphasis is on climate finance management and states that climate change should be integrated into all sectoral plans at all levels and proposes a Climate Change Budget Code (Clause 8.12-c). The policy also states that mobilization of finance from private sector will be encouraged through Green Bond, Blended Finance, Result-based Financing, Carbon Offset and Corporate Social Responsibility (8.12-e).
National Water Policy, 2020	C1: The NWP does not mention NbS but explicitly recognizes the need for planning at scale. The policy recognizes watershed or basin scale planning to design measures to limit water-induced disasters, and Integrated Water Resource Management (IWRM) as a tool for sustainable watershed and river basin management. C2: One of the strategies under the policy is to encourage the participation of stakeholders (Strategy 4). The policy also aims at defining the roles and responsibilities of local institutions that represent communities (Clause 7). C3: The policy suggests development of water resources with minimal negative impact on the environment, particularly ecosystems, is one of the objectives of the Water Policy (Clause 6). Also, watershed and basin approach is promoted for reduction of water induced disasters that require integration of various sectors (Clause 11). C4: The policy proposes the establishment of coordination mechanisms among various government agencies, private sector entities, and civil society organizations (Section 4.1). It includes forming committees and working groups at local, provincial, and national levels to ensure effective collaboration and information sharing.

Policy	NbS recognition and enabling provisions
	C5: Study and research in the water sector shall be increased with assimilation of knowledge and data analysis (Chapter 5). The policy promotes the creation of a centralized database for water resources, which will be accessible to all relevant stakeholders (Section 5.1). This aims to improve decision-making processes by providing accurate and up-to-date information on water availability, quality, and usage. C6: Market-based innovative financing is not mentioned. However, it does mention cost-sharing mechanism for operations and maintenance of water supply and suggests that a workable mechanism for cost-sharing must be implemented and followed to, a) maintain the existing water supply schemes; and b) ensure financing for the expansion, rehabilitation and improvement of water supply schemes (Section 5.10.2.1 - Drinking Water and Sanitation). Further, it recognizes that one aspect of sustainable financing for ongoing O&M costs for irrigation to be paid from the increased incomes that farmers realize following the development of irrigation projects (5.10.2.2 Irrigation).
National Policy for Disaster Risk Reduction, 2018	C1: The National Policy for Disaster Risk Reduction does not recognize NbS in any form. There is only some acknowledgment that complex interactions between ecosystems and people influence disaster risks. However, the policy does not mention the potential for better ecosystem management to reduce disaster risks. Efforts to reduce the loss of environmental assets and increase their resilience by implementing DRR measures is in the Mission under the policy (3. Mission); C2: Disaster risk assessment and mapping system will be developed, identifying vulnerable communities, and capacity development activities will be conducted for them (7.4 Policy). The policy ensures participation of local users, community organizations and people in the formulation, implementation and management of plans for irrigation, river control, forest management, school, health institutions, drinking water and sanitation (Clause 7.16). There is further emphasis on inclusivity of community for disaster risk reduction at all stages. C3: There is no discussion on biodiversity, ecosystem connectivity and net biodiversity benefits under this policy. C4: The policy acknowledges the need to mainstream disaster risk reduction in all development processes by integrating it with climate change adaptation activities (Section 5.3). The policy establishes various institutional frameworks, such as the National Disaster Risk Reduction and Management Authority (NDRRMA), which is responsible for coordinating DRR activities across different sectors and levels of government. This includes collaboration with ministries responsible for agriculture, water resources, forestry, and other relevant sectors. However, clear protocols and mechanisms for collaborative planning, implementation, and monitoring of DRR activities are needed to ensure effective coordination.

Policy	NbS recognition and enabling provisions
	C5: Natural hazards like flood, landslide, drought, thunderbolt, windstorm, hot wave, cold wave, fire, epidemics and glacier lake outburst will be monitored and forecasted regularly and Forecast-Based Preparedness and Response Plans will be developed and implemented by developing early warning system (Clause 7.42). The policy outlines the establishment of a robust monitoring and evaluation (M&E) framework to track the progress of DRR activities. This framework is designed to ensure that the objectives of the policy are being met effectively and that any necessary adjustments can be made based on the findings. However, the policy lacks detailed guidelines and indicators for monitoring the effectiveness of DRR measures. Clear protocols and metrics should be established to track the progress and impact of implemented strategies. C6: The role of market-based mechanism in funding disaster preparedness is not recognized. Funding is proposed through traditional strategies and includes establishment of a Disaster Management Fund to mobilize resources at all the levels of governance (Clause 7.47), and a percentage of the budget is also to be allocated by the national and local governments (Clause 7.40). The policy also encourages the private sector, bank and financial institutions, insurance company, development partners and donor agencies for the investment on DRR (Clause 7.35).

4.2.4 Pakistan

Table 5 NbS enabling provisions in selected policies in Pakistan

Policy	NbS recognition and enabling provisions
Updated Nationally Determined Contributions (NDC), 2021	C1: Pakistan's updated NDC explicitly mentions NbS as a key approach, particularly for its role in enhancing green livelihood opportunities and climate change mitigation. The NDC identifies several large-scale national-level NbS flagship projects for achieving policy targets, including a) Ten Billion Tree Tsunami (TBTT) program is cited as a flagship NbS project. The NDC estimates the TBTT program will help sequester 148.76 Mt CO_2 in the next 10 years.; b) Recharge Pakistan, in the Indus basin for flood risk mitigation and enhanced water recharge; c) Expansion of protected area (PA) cover from 12% to 15% by 2030. C2: The NDC, 2021 states that the government will implement ecological management plans and governance through community-led conservation funds (5.2 Adaptation Actions). The actions also include community-based disaster risk management (CBDRM), and climate-resilient livelihood options, community-focused management plans for protected areas (Clause 5.3 Contributions to Adaptation). It also proposes creation of a Gender and Climate Action Plan, which promotes capacity-building for women in new agricultural technologies and integrates climate considerations into the National Commission on the Status of Women's portfolio at all regional levels.

Policy	NbS recognition and enabling provisions
	C3: The NDC identifies several large-scale NbS projects that contribute to biodiversity conservation and ecosystem integrity. These include, a) Ten Billion Tree Tsunami (TBTT) that aims to restore and enhance forest cover, sequester carbon, and improve biodiversity, b) Recharge Pakistan: focuses on flood risk mitigation and enhanced water recharge in the Indus basin, promoting ecosystem connectivity and resilience, and c) Expansion of Protected Areas: Plans to increase the protected area cover from 12% to 15% by 2030, ensuring the conservation of critical habitats and biodiversity. However, the NDC lacks detailed indicators and metrics for monitoring the effectiveness of biodiversity conservation and ecosystem management efforts. Establishing clear indicators for ecosystem connectivity and integrity would help to track progress and ensure that policy objectives are being met.
	C4: Policy promotes cross-sectoral collaboration and provides a mechanism for this. At the federal level, the Ministry of Climate Change & Environmental Coordination (MoCC&EC) steers the NAP process, conducts climate research, and mobilizes finance for adaptation. The Ministry of Planning, Development & Special Initiatives (MoPD&SI) coordinates planning and budgeting with provincial departments. Line ministries integrate climate considerations into their policies. At the provincial and district levels, local governments develop and implement tailored adaptation plans. Key advisory bodies include the Pakistan Climate Change Council (PCCC) and an Expert Group (EG) that monitors and evaluates progress.
	C5: NDC 2021 provides monitoring and verification across the sectors critical for climate mitigation and adaptation. It puts emphasis on evidence-based decision-making, propelling policy adjustments that truly reflect the needs and aspirations of those most affected by climate change. It calls for the government agencies, communities, technical experts, and private sector to be actively involved in the monitoring and evaluation process. This inclusive approach will foster collaboration, transparency and generate a wealth of invaluable insights and experiences (5.2.1 Importance of NAP M&E).
	C6: The Updated Nationally Determined Contributions (NDC), 2021 of Pakistan recognizes the importance of financial innovation and the development of market-based mechanisms to mobilize resources for climate action, including NbS. The policy includes several provisions that highlight the need for innovative financial approaches. It proposes that the climate proofing risk assessment will be carried out for new public/private sector finance projects. The NDC calls for Sustainable Finance Framework (Chapter 9). This framework aims to integrate climate considerations into financial decision-making processes and promote investments in sustainable projects. However, clear protocols and frameworks should be developed to operationalize these mechanisms effectively.

Policy	NbS recognition and enabling provisions
National Climate Change Policy (NCCP), 2021	C1: Promoting tree plantation, conservation of natural resources, NbS and long-term sustainable use recognized (Policy Objective No.14), and there is an emphasis on investing in cost-effective, no regret NbS to reduce vulnerability to disaster risk reduction (Clause e). C2: The NPCC, 2021 calls for an inclusive pro-poor, gender sensitive adaptation (2. Policy Objective). The policy aims to enhance resilience of vulnerable communities and ensure that the development process is sustainable and caters to the needs of the poor (Clause 4.8). There is emphasis on access to technology and advocacy measures. However, it does not envisage their representation in institutions or role in decision making with respect to any of the measures. C3: Recognizes biological diversity as intrinsically important due to its contribution to the functioning of ecosystems and promotes Ecosystem- based Adaptation involving the conservation, sustainable management and restoration of ecosystems that can help people adapt to the impacts of climate change (Section 4.4, 4.5). Envisaged biodiversity related vulnerabilities to be integrated into national policies. Further the Plan envisages conservation of biodiversity shall be increased with proper management plans, legislative interventions as well as standardized eco-infrastructure designs. The policy links with the NDC, by identifying flagship programs like the Ten Billion Tree Tsunami Programme, Urban Forest Project, Clean Green Pakistan Movement, and Protected Areas and National Park Initiatives to reduce the impact of climate change. The policy targets the development of fifteen model protected areas across the country and ensures that at least 15% of the country's landmass is under some form of formal protection by 2023. C4: Does not have a clear or specific mandate on cross sectoral collaboration. Only recognizes the role of other sectors for achieving the policy objectives (Preamble). It suggests the establishment of Climate Change Cells in federal and provincial ministries (Institutional Mechanisms, a), and improve inter-ministerial and inter-departmental decision-making and coordination mechanisms on climate change issues both at provincial and federal levels to develop Pakistan's stand on various international policy issues relating to climate change. C5: Provides for capacity building that includes monitoring, reporting and verification system (Clause 6). The climate change implementation committees are established at the federal level for monitoring and verification purposes (Clause 11). C6: The National Climate Change Policy (NCCP) of Pakistan recognizes the importance of financial innovation and the development of market-based mechanisms to mobilize resources for climate action, including NbS. The funding of measures is through dedicated "Ecosystem Restoration Fund" as a comprehensive financial mechanism to finance projects and programs under the Ecosystems Restoration Initiative (Clause 4.5 (2)). The policy recognizes the need to create an enabling environment for private sector engagement in CC adaptation and mitigation activities that will catalyze greater and more frequent investments and accelerate the replication of climate-resilient technologies and approaches in core development sectors. Green bonds are specifically mentioned as a tool to finance sustainable infrastructure and NbS initiatives, providing a reliable source of funding for long-term projects.

Policy	NbS recognition and enabling provisions
National Water Policy Pakistan, 2018	C1: The NWP, 2018 in Pakistan recognizes climate impacts of water resources and its consequences on ecosystems such as glaciers and mountains. The objectives under the policy include improving watershed management through extensive soil conservation, catchment area treatment, preservation of forests and increasing forest cover (Clause 2.10 Policy objectives) C2: The policy outlines an integrated water management strategy that can optimize the economic, social and environmental returns on water resources, ensure equitable allocation among its competing demands as well as its judicious use by consumers and safe disposal of post-use effluents. Pakistan's Water Charter does not mention the community's role in decision making. C3: The Water Policy generally recognizes the need for improving watersheds and preservation of forests, increasing forest cover and restoring the health of related ecosystems (Clause 2.10). C4: Has a clear mandate for improving watershed through soil conservation, catchment area treatment, preservation of forests and increase in forest cover that requires collaboration sector specific agencies (2. Policy Objectives 2.10). Also calls for basin level planning that also requires cross-sectoral collaboration. C5: The policy pledges sustaining and upgrading the environmental integrity of the basin (Clause 6.1). Monitoring and upgrading of information is also mentioned. While the policy emphasizes the importance of identifying and sharing best practices, it does not provide specific mechanisms for upscaling these practices. Detailed action plans and frameworks should be developed to facilitate the wider adoption of successful water management strategies. C6: Aims at making water delivery services economically and financially sustainable (Clause 25.1). The NWP emphasizes the need for water pricing to reflect its economic value. It suggests that water charges should be determined on a volumetric basis and should cover at least the operation and maintenance costs of providing water services. However, the recognition and development of market-based mechanisms for mobilizing resources for NbS are limited.
National Disaster Management Plan (NDMP), 2012	C1: The NDMP does not explicitly mention NbS or include ecosystem-based approaches for DRR. The objective is to develop resilience in society by mitigating recurring floods, droughts, landslides (Objective 5.1). Overall, it aims to protect life and civic properties from disaster impacts and early post-disaster recovery. C2: The NDMP recognizes the need to reduce the vulnerability of poor and vulnerable communities and the recognition does not extend to their role in decision making (Clause 5 (iii)-objectives). The policy recognized that better knowledge of disaster risk at local and community levels, and monitoring of the situation concerning the rights of local communities, are needed for DRR. C3: Disaster preparedness under this plan is sought to be achieved by addressing other concerns such as water resources, agriculture, forestry, coastal areas, biodiversity and various vulnerable ecosystems. (2.2.1 Climate Change).

Policy	NbS recognition and enabling provisions
	C4: The plan does not provide a clear mandate for cross-sectoral collaboration and its role in the implementation of policy. C5: Aims at establishing better knowledge of disaster and risk at the local level. Promotion of Training, Education and Awareness in Relation to Disaster Management is also envisaged (4.7 Intervention). C6: No linkages with innovative and new age financing to support the investment required for disaster management. The main souce will be the government, with the national governments providing resources to local governments through a mix of fiscal and monetary measures (Clause 5.4.2).

4.3 Summary of gaps in policies

The policy review highlights several enabling provisions, as well as gaps in the policies for improvement, and these are summarized below.

1. Recognition of NbS and/or Ecosystem Management in Building Water Resilience

Common Gaps: Many policies recognize ecosystem management but lack explicit mention of NbS. Enhancing the integration of NbS into policy frameworks is needed to set a strong foundation for water resilience and climate adaptation. The linkages between water risk resilience and biodiversity in the policies are underdeveloped. Water policies in all four countries focus more on the impacts of natural and anthropogenic activities on water ecosystems rather than how ecosystems can support biodiversity, which in turn can enhance resilience and well-being.

Bangladesh: The Nationally Determined Contributions (NDC) 2021 recognizes ecosystem management but does not explicitly mention NbS. It includes targets for reforestation and forest restoration. The National Adaptation Plan (NAP) 2023-2050 explicitly recognizes NbS and prioritizes ecosystem restoration, including wetlands and biodiversity conservation. The National Water Policy (NWP) 1999 and National Plan for Disaster Management (NPDM) 2021-2025 do not mention NbS but acknowledge the importance of maintaining ecosystem health for long-term water security and disaster risk reduction.

India: The NDC 2021 recognizes NbS for climate mitigation, aiming to increase forest and tree cover. However, it lacks specific targets for wetlands. The National Water Mission (NWM) 2008 (Revised 2021) and National Water Policy (NWP) 2012 recognize ecosystem considerations and traditional water management systems but do not explicitly mention NbS.

Nepal: The NDC 2020 recognizes NbS with targets for increasing forest cover and community-based forest management. It also promotes nature-based tourism. The National Climate Change Policy (NCCP) 2019 does not explicitly mention NbS but acknowledges ecosystem-based approaches and the importance of building ecosystem resilience.

Pakistan: The NDC 2021 explicitly mentions NbS, highlighting flagship projects like the Ten Billion Tree Tsunami and Recharge Pakistan. The National Climate Change Policy (NCCP) 2021 promotes NbS for DRR and emphasizes long-term sustainability through nature-based approaches.

2. Community Inclusion in Decision-Making

Common Gaps: On community participation and inclusion, policies generally acknowledge vulnerabilities and provide for some level of participation in ecosystem management and improvement. However, most policies lack clear procedures for including communities in decision-making processes. An inclusive and gender-responsive NbS policy would require mandatory consultation with communities at the planning stage of NbS. The policies should include specific guidelines for engaging local communities in planning, implementing, and monitoring NbS initiatives to ensure local ownership and effectiveness.

Bangladesh: The NDC 2021 promotes community inclusion through collaborative forest management and social forestry programs. The NAP 2023-2050 emphasizes locally led adaptation and community-based afforestation, ensuring that local preferences and knowledge are central to adaptation strategies. The NWP 1999 mandates regular consultation with local communities for water management issues.

India: The NDC 2021 lacks provisions for community inclusion but promotes sustainable living through the 'Lifestyle for Environment - LIFE' initiative. The NWM 2008 promotes active community participation in groundwater recharging and water harvesting.

Nepal: The NDC 2020 outlines several mechanisms for community participation, including Local Adaptation Plans of Action (LAPAs) and community forestry programs. It emphasizes the leadership and participation of women, Indigenous Peoples, and youth in climate change forums. The NCCP 2019 aims at mainstreaming gender equality and social inclusion but lacks detailed mechanisms for ensuring active community participation in decision-making processes for NbS initiatives.

Pakistan: The NDC 2021 promotes community-led conservation funds and community-based disaster risk management. It emphasizes the importance of engaging local communities in planning and implementing DRR measures. The NCCP 2021 calls for pro-poor, gender-sensitive adaptation but lacks representation of communities in decision-making processes.

3. Recognition of Biodiversity, Ecosystem Integrity, and Connectivity

Common Gaps: Recognition of ecosystem connectivity is often missing. Policies should emphasize the importance of connected ecosystems for achieving climate resilience and include indicators for monitoring ecosystem connectivity.

Bangladesh: The NDC 2021 does not recognize ecosystem connectivity but acknowledges the role of ecosystems in adaptation and mitigation. The NAP 2023-2050 prioritizes ecosystem restoration and conservation but lacks indicators for monitoring ecosystem connectivity.

India: The NDC 2021 does not mention ecosystem connectivity but recognizes the role of forest and tree cover in achieving mitigation targets. The NWM 2008 promotes basin-level Integrated Water Resources Management (IWRM) as a key strategy for water resilience.

Nepal: The NDC 2020 recognizes the role of biodiversity in achieving policy objectives but does not mention ecosystem connectivity. The NCCP 2019 emphasizes integrated watershed management and the importance of coordinated efforts to manage land, water, and forest resources within watersheds.

Pakistan: The NDC 2021 recognizes the importance of biodiversity and ecosystem-based adaptation. It highlights flagship programs like the Ten Billion Tree Tsunami and the expansion of protected areas. The NCCP 2021 promotes ecosystem-based adaptation but lacks explicit mention of ecosystem connectivity.

4. Promotion of Cross-Sectoral Collaboration

Common Gaps: Specific guidelines for cross-sectoral collaboration are often missing.

Policies should provide clear frameworks for joint decision-making among different sectors to leverage synergies and work towards a joint vision for climate resilience.

Bangladesh: The NDC 2021 promotes cross-sectoral collaboration but lacks clear mechanisms for implementation. The NAP 2023-2050 seeks collaboration across sectors such as infrastructure, agriculture, and socio-economic sectors but acknowledges the need for stronger institutional arrangements.

India: The NDC 2021 highlights the need for international collaboration but lacks specific guidelines for cross-sectoral collaboration. The NWM 2008 promotes basin-level IWRM involving multiple sectors and proposes the formation of inter-sectoral advisory groups.

Nepal: The NDC 2020 recognizes the need for cross-collaboration and includes the formation of inter-ministerial committees and working groups focused on climate action. The NCCP 2019 promotes cross-sectoral collaboration through multi-stakeholder coordination mechanisms but lacks specific guidelines for joint decision-making.

Pakistan: The NDC 2021 outlines roles for various ministries and advisory bodies to ensure effective collaboration. The NCCP 2021 suggests establishing Climate Change Cells in federal and provincial ministries to improve inter-ministerial and inter-departmental coordination.

5. Mechanisms for Regular Monitoring and Upscaling of Best Practices

Common Gaps: Robust mechanisms for monitoring and upscaling best practices are often lacking. Policies should incorporate adaptive management practices, establish centralized databases for tracking best practices, and promote community involvement in monitoring and upscaling efforts. Strengthening cross-sectoral approaches in water-related policy and planning frameworks is essential. NbS are typically reflected in climate change policies (e.g., Pakistan) and adaptation plans (e.g. Nepal), but mechanisms for interaction between sector-specific policies, such as water and disaster management, needs elaboration to increase opportunities for accelerating NbS integration across sector for water management.

Bangladesh: The NDC 2021 mentions the need for a Monitoring, Reporting, and Verification (MRV) system but lacks detailed mechanisms. The NAP 2023-2050 provides detailed guidance on monitoring and evaluation (M&E) and upscaling best practices, including a web-based monitoring system and communication tools for sharing best practices.

India: The NDC 2021 recognizes the need for monitoring but lacks specific mechanisms. The NWM 2008 includes provisions for monitoring and upscaling through the Bureau of Water Use Efficiency (BWUE) and the Jal Sanchay Jan Bhagidari Dashboard, which facilitates real-time tracking and reporting of water management activities.

Nepal: The NDC 2020 promotes data-driven tracking of NDC targets and includes mechanisms for strengthening data generation and validation frameworks. The NCCP 2019 includes provisions for regular monitoring through LAPAs, which are designed to be flexible and adaptive.

Pakistan: The NDC 2021 emphasizes evidence-based decision-making and inclusive monitoring and evaluation (M&E). It calls for the involvement of government agencies, communities, technical experts, and the private sector in the M&E process. The NCCP 2021 includes monitoring and verification systems for climate change adaptation and mitigation measures.

6. Recognition of Financial Innovation and Market-Based Mechanisms

Common Gaps: Recognition of financial innovation and market-based mechanisms is often limited. Even in cases it is recognized, there is not much guidance on how to operationalise these. The policies should explore and promote innovative financial mechanisms such as carbon trading, green bonds, and public-private partnerships to mobilize resources for NbS implementation and enhance climate resilience.

Bangladesh: The NDC 2021 identifies the need for market-based mechanisms like REDD+ and resilience bonds. The NAP 2023-2050 highlights the role of public-private partnerships and innovative financing instruments such as green bonds and blended finance.

India: The NDC 2021 does not mention financial innovation or market-based mechanisms. The NWM 2008 is silent on market-based mechanisms but acknowledges the need for innovative approaches to tackle water resource management issues.

Nepal: The NDC 2020 proposes the formulation of a Climate Finance Strategy and mentions the potential use of carbon markets. The NCCP 2019 promotes innovative financial mechanisms such as green bonds, blended finance, and result-based financing to support climate change mitigation and adaptation measures.

Pakistan: The NDC 2021 calls for a Sustainable Finance Framework and emphasizes the need for climate-proofing risk assessments for new public and private sector finance projects. The NCCP 2021 includes an Ecosystem Restoration Fund but lacks specific market-based mechanisms.

By addressing these gaps, policies can be more effective in promoting sustainable and resilient climate change adaptation strategies, ensuring that NbS are integrated into national frameworks, community participation is enhanced, ecosystem connectivity is recognized, cross-sectoral collaboration is facilitated, robust monitoring and upscaling mechanisms are established, and innovative financial mechanisms are utilized.

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CHAPTER 5

Financing for Ecosystem Protection and Water Resilience in South Asia

The chapter provides an overview of financing strategies adopted by South Asian countries including the development of national-level environment and disaster management funds, testing of Payment for Ecosystem Services (PES), and policy measures for enhancing private sector engagement and funding for ecosystem-based approaches to water resilience.

It is estimated that South Asia will need 2.727 trillion USD to meet their NDC targets (ADB 2023). However, in 2019, only 21.9 billion USD was available for South Asia, indicating a substantial gap between finance needs and availability.

At the national level, Nepal's latest NDC communication estimates that the cost to achieve conditional mitigation targets for Nepal will be 25 billion USD by 2030. Similarly, for Pakistan, the World Bank Country Climate and Development Report estimates an investment of 348 billion US dollars, or 10.7% of the country's GDP by 2030 for adaptation and decarbonization efforts. As per NAP 2021 Bangladesh, the country is spending approximately 1 billion USD annually, which is about 6 to 7 percent of its annual budget, on climate change adaptation (CCA). However, the World Bank estimates that by 2050, the country will need 5.7 billion USD for adaptation finance, over five times the current expenditure. Three-quarters of the climate change funding comes from the government, with the remainder sourced from international development partners, including bilateral, multilateral, and private funding. However, there is a clear acknowledgement that the full implementation of the strategic mitigation actions identified under NDC is conditional on the support of international stakeholders.

The section below provides examples of financing strategies adopted by South Asian countries and suggestions on how they could better align to support the design and implementation of NbS projects for water sector resilience.

5.1 National-level environment and climate protection funds

Most countries in South Asia have established dedicated environment and climate protection funds within their climate, environment, or disaster management policy frameworks. Some examples include the Bangladesh Climate Change Trust Fund (BCCTF), established under the Ministry of Finance, with the funds used to promote sustainable development practices and the implementation of the climate change strategy of Bangladesh. Similarly, Pakistan, supported by the World Bank, initiated the Ecosystem Restoration Fund (ESRF) at the United Nations Framework Convention on Climate Change (UNFCCC). The fund is supporting large-scale afforestation, as a part of the Recharge Pakistan and other similar initiative in the Indus Basin, and investments are linked to multiple benefits such as biodiversity conservation, land restoration, and ecotourism development. These investments serve as a catalytic entry point for scaling up NbS, directly contributing to national water resilience, climate adaptation, and sustainable development targets.

5.2 Payment for Ecosystem Services (PES)

The concept of PES is gaining recognition across South Asia, and governments are actively exploring approaches to leverage it as a nature conservation, poverty reduction, and benefit-sharing strategy. The primary objective of PES is to establish a mechanism that ensures tangible and equitable sharing of the benefits derived from ecosystem services with local communities.

In the forestry sector, Bhutan, India, and Nepal have been implementing several pilot initiatives to test methodologies for operationalizing the PES schemes. The analysis of ongoing PES schemes suggests that these initiatives mostly target forest protection and aim to incentivize forest-dependent local communities to engage in watershed and irrigation management.

The Bhutan For Life initiative is supporting the renewal of existing PES contracts between communities and water users. In 2020, the Yakpugang community renewed a 20-year agreement with Mongar Municipality, and additional PES agreements are planned in Paro and other watersheds throughout the country.

In 2022, India's state government of Meghalaya launched a PES scheme across the East Khasi Hill Districts, home to forest-dependent Indigenous Khasi communities. This scheme provides financial support of approximately 100 USD/hectare/year (8000 INR) for forest protection and provides an additional 20 USD/hectare/year for areas that are traditionally recognized and protected as sacred groves or are located in eco-sensitive zones around protected areas including national parks, wildlife sanctuary or wildlife corridors (MBMA, n.d.).

Although these initiatives have high potential, it is crucial to note that they are still in their early stages of development, signifying the ongoing evolution of PES as a sustainable and inclusive strategy for sustainable financing of NbS across the South Asia region.

Countries must document and share the lessons learned from these new NbS financing approaches at the national and regional levels. This will allow them to enhance their capacity for PES implementation and develop robust strategies to support the sustainable financing of NbS for water resilience.

5.3 Private sector financing

Collaborations between public entities and private companies can unlock funding for climate-related water projects. Public-Private Partnerships (PPPs) provide a mechanism for sharing costs, risks, and expertise to tackle complex water challenges more effectively. There is currently very limited private sector contribution to climate and nature financing, providing an opportunity to build partnerships and scale-up efforts.

In recent years, several financial innovations have emerged to de-risk and catalyze investments in climate and nature. The GCF Guidance on Blended Financing instruments identifies five categories of instruments to catalyze finance for NbS through the private sector (Glemarec, Y et al. 2023).

1. Information instruments, including action plans for promoting behavioural changes in the financial and corporate sectors, such as policies requiring the disclosure of the impact of private sector production patterns and supply chains on nature and ecosystem degradation, or tools informing consumers of sustainable products, such as deforestation-free commodities.
2. Control and regulatory instruments to limit the unsustainable extraction of natural resources, such as permits and licenses such as fishing quotas and logging regulations.
3. Economic and market instruments that act as incentives or disincentives to shape a firm's decision-making and consumer preferences. These include biodiversity-positive carbon credits, biodiversity offsets, payments for ecosystem services, and nature certificates. Green procurement and certification schemes are also included in this category, acting in tandem with information instruments.
4. Institutional instruments, such as the creation of market regulating agencies, environmental agencies, green banks, investors' coalitions, green courts or tribunals, and associations of central bank regulators.
5. Direct public investments, often to demonstrate proof of concept and potential scalability or replicability of NbS. This includes grants in the form of fiscal transfers, official development assistance, private philanthropy, or individual grants.

The application of these instruments is visible in countries like India, which is one of the world's largest economies and has a vibrant private sector. India amended the Indian Companies Act in 2013, requiring a minimum of 2% of the annual profits of a company to be redirected to corporate social responsibility, and providing an opportunity to redirect redirect funds to enhance social development throughout the country. In 2022, India's corporate social responsibility spending on ecosystem management and conservation-related issues more than doubled to 34.6 million USD (2,837 crores INR), making it the largest recipient of such funds after health and education, according to data released by the Ministry of Corporate Affairs (Pattanayak, B. 2023). This substantial increase reflects a growing commitment in the private sector towards environmental sustainability and should be shared among the private sector and governments in other countries in South Asia.

In 2022, the Government of India announced the development of sovereign green bonds and thematic funds as instruments for mobilizing resources for climate action and enhancing the role of the private sector in mobilizing climate finance. Green bonds and thematic funds are issued by sovereign governments and the private sector to raise money for initiatives that contribute to environmental and climate benefits. The proceeds from these bonds are earmarked for green projects, providing an entry point for the government to mobilize resources for resources for large-scale NbS for water resilience projects jointly with the private sector. Green projects funded through the bonds can include initiatives for sustainable water management including water recycling, sustainable land use initiatives for sustainable forestry and agriculture, afforestation, and biodiversity conservation.

5.4 Multilateral funding mechanisms

There are several multilateral funding mechanisms available to national governments in South Asia, such as REDD+ under the UNFCCC, as well as the Green Climate Fund (GCF), and Global Environment Facility (GEF) which provide funding for large-scale NbS projects. Other than these multilateral funding windows, several banks and development agencies, such as the World Bank (WB), Asian Development Bank (ADB), and Japan International Cooperation Agency (JICA) are promoting investments in NbS.

5.4.1 Reducing Emissions from Deforestation and Forest Degradation (REDD+)

Reducing emissions from deforestation and forest degradation in developing countries (REDD+) is one of the well-established funding mechanisms regularly accessed by countries in South Asia. REDD+ was developed under the UNFCCC to incentivize forest-dependent communities to conserve forest landscapes.

Across the South Asia region, several REDD+ projects are ongoing or in the readiness phase working on the analysis of carbon storage and sequestering the potential of forest landscapes, land rights, data storage and monitoring mechanisms. India has over 40 ongoing projects listed in the REDD+ database, showcasing a diverse range of projects, from the expansive Himachal Pradesh Reforestation Project, covering an area of 222,951 hectares, to smaller-scale initiatives like the Plantation Project on Wastelands by Sun Plant Agro Limited, implemented on less than 1,000 hectares. The Himachal REDD+ project focuses on reforesting degraded land in the watersheds of the Mid-Himalayan region by planting native tree species to enhance the health of catchment areas, and its biomass production potential for increasing carbon stock. The Sun Plant Agro project aims to establish and manage commercial timber and Jatropha plantations, with its primary objective to create a carbon sink through forest planting in barren lands, aiming for carbon-verified sequestration and subsequent monetisation through the REDD+ mechanism.

5.4.2 Green Climate Fund (GCF)

The GCF has emerged as one of the main sources of climate financing in South Asia, particularly in Bangladesh, Bhutan, Nepal, and Pakistan. In its more than two decades of existence, the GCF has funded 29 projects in the region, totalling 1.344 billion USD. Some examples of externally funded projects promoting NbS for water sector resilience in South Asia are discussed below. Different countries are using the GCF funding to build readiness or to support large-scale national and basin-level initiatives.

In Nepal, the Gandaki River Basin supports the livelihoods of approximately 19.3% of the population, including representatives of more than 40 indigenous groups (GCF 2020). These local livelihoods are increasingly impacted by extreme climate-related events, including landslides and floods. To increase their resilience, Nepal is implementing the GCF-funded *Improving Climate Resilience of Vulnerable Communities and Ecosystems in the Gandaki River Basin, Nepal,* project, with a total budget of 25 million USD. The project adopts an ecosystem and community-based approach for reducing climate vulnerability, by addressing issues including a lack of information about climate change, and weak regulatory frameworks and institutional structures. This initiative will strengthen the conservation and restoration of ecosystems to enhance their climate resilience and reduce the impacts of landslides and floods. The project also provides technical assistance to increase the capacity of communities to maintain and support climate-resilient ecosystems. The results of the project can help to strengthen community engagement in the design and implementation of ecosystem-based adaptation and forestry management, linked to Criterion 5 on inclusive, transparent, and empowering governance processes, and Criterion 2 on design at scale, and can inform similar initiatives across the Himalayan region at the basin-level.

5.4.3 Development banks

Development banks are increasingly recognising the benefits of NbS, and allocating funding for NbS development, implementation, and upscaling. In 2017, the World Bank initiated its NbS program for disaster risk management to ensure that Bank's operational teams and clients make use of natural and modified ecosystems to reduce risks associated with natural hazards and achieve other development objectives (WB 2017). The initiative aims to promote the identification of NbS investments across the World Bank portfolio and support the mainstreaming of NbS among clients, management, and operational staff by providing technical guidance and conducting pilot projects. During 2012 and 2023, the World Bank approved over 200 projects with NbS components. The financing committed to NbS components in these projects together exceeds $10 billion. A significant portion of this lending is linked to technical support from the Global Program on Nature-based Solutions for Climate Resilience (GPNBS), which is hosted by the Global Facility for Disaster Reduction and Recovery (GFDRR).

At the regional level, the Asian Development Bank launched the Nature Solutions Finance (NSF) hub for Asia and the Pacific in 2023. The hub aims to support nature-based climate actions and biodiversity conservation projects which are prioritized by national governments. The NSF Hub's goal is to increase investments in NbS projects across Asia and the Pacific by developing scalable and bankable NbS demonstration projects (ADB 2023b).

Development banks are also supporting national-scale NbS. In the state of Meghalaya in India, the World Bank is supporting the Meghalaya Community-Led Landscape Management Project (MCLLMP) with a 60 million USD loan (MBDA 2018). The project aims to capacitate rural communities and traditional institutions to manage their natural resources through the design and implementation of community-led natural resource management plans at village and block levels. Strengthened community management will reduce deforestation, mining, and shifting agriculture which has been a cause of environmental degradation and increased the vulnerability of communities to climate change. Following a similar strategy, the JICA is supporting the Sustainable Catchment Forest Management project in Tripura (SCATFORM) in India, through a Japanese Overseas Development Assistance (ODA) loan of 16 million US dollars (1000 crore INR). The 10-year project began in 2018 and will be implemented by the Tripura Forest Department, in North Tripura, Unakoti, Khowai, West Tripura, Sepahijala, Gomati, and South Tripura. The project aims to establish 450 new Joint Forest Management Committees and 1,350 self-help groups to promote sustainable forest management and measures to improve the livelihoods of forest-dependent communities (Tripura Forest Department, n.d.).

The development of these large-scale initiatives supported by detailed climate analyses and project feasibility studies has contributed to strengthened community engagement in landscape management. The learnings from these projects should be documented and shared at the national and regional levels to enhance community engagement in the implementation of NbS measures.

Photo Credit: IUCN Pakistan

CHAPTER 6

Analysis of Barriers and Strategies to NbS Implementation at Scale

This chapter presents a summary of key challenges to implementing NbS at scale, and the strategies that governments are applying to address them.

During the regional surveys, government officials identified several barriers to implementing NbS at scale and provided insights into how the national governments are addressing these barriers through examples of some flagship national initiatives. The barriers to NbS implementation range from inadequate understanding of NbS, policy and institutional silos, poor engagement of indigenous and rural communities, inconsistencies in monitoring and sustainable financing. These are discussed under challenges in translating enabling policy provisions (section 6.1) and some examples of how countries are trying to address these challenges (section 6.2).

6.1 Challenges in translating enabling policy provisions

The review of policies in Chapter 4 highlights several NbS-enabling features and includes strategies such as Forest Landscape Restoration (FLR) for river basin restoration, and Ecosystem-based Adaptation (EbA) for addressing floods and droughts. Despite this, there are several barriers to implementing the policies, as highlighted by government officials during the regional survey and discussed below.

a) Inadequate awareness and understanding of the multiple benefits of NbS

Limited understanding of NbS and its benefits among policymakers and stakeholders is a significant challenge. NbS is often seen as a new mitigation approach rather than a comprehensive strategy, leading to poor integration into existing policies. For example, in India, Pakistan, and Nepal, participants identified inter-basin water transfers as a viable strategy to manage climate-induced droughts and floods, despite several studies demonstrating that such transfers can have disastrous effects on basin ecosystems and local communities.

There is a lack of comprehensive training and capacity-building programs focused on NbS. While there have been some initiatives on water governance and natural resource management, there is no example of a national-level NbS capacity-building initiative in the South Asia region. This gap in capacity building means that stakeholders are not adequately equipped with the knowledge and skills needed to implement NbS effectively.

Due to a poor understanding of the multiple benefits of NbS and its role in addressing water resilience issues, the water and disaster management planning continues to prioritize investments in large-scale engineering solutions, such as the construction of canals for inter-basin transfers and dams for water storage. During the government survey, more than two-thirds of respondents indicated that NbS generally takes longer to provide benefits, whereas grey infrastructure can be implemented with certainty about the type and timescale of benefits. This preference for traditional engineering solutions over NbS is a significant barrier to the mainstreaming of NbS.

To create better awareness, there is a need to invest in data and analysis. Accurate and up-to-date data on rainfall patterns, river flows, and groundwater levels are crucial for planning and implementing climate change adaptation strategies. Inadequate data hinders effective decision-making.

Green-grey infrastructure solutions are crucial for the maintenance of ecological connectivity at scale sustaining urban growth, meeting the demand for water from different sectors, and mitigating the impacts of climate change. However, there is a poor understanding of how green-grey infrastructure can be applied in specific contexts. Additionally, NbS implementation is perceived as costly, which affects its adoption and limits its application at scale, reducing overall effectiveness. More research and sectoral guidance are needed.

b) Lack of integrated policies creates institutional silos

NbS application at scale require spatial planning and cross-sectoral collaboration, but existing policies are frequently siloed, leading to fragmented efforts and inefficiencies. This lack of coherence among sectoral policies is a major challenge identified by government officials in mainstreaming NbS as a strategic tool in water resilience planning. It also leads to a lack of clarity on the roles and responsibilities of different levels of government regarding NbS, making it difficult to develop a joint vision on how NbS can be integrated into national, district, and block-level plans.

Climate change, disaster risk, and water management are often treated as separate issues rather than interconnected challenges. There is a lack of cross-sectional policies that synergize various environmental agendas. Multiple programs exist, but a cohesive policy framework is missing at the national level. NbS is relatively new and not fully integrated into government strategies, and sectoral guidance and monitoring frameworks are underdeveloped. Additionally, there is no intergovernmental mechanism with a clear mandate for promoting NbS.

Limited synergies between the climate, environment, water, rural development, and planning agencies make it challenging to mainstream NbS into water management strategies. For example, water policies recognize the role of IWRM as an important strategy for managing climate impacts on water resources. This provides a good entry point for the development of NbS projects, however, the approach remains underutilized due to conflicting institutional priorities, lack of interagency coordination mechanisms, and insufficient data to build economic arguments for NbS mainstreaming using IWRM.

To promote policy coherence, countries need to invest in the development of long-term visions and strategies that recognize the importance of NbS. However, the challenge remains in raising awareness among policymakers about the benefits of NbS and ensuring that these strategies are implemented effectively. The lack of coordination among different agencies and stakeholders further complicates the integration of NbS into water management and climate resilience planning.

Institutional coordination is needed for achieving transformational changes through NbS. While policies promote cross-sectoral collaboration and data sharing, there are still large gaps in coordination and integration among sectors. Currently, none of the countries in the region have identified and mandated any institution for coordinating interagency dialogue and programmatic alignment to contribute to NbS. For improving data consolidation and management, India has established the Water Resources Information System (WRIS) under NHP for standardized data and information management. To deal with water stress, the water policy emphasizes the need for groundwater development and recharge mechanisms, as well as comprehensive flood management programs under the National Hydrology Project (NHP). In India, government agencies like NIUA are working on the River Cities Alliance (RCA) and NbS Forum for Cities to promote cross-sectoral collaboration for mainstreaming of NbS in urban planning.

Policies should encourage stronger interdepartmental cooperation and establish dedicated platforms to convene and facilitate the process. For example, Pakistan's Climate Action Plan proposes the establishment of "Climate Change Policy Implementation Committees" at the federal and provincial levels, which are responsible for regularly monitoring and updating the Climate Action Plan. Expanding the scope of the existing platform to include water, disaster, and agriculture ministries can better support integration in planning and policy development and decrease redundancies in national project budgeting.

c) Strategic engagement of rural and indigenous communities

Climate change and other anthropogenic drivers have impacted the functioning of ecosystems and the services they provide. Anthropogenic factors, such as unsustainable use, illegal trade in wildlife, conversion of habitats, invasive alien species, pollution, and climate change, are combined with indirect drivers such as socioeconomic and demographic changes to create stress on the ecosystems (IPBES 2018). Climate change will exacerbate these impacts, especially among Indigenous and vulnerable communities, and inclusive environmental governance strategies and targeted policies can help arrest this loss. Therefore, involving local communities in project planning and execution is vital for sustainability. Raising awareness about climate change impacts and water conservation can enhance community engagement.

Despite the important of indigenous community in NbS design and implementation, there are significant gaps in the strategic inclusion of communities and their traditional wisdom. At the local level, Indigenous knowledge and customary laws play an important role in the sustainable management of ecosystems, however, co-management strategies with Indigenous Peoples and Local Communities (IPLCs) remain underutilized, leading to resource conflicts. In many landscapes dominated by Indigenous People, national policies often contradict Indigenous land-use practices, resulting in a lack of adherence and cooperation.

The policies reviewed recognize the importance of involving IPLCs in project decision processes, however, there is often a gap between recognizing the importance of community engagement and developing comprehensive mechanisms for implementation. Policies should provide more detailed guidance on how to engage IPLCs in NbS project design and implementation. The Nepal Climate Change Policy is a strong example, suggesting several strategies to promote community engagement.

There is a need to invest in a better understanding of community engagement mechanisms in the specific ecosystem and social contexts. In South Asia, there are several instances of design and implementation of forest and landscape management strategies in the lands traditionally managed by the Indigenous communities without much local consultation. The lack of involvement of Indigenous communities and failure to recognize their historical role in managing a landscape, forest or wetland can lead to conflict and lack of achievement of the project objectives. Therefore, the studies should be designed to document and provide recommendations to support the design of NbS projects for water resilience in an inclusive manner.

d) Poor monitoring and evaluation

NbS projects need long-term monitoring and evaluation to assess their effectiveness and make necessary adjustments, however, these are often underfunded and lack the necessary technical capacity. Ensuring that there are sufficient resources for ongoing monitoring and evaluation can be challenging. Effective monitoring and evaluation frameworks are essential to track the progress and outcomes of NbS projects, as well as for the design and implementation of adaptive management practices. Adaptive management involves adjusting strategies and actions based on new information and changing conditions.

Without detailed monitoring and evaluation, it is difficult to highlight the benefits provided by NbS interventions and compare the approaches against traditional grey infrastructure. This also limits the countries' ability to access global climate funding, which requires robust monitoring and evaluation systems. This gap often leads to the failure of NbS projects to achieve their intended outcomes. For instance, many NbS initiatives are project-focused and lack strategic orientation, resulting in fragmented efforts that do not contribute to long-term water resilience. Without systematic monitoring, it is challenging to assess the effectiveness of these solutions, leading to missed opportunities for learning and improvement.

A notable example is the implementation of NbS in South Asia, where water-related disasters such as flash floods and landslides are prevalent. Despite having a well-established institutional architecture for disaster management, the integration of NbS remains limited due to inadequate monitoring mechanisms. It is therefore important to develop and implement comprehensive monitoring and evaluation frameworks that are integrated into NbS policies from the outset. This includes setting clear objectives, indicators, and methodologies for assessing the performance of NbS projects. By doing so, governments can ensure that NbS initiatives are not only effectively implemented but are also continuously improved based on empirical evidence.

e) Funding constraints

Securing adequate funding for water sector projects is a significant challenge. Both the government survey and online literature review suggest that national governments are the main source of funding for projects related to climate change adaptation and mitigation, however, these are not sufficient. In Bangladesh, government representatives shared that the country currently spends about 0.8% of GDP on the BDP 2100 program, but the estimated need is around 2% of GDP. Projections suggest that Bangladesh should aim to secure at least 0.5% of GDP per year from the private sector for water resource management projects.

In Nepal, survey participants indicated that, to fund climate resilience initiatives, the government relies primarily on national budget allocations, with additional support from development partners and international funding sources such as GCF, GEF, ADB, World

Bank, JICA, and others. The Environment Protection Act 2019 established an Environment Protection Fund (EPF) to finance climate resilience activities. In India, national governments provide funding through multi-million dollar and multi-year flagship programs such as the National Mission on Clean Ganga and established the Ecosystem Restoration Fund (ERF).

The NDCs of India, Nepal, and Pakistan indicate a strong recognition of the need for private sector funding and include mention of carbon markets, blended financing, and other market-based mechanisms for enhancing the private sector's role in financing climate change adaptation and mitigation measures. India is currently developing platforms where companies can buy and sell carbon credits at the national level. In Pakistan, the government has included large scale ecosystem-based adaptation initiatives in its NDCs such as the 10 Billion Tree Tsunami (TBTT) and Recharge Pakistan. These help the government in raising funds with a long-term perspective and diversity of sources.

In India, the state of Meghalaya is pioneering the concept of "Nature Banks" to mobilize funds for conservation efforts and integrate environmental risks into capital markets. Meghalaya is also developing a "Carbon Farming Act" to establish a legislative framework to promote regenerative agriculture and retain soil carbon.

Despite these efforts, the private sector investment in NbS remains weak, due to the perception that NbS requires high initial costs, deterring potential investors from engaging in the financing of such projects. This challenge can be addressed by developing capacity-building programs for private sector banks and foundations on the principles and multiple benefits of NbS, strengthening cost-benefit analyses for NbS projects, and publishing the results in a national database of NbS projects that is readily available to the private sector. The private sector also faces difficulties in quantifying the benefits of NbS projects. Unlike traditional infrastructure projects, the multiple benefits of NbS are not documented and are not easily translated into monetary terms. This limits the assessment of the return on investments of NbS projects, which are the primary considerations for the private sector before providing finance to such initiatives.

Therefore, there is a need to invest in cost-benefit analysis of NbS in terms of time needed for implementation of a particular type of solution and the return on investments. This will promote the mainstreaming of NbS in policy and infrastructure projects on water resilience and support the development and testing of methodologies to quantify the economic benefits of NbS integration. The research outputs will feed the development of NbS models that can attract both public and private investments, and application of blended financing tools such as green bonds, climate funds, and public-private partnerships.

To address the challenges in securing the long-term financing of nature, countries in the region are testing different strategies, including Project Finance for Permanence (PFP) approach. The PFP approach include designing long-term project to secure the policies, capacity, institutional arrangements and full funding for the effective and long-lasting protection of natural resources and climate change adaptation. Some examples of PFP initiatives from the region are discussed in the next section.

6.2 Examples of flagship initiatives promoting NbS application at scale

6.2.1 Coastal Embankment Improvement Project (CEIP), Bangladesh

In Bangladesh, the Coastal Embankment Improvement Project (CEIP) is executed by the Bangladesh Water Development Board (BWDB) and aims to enhance the climate resilience of coastal areas through a combination of grey and green infrastructure approaches. The project is supporting the upgradation of embankments to protect against tidal flooding and storm surges while incorporating NbS such as mangrove afforestation to stabilize soil and reduce erosion.

This combination of engineered structures and natural systems has helped in improving the physical resilience of the coastal areas but also enhanced biodiversity and ecosystem services, contributing to long-term sustainability. The project is supported by a mix of funding sources, including International Development Association (IDA) credit of of 375 million USD and a Pilot Program for Climate Resilience (PPCR) grant of 25 million USD. This blend of traditional loans and climate resilience grants demonstrates a strategic approach to mobilizing financial resources for large-scale infrastructure projects. The project also emphasizes community involvement in maintenance and monitoring, which helps ensure the sustainability of the investments and fosters local ownership. By leveraging both green and grey infrastructure and employing innovative financing mechanisms, the CEIP sets a precedent for future climate resilience projects in Bangladesh.

6.2.2 Improving Climate Resilience of Vulnerable Communities and Ecosystems in Gandaki River Basin, Nepal

In Nepal, the Gandaki River Basin supports the livelihoods of approximately 19.3% of the population, including representatives of more than 40 indigenous groups (GCF 2020). These local livelihoods are increasingly impacted by extreme climate-related events, including landslides and floods. To increase their resilience, Nepal is implementing the GCF-funded Improving Climate Resilience of Vulnerable Communities and Ecosystems in the Gandaki River Basin, Nepal, project, with a total budget of 25 million USD. The project adopts an ecosystem and community-based approach for reducing climate vulnerability, by addressing issues including a lack of information about climate change, and weak regulatory frameworks and institutional structures. This initiative will strengthen the conservation and restoration of ecosystems to enhance their climate resilience and reduce the impacts of landslides and floods. The project also provides technical assistance to increase the capacity of communities to maintain and support climate-resilient ecosystems. The results of the project can help to strengthen community engagement in the design and implementation of ecosystem-based adaptation and forestry management, linked to Criterion 5 on inclusive, transparent, and empowering governance processes, and Criterion 2 on design at scale, and can inform similar initiatives across the Himalayan region at the basin-level.

6.2.3 Metro Colombo Urban Development (MCUDP), Sri Lanka

Funded by the World Bank, with a budget of 32 million USD, promoted the use of wetlands as a key solution for the flood mitigation in Colombo. During the last 30 years, an estimated 40% of the wetlands in Colombo has been lost due to direct and indirect impacts of urbanization. In 2010, the city was severely impacted by floods that displaced residents from their homes and submerged many important government buildings including the parliament. The economic concerns about flooding – with estimates suggesting that Colombo could lose one percent of its GDP on average every year due to a major flood, this created recognition among the government on the need for preserving wetlands, which subsequently protects humans.

The project funded selective partial dredging of the lake in 2017 following extensive stakeholder consultations and scientific study. The projects like MCUDP, are helping countries integrate "green" infrastructure – forests, wetlands and mangroves – into human built "gray" infrastructure to better address climate and ecosystem risks and enhance human health and well-being.

The project supported the restoration and protection of 20 square kilometres of freshwater lakes and wetlands, and the creation of additional grasslands and swamps in the city as a buffer for flash floods. The project documents suggest that the wetlands and forests of the city absorb up to 90 percent of Colombo's greenhouse gas emissions and also provide other benefits like food and supplementary income for the urban poor. The project integrates natural solutions with engineered infrastructure, such as pumping stations and tunnels, to manage floodwaters effectively. This hybrid approach has benefited 2.5 million residents by enhancing flood resilience and improving urban living conditions. The success of this initiative has inspired the Sri Lankan government to develop 50 additional plans for wetland conservation across the country, demonstrating the project's scalability and potential for broader impact. Overall, the Colombo wetlands project highlights how NbS can address urban flood risks while providing multiple co-benefits, such as climate mitigation, conservation of biodiversity, and socio-economic benefits for vulnerable communities.

6.2.4 River Rejuvenation Program, India

India launched the nationwide river rejuvenation program in August 2022 to address reduced ground water availability and increasing riverbank erosion, with a total budget of more than 2.2 million USD over the next five years (PIB 2022). The program targets 13 rivers across the country, including international rivers such as the Sutlej, Chenab, and Brahmaputra, as well as national rivers such as the Narmada, Godavari, and Mahanadi. These 13 rivers cover a total basin area of over 16,11,149 square kilometers representing more than 49% of the geographical areas of the country. The cumulative length of all the 13 rivers, including 202 tributaries is 44,854 km (ICFRE 2022b).

The initiative will implement a large-scale afforestation program to restore river catchments to address growing water stress and support the achievement of national targets on climate change. The program is also implementing site-specific interventions such as plantations for soil and moisture conservation, and ecotourism development along the riverfront to subsidize local livelihoods. India can apply the criteria and indicators from IUCN Global Standard for NbS during the design and monitoring of these initiatives to ensure they maximise benefits to society and biodiversity, and maladaptation or negative impacts on local communities and river ecosystems.

6.2.5 Bhutan for Life (BFL), Bhutan

Bhutan for Life (BFL) is one example of PFP, and a 14-year long, 118.3 million USD initiative, GCF-funded program. It aims to secure long-term financing for the conservation and sustainable management of Bhutan's protected areas, which comprise more than 50% of the country (GCF 2017). Many of Bhutan's protected areas are under pressure due to increasing economic development, illegal extraction of resources, and the adverse impacts of weather-related events such as landslides, floods, and forest fires. The initiative started in 2017, aims to address these challenges using a collaborative model with donations and grants from the GCF, the Royal Government of Bhutan, the World Wildlife Fund (WWF), and the Bhutan Trust Fund for Environmental Conservation (BTFEC). One of the stated objectives of BFL is to support the implementation of Bhutan's NDC commitment to remain carbon neutral, by ensuring that at least 60% of the country remains under forest cover or protected areas (PAs). The initiative supports capacity building for forest officials and strengthens community co-management of PAs through the development of policy frameworks, implementation of Payment for Ecosystem (PES) schemes, and other EbA interventions. As per the 2020 annual project report, 79% of the population (5,936 out of 7,500 people) living within the PAs were trained in environmental conservation and waste management (GCF 2021). The project is also providing support to existing PES mechanisms, including the agreement between the Yakpugang community and the Mongar municipality, which has been in operation since 2010 and aims to incentivize the members of Yakpugang Community Forest to protect the watershed that acts as the water source for Mongar city. In return, the community receives a payment of 0.4 to 0.6 cents USD (Nu 30 to 50) per unit of water used by Mongar municipality. The learning from these PES sites can inform the development of similar long-term financing strategies for NbS projects, at the national and regional level and can serve as a case study for Criterion 4 on making NbS economically viable for the long term and consider a portfolio of resourcing options to support NbS implementation.

6.2.6 Living Indus Basin Initiative, Pakistan

Launched in 2022, the Living Indus Basin is included as one of the initiative under NDC, and an example of PFP model. The initiative is designed as an umbrella initiative and aims to secure long-term financing for 25 priority projects identified through multi-stakeholder consultation process. The priority projects include creation of a basin level institution in the form of an Indus Council, and application of innovative financing tools such as Climate and Nature bonds. The initiative has created a Trust Fund, which focuses on securing funding from diverse sources and institutional support for conservation projects. The initiative is designed as a multi-year umbrella initiative and aims to restore 25 million hectares by 2030. The initiative incorporates a variety of NbS, such as reforestation, wetland restoration, and the conservation of natural habitats. The overall cost of implementing the 25 identified interventions is estimated to be between 11-17 billion USD, and the project is applying a multipronged funding approach with the following key elements of its funding strategy:

- **International Support and Recognition:** The initiative has been recognized as a World Restoration Flagship, making it eligible for additional technical and financial support from the United Nations.
- **Innovative Financial Instruments:** The initiative plans to pilot new financial vehicles such as Climate and Nature Performance Bonds. These bonds are designed to attract investment by linking financial returns to environmental performance.
- **Public and Private Sector Collaboration:** Extensive consultations with public sector entities, private sector stakeholders, experts, and civil society have been conducted to consolidate efforts and resources.
- **Crowdsourcing Knowledge and Resources:** The initiative includes the establishment of the Living Indus Knowledge Platform and Indus Trust Fund to crowdsource knowledge and pooling of resources.

Photo Credit: Sarovar Alam

CHAPTER 7

Recommendations for Mainstreaming NbS for Water Resilience

This chapter provides recommendations for promoting the mainstreaming of NbS in water resilience initiatives, starting from the constitution of the National-level NbS Task Force (NTF) multi-agency and multi-disciplinary platform of different government agencies, including those who need to work together to design a national NbS strategy for water management and address the current knowledge, capacity, and funding gaps.

This chapter aims to provide national governments with a guidance in the form of recommendations to accelerate the adoption of NbS for long-term water resilience and sustainable development. The recommendations provide the strategic directions for the federal governments and build on the previously identified barriers through the regional government survey. The guidance also integrates recommendations from IUCN Global Standard for NbS, tailored to the South Asia context. Based on the review and analysis, the recommendations are organized as general and specific insights and recommendations. The specific recommendations are contained under five action points and they aim to address policy, institutional, and knowledge gaps, as well as discuss examples of how countries are trying to address these gaps.

By initiating the implementation of recommendations, countries can enhance the implementation of current policy provisions, through which, the national governments will be able to define objectives and indicators specific to the country's needs. The national strategy will then provide the basis for the development of the sub-national or landscape-level strategy with specific approaches to address water-related climate risks prioritized by the local communities.

The recommendations are designed to last for up to five years, as the understanding of NbS is changing quickly, and additional case studies and lessons learned will contribute to the application of this guidance. Financing for NbS is rapidly changing, with new approaches to engage the private sector and new funding mechanisms being developed as investors become aware of the benefits of NbS.

7.1 Constitute national NbS Task Force to lead the development of a NbS strategy for water management at scale

The regional government survey highlighted the absence of a national-level guidance and interagency coordination platform as a hindrance to promoting NbS in water resources planning at scale. Currently the inclusion of NbS is project-specific and is often applied as an afterthought, following business-as-usual development, and tends to lack strategic orientation. The NbS design and implementation for water resilience requires coordination among various government agencies working at different levels and in different sectors, and engagement of academia, private sector, local communities and non-governmental organizations (NGOs). As highlighted by the regional survey and water policy review, governments are promoting IWRM and basin approach to better deal with the impacts of climate change. However, its implementation has been challenging. The implementation of river basin or landscape-level IWRM plans require coordination and joint visioning among several agencies working at different levels.

It is recommended that the national government create a national inter-ministerial NbS Task Force (NTF) to lead the development of a national strategy for accelerating the adoption of NbS in water management across sectoral policies, programs, as well as national accounting methods and standards. The strategy should summarize key water-related issues in the country, the priority sectors as well as highlight NbS examples and how these contribute to national policies and targets. The national NbS strategy should then define country-level objectives, targets, and indicators for mainstreaming NbS, along with the preferred strategies for the country, which should be harmonized with different sectors and approaches.

The strategy should map the multiple benefits provided by NbS implementation to different sectors and its linkage to the commitments made by the governments under Multilateral Environment and Climate Agreements to motivate participation from wider group of stakeholders. The strategy should include a sub-section highlighting sector specific barrier and opportunities for cross-sectoral collaboration and stakeholder engagement with a particular focus on IPLCs and build on their local knowledge to scale up NbS. It should also highlight research and capacity needs and include a funding strategy with an estimated budget for implementation and the identification of potential sources of financing. The strategy could then adopt a sub-national or ecosystem specific approach with recommendations for specific approaches on key priority landscapes in the country.

NTF and its composition and function

The NTF should be a multi-agency and multi-disciplinary platform hosted by a suitable national Ministry, and should be mandated to gather representatives from all agencies responsible for water management and related sectors, including environmental, agricultural, fisheries, urban development and energy-related agencies. All the government agencies mentioned by officials during the regional government survey, that are responsible for building water sector resilience to climate change from national to local level, are listed in Annex II (Summary from Regional NbS Survey). A review of the list suggest that several ministries are important when it comes to building water resilience, including those responsible for the management of water resources, disaster relief, environment, land-use, agriculture, finance, planning and rural development. These government agencies were identified by survey participants as key players in enhancing water sector resilience, at the national, provincial, and local levels. The list, therefore, can be used as a starting point by the national governments in these countries while discussing the roles and mandates of each agency within the NbS Task Force.

Based on the specific institutional mandate and policy context, each country can assign a relevant ministry or its relevant agencies to host the Task Force, with a clear mandate and defined roles and responsibilities of participating ministries (or their agencies). The task force will include sub-committees (see Table 6), and the organization of these sub-committees could build on existing roles and responsibilities of the national ministries under the National Action Plan. For example, National Action Plan on Climate Change (NAPCC), India identifies eight national missions on climate change, and the Department of Science & Technology of the Ministry of Science & Technology is entrusted with the responsibility of coordinating two missions, a) National Mission for Sustaining Himalayan Ecosystem (NMSHE), and b) National Mission on Strategic Knowledge for Climate Change (NMSKCC). Therefore, building on the already assigned roles, such as the Ministry of Science and Technology to lead on thematic committee within the task groups, and lead on research and knowledge generation linked to NbS effectiveness, its implementation at scale and its cost-effectiveness. In this role, the ministry can convene all other sector specific ministries on a thematic basis and form theme specific inter-sectoral ministerial groups for better coordination. For example, vulnerability assessment, adaptation policy research and pilots can be co-owned by two or more ministries whereas the supervisory functions are looked after by the DST or one of its designated institutions as an administrative secretariat.

By developing the Task Force, as discussed, countries can address the policy, institutional and knowledge barriers hindering the design and implementation of integrated water management strategies at a landscape or basin scale. It is understood that the Task Force will require government leadership, but it will also be important to build space for the representation of non-government actors including civil society organization and academia, who can guide the design of NbS strategies at scale and share knowledge and concerns from stakeholders. This aligns with the recommendation of Criterion 5 of the Global Standard for NbS, that "NbS is based on inclusive, transparent and empowering governance processes".

The proposed key roles of the NbS Task Force on water management include:

- Ensure policy and planning coherence for water management including the development of a national strategy for NbS for water management and the mainstreaming of NbS within key national plans and targets (e.g. NAP, NDCs);
- Identify capacity-building needs and strategies;
- Consolidate relevant information on historical climate data and projections, land use, basin strategies, and future development plans;
- Inventory national best practices and existing knowledge;
- Identify financing opportunities for NbS for water management;

In countries with federal structure, the national-level NTF should develop mechanisms to engage and work closely with the provincial and local level technical groups in gathering data and be supported by the scientific committees, user groups and NbS investment committees. Working through the provincial groups and committees, the Task Force will promote research and development of technical standards and guidelines for NbS mainstreaming at different levels, including mobilization of resources for the financing of NbS.

Suggested mandate, powers and responsibility of NTF for water management:

The mandate of NTF will be to develop NbS strategy for water management with recommendations for integration of NbS approach into national and sub-national (state, provincial) policies, and strategy for funding its implementation through national and international sources and including the role of innovative finance mechanisms to attract private sector investment and de-risk NbS project.

To effectively facilitate the NTF, following are suggested for inclusion in the powers of the NTF, and should guide the identification of Ministry (or its Agency) to host and facilitate the NTF.

- Empower the NTF to develop and implement a national NbS strategy for water resilience, including setting objectives, targets, and monitoring indicators for expanding the use of NbS in sectoral policies and plans.
- Empower the NTF to convene and coordinate among various government agencies, including environment, agriculture, fisheries, infrastructure, energy, and disaster management sectors at the national and state level.
- Empower the NTF to consolidate and manage relevant data on climate projections, land use, and water management strategies, serving as a central knowledge hub.

The National NTF will be advised by the work of the technical groups as discussed in the table below:

Table 6 | Proposed members and role of the technical groups within NTF

Technical group	Members	Roles
NTF Scientific Committees	• Experts from academia and civil society on ecosystem services and restoration, climate change adaptation • Policymakers (from relevant technical agencies and think tanks under the disaster, environment, water, science and technology ministries)	• Document best practices • Support the design of monitoring and evaluation frameworks for NbS to gauge effectiveness • Identify research needs and develop technical recommendations linked to NbS cost-benefit analysis • Draft policy recommendations • Develop action plans and guidelines for community participation
NTF Thematic or User Group	• Members of state disaster management and climate change authorities, • Indigenous Peoples, women's groups, youth groups, and community representatives • This could be organized thematically based on the priority areas identified jointly by National and State NTF;	• Provide feedback on the application of NbS, challenges, and opportunities to the scientific committees • Provide inputs on the development of national and sectoral guidelines for better integration of NbS

NTF Investment Committee	• Representatives from the Disaster Management Fund, Climate Fund, Ecosystem Protection Fund, and Rural Development Funds • Private Sector Lenders	• Develop proposals for external financing support for large-scale NbS measures • Identify opportunities for domestic financing of NbS measures from existing sectoral budgets • Strengthen engagement with the private sector and identify synergies for funding NbS

The NTF can build on the existing climate, urban development, disaster management governance architecture for developing state or provincial-level NTF. These state or provincial NTFs will provide a mechanism for the engagement of IPLCs in the design and implementation of capacity-building programs, gathering feedback on locally relevant NbS strategies, and identifying priorities for the development of funding proposals.

The state or provincial NTFs will nominate members to the Thematic or User Groups and also identify local practitioners and community representatives for representation in the user group. Due to this reason, during the selection of agency hosting of NTF, capacity to work with and engage non-government actors such as civil society organizations, academia, and local communities in the NTF shall be considered.

7.2 Strengthen research and knowledge on NbS for water

As indicated by government surveys and policy analysis, further research is needed to promote a better understanding of the interactions between people, the economy, and the ecosystem to design successful NbS projects.

IUCN Global Standard for NbS highlights a need for a "systems" framing in the design of NbS to ensure durable and sustainable solutions. The systems framing requires the acknowledgment of different types of interactions between ecosystems, people, and the economy and builds them into the decision-making process. The studies are also needed to provide inputs to the development of a structured approach for engaging IPLCs in the co-management of ecosystems and the design of national and local guidelines and frameworks.

Below are the priorities identified by stakeholders for research and development:

7.2.1 Scientific assessment of the status of ecosystem services in representative landscapes

The ecosystem services (ES) assessment is needed to fill gaps in the current understanding of the status and trends in ecosystem services, particularly in the regions most impacted by climate change. To ensure that the ES study informs the design of NbS, the ES assessments must consider the linkages between social and economic factors and stakeholder power dynamics.

Criterion 3 of IUCN Global Standard for NbS highlights the need to design NbS with a comprehensive understanding of the current state of the ecosystems. It further suggests that baseline assessments should be broad enough to characterize the ecological state, drivers of ecosystem loss, and options for improvements, while making use of both local knowledge and scientific understanding of the drivers and the impacts on the local community.

To complete the assessments, countries can begin with ecosystem services assessments in representative landscapes with different types of ecosystems and anthropogenic drivers. Considering the impacts of climate change are manifested through water, a river basin or watershed provides the right scale to understand the linkages between ecosystem services and their role in increasing or reducing vulnerability to water security and disasters. The results of the ecosystem services assessment will also provide inputs for the development of integrated river basin management strategies which are being prioritized by Bangladesh, India and Nepal.

NbS initiatives can build on ongoing research, such as the global Natural Capital Accounting and Ecosystem Services project. India is part of the study with the Ministry of Statistics and Programme Implementation (MoSPI), in partnership with the Ministry of Environment and Climate Change and the National Remote Sensing Centre of India is implementing the project. The MoSPI is facilitating stakeholder consultations to connect diverse parties, including producers and policymakers to develop a methodology and analyse the economic value of ecosystem services.

Several methodologies exist to capture key ecosystem services, at different levels, including national and landscape levels. The Toolkit for Ecosystem Service Site-Based Assessment (TESSA) is an example of a methodology that could be applied nationally to help frame key ecosystems and their services in countries (TESSA 2024).

7.2.2 Cost-benefit analysis of NbS compared to other measures in a specific context

The regional survey identified the need for cost-effectiveness studies to support the choice of NbS including the likely impact of any relevant regulations or subsidies. In the absence of cost-benefit analyses on NbS for water resilience, governments, and the private sector tend to prioritize grey infrastructure measures, such as the creation of dams and concrete embankments to manage floods and riverbank erosion. Strengthening stakeholders' understanding of the cost and benefits of NbS will contribute to reducing the use of grey infrastructure in the adaptation strategy, allowing countries to design innovative NbS or green-grey solutions with increased benefits to biodiversity and ecosystem services.

This is aligned with Criterion 4 of IUCN Global Standard for NbS: NbS are economically viable. A cost-effectiveness study will enable an examination of the upfront and recurring costs against the anticipated longer-term benefits of the proposed NbS interventions over time but also allow assumptions to be made explicit, tested, and verified. The direct and indirect benefits and costs associated with the NbS, who pays and who benefits, will be identified and documented.

The availability of cost-benefit analysis will facilitate the development of tools by banks and financial institutions to assess the risks of an NbS project and calculate monetary benefits, creating an enabling environment for the engagement of project developers and financiers from the private sector for financing large-scale NbS projects.

The World Bank publication, *Assessing the Benefits and Costs of Nature-based Solutions for Climate Resilience: A Guideline for Project Developers,* provides guidance on the main considerations for defining benefits and costs (Van Zanten et al. 2023). The control of flooding, erosion, droughts, water flow regulation, and landslides are some of the risk reduction benefits of NbS that the cost-benefit analysis needs to consider. The WB guidance lists NbS-related costs and benefits under the following five types of costs linked to the design and implementation of an NbS project, a) Capital expenditure (CAPEX); b) Operational expenditure (OPEX); c) Transaction costs; d) Opportunity costs; e) NbS externalities (trade-offs).

7.2.3 Facilitate the documentation of NbS-relevant traditional knowledge and community engagement strategies

The results of the government survey indicate a gap in understanding, valuing, and integrating the role of IPLCs in implementing NbS. There is a need to conduct a thorough analysis of the effectiveness of biodiversity and resource management strategies employed by these stakeholder groups and recognize their overall benefits to conservation and NbS. The design and implementation of successful NbS also require the identification of stakeholders who will be directly and indirectly affected by the NbS and their engagement in all stages of the NbS intervention. The stakeholder mapping and power analysis can support the identification of stakeholders who may be directly or indirectly, positively or negatively, affected by the NbS. This provides an opportunity to engage the affected stakeholders in the design and implementation and ensure that the rights and interests of the affected communities are considered, reducing the risk of marginalization of rights holders.

Table 7 | Costs and benefits of NbS

CAPEX	OPEX	Transaction costs	Opportunity costs	NbS externalities
• Design and planning • Land acquisition and/or rental cost • Construction of grey infrastructure as a part of NbS	• Project monitoring • Labour cost for project implementation • Tree plantation and maintenance of vegetation	• Scoping/ feasibility study • Community engagement • Goal setting and prioritization	• Value of using land for other purposes	• Negative impacts and trade-offs

Source: Adapted from Zanten V et al. 2023

7.3 Design and implement national capacity-building programs on NbS tailored to the needs of stakeholder groups

The lack of comprehensive understanding of the technical aspects of NbS, such as criteria for the design of NbS projects and how to integrate it in sectoral policies and project cycle management are the main barriers to the mainstreaming and upscaling of NbS for enhancing water resilience in South Asia.

As per regional government surveys, the capacity gaps are most pronounced at the provincial and local level. Therefore, governments should assign resources for the task force to initiate comprehensive programs to strengthen the technical capacity of relevant government agencies. The content of such program should include tools and steps for the design and implementation of projects that leads to enhanced water security and DRR. The capacity building program should aim to help the beneficiaries better contextualise and document the multiple social and economic benefits of NbS (economic use of wetlands, biodiversity conservation, creation of additional carbon sinks, human well-being, and job creation) from provincial, and river basin level perspective.

Capacity building is also needed on strategies to leverage finances to implement NbS interventions, particularly how different sectors can work together to design joint projects and allocate or mobilize resources from their current budget. The review of policies and initiatives from national governments in South Asia highlight several opportunities for mobilisation of resources of NbS related actions, particularly in the context of floods and droughts. In India, the Mahatma Gandhi National Rural Employment Guarantee Act (MGNREGA) includes provisions for using funds to support ecosystem protection, such as reforestation and watershed management. With an annual budget of approximately $10.4 billion USD (2024-25), MGNREGA mandates that 60% of its expenditure be directed towards natural resource management, including water conservation, afforestation, and soil protection. Additionally, the National River Conservation Plan (NRCP) aims to rejuvenate polluted river stretches across the country. The NRCP has covered polluted stretches on 38 rivers in 82 towns, with a total cost of $1.1 billion USD, creating a sewage treatment capacity of 2,910.50 million liters per day.

In Pakistan, the Living Indus initiative aims to restore more than 30% of the Indus River Basin by 2030. This initiative, recognized as a UN World Restoration Flagship, focuses on community-led, gender-responsive, and transparent NbS to enhance the resilience of the Indus ecosystem. The initiative identifies 25 projects estimated to cost up to $17 billion USD, with plans to restore 25 million hectares of the river basin. Similarly, Bhutan for Life (BfL) is an example of Project for Permanence (PFP) which allows for long-term financial and resources mobilisation for the NbS project and the experiences from similar project can inform the design of capacity building on NbS financing.

Similarly, Asian Development Bank (ADB) has launched the Nature Solutions Finance Hub for Asia and the Pacific, aiming to attract significant investments into NbS projects, with a goal of catalysing at least $5 billion USD in capital flows, including contributions from the private sector. These initiatives highlight the potential of these initiatives to fund NbS projects for water resilience.

Speaking about the multiple benefits of NbS and its potential role in poverty reduction, a recent IUCN, UNEP and ILO joint report, ‘Decent work in Nature-based Solutions 2022’ highlights that additional 20 million jobs could be generated worldwide if investment in NbS were tripled by 2030. This is also important for achieving biodiversity, land restoration and climate goals and provides additional motivation for prioritising national governments and international donor funding for NbS.

For the development of capacity building program governments can start by mapping relevant government agencies, initiatives and academic institutions for the development of sector and context specific training. The country survey report (Annexe II) lists several agencies mentioned by the officials that can be prioritised for inclusion in the Scientific Committee. These agencies should be tasked with specific roles and responsibilities related to the design and implementation of the capacity building program and mainstreaming of NbS in their sector. The capacity building program will build on the comprehensive data collection and assessment of the status of ecosystems, biodiversity, how these are impacted by current infrastructure and landuse planning with a specific landscape and river basin context and emphasising their links to climate and water risks.

The design of the capacity building program will be informed by the needs of the target stakeholder groups. Countries can create a network of government and academic institutions and experts linked to the NbS Scientific Committees and NbS User Groups discussed earlier. The priority themes for capacity building of the government and the local communities, including women and youth, and private sector include:

- Understanding of the values and trends in the ecosystem services and how they impact long-term water security and disaster at the river basin, local, and national levels considering that ecosystem services are the basis for the design of NbS. A better status and trends of local ecosystem services are needed for the designing intervention to better manage and enhance relevant ecosystem services, such as wetlands protection for buffering against the floods.
- Enabling provisions of NbS in current climate, water, and disaster management policies.
- Understanding and documenting traditional knowledge and indigenous governance systems in the design of NbS at the landscape level.
- Identifying strategic opportunities for restoration of ecosystems in different contexts, for example, wetlands-based adaptation for floods and drought resilience, and forest landscape restoration.
- Developing financing strategies, including green bonds, blended finance and private sector funding, for NbS projects.

Table 8 summarizes key national level capacity objectives identified for different stakeholder groups. The national capacity-building initiatives can be funded through the existing environmental protection, climate change, and disaster management funds established by countries at the national and local levels. Furthermore, the climate action policies of the countries in South Asia include capacity building and climate research components providing an entry point for the design of a capacity building program on NbS for water resilience. An example is, the National Action Plan on Climate Change (2008) in India, which includes eight national missions, including 'National Mission on Strategic Knowledge for Climate Change (NMSKCC)'. The objective of NMSKCC is to build human and institutional capacities on climate change, as well as documentation of strategic knowledge on climate change adaptation and mitigation.

Table 8 | Target groups for national capacity NbS building programme

Target groups	Objectives
National/ Provincial Government	• Improve understanding of NbS in national and local policies, and its operationalization through a multi-agency and multi-level approach. • Developing participatory approaches to involve stakeholders in NbS projects. • Learning methods for valuing ecosystem services and integrating these valuations into financial planning and investment decisions. • Understanding of cost-benefits and the economic feasibility of NbS. • Criteria for the design and verification of NbS projects. • Understanding and leveraging financing options, including green bonds, blended finance, and private sector funding for NbS projects.

Civil Society and Community-Based Organisations	• Enhanced skills to engage and mobilize local communities in NbS initiatives and development of project-specific action plans for ensuring IPLC engagement in the design and implementation of NbS. • Participatory monitoring and evaluation techniques to ensure community involvement in tracking the progress and impact of NbS projects. • Skills to access funding, including grants, donations, and partnerships with private sector entities for community-led watershed level NbS projects.
Private Sector	• Understating criteria and guidelines for implementing NbS, including IUCN Global Standard for NbS. • Cost-benefit analysis and economic feasibility of NbS mainstreaming in supply chain and business operations. • How to integrate NbS into corporate sustainability reporting and align with environmental, social, and governance (ESG) criteria. • Financing options such as Corporate Social Responsibilities, green bonds and blended finance to support NbS projects.

Several international organizations have developed training on NbS and these provide starting point for developing national capacity building program for NbS. While the government cannot fully rely on these to meet the specific capacity needs of all stakeholders at the national and local level, they remain useful to build the capacities of NbS champions and train potential master trainers who then can develop and lead national capacity-building programs. These may include the following:

- IUCN Academy has developed an online platform, bringing together the wealth of IUCN capacity development activities, and offering a catalogue of courses, including professional certification programs on IUCN Global Standard for NbS, Measuring and Monitoring Urban Nature and Finance for Nature. These courses aim to build the capacities of citizens and professionals to make their contributions to the field of nature conservation more strategic.
- ADPC Academy provides specialized disaster and climate resilience training for governments and international and non-governmental organizations. The courses offered by ADPC Academy cover key areas in DRR and climate change adaptation through diverse and customized approaches. ADPC offers training on NbS for Disaster and Climate Resilience often in cooperation with IUCN and NbS for DRR in Hindukush-Himalayan Region in collaboration with ICIMOD.
- ADPC carried out a detailed sectoral Capacity Needs Assessment in three countries as part of the CARE for South Asia project, including the water sector. Analysis of the sectoral engagement with water sector stakeholders brought forward the need for training on NbS as technical personnel trained in implementing NbS are limited. The focus areas of training suggested were fundamentals, planning, and implementing NbS in the water resources sector, orientation on locally applicable NbS and bio-engineering solutions and best practices to manage flood and landslide risks, use of NbS and Eco-DRR approaches in environment conservation plans and support in

development of reference document for NbS and Eco-DRR approaches for water resources management.

- The specific training on "Promoting Nature-based Solutions for Climate Resilience in Water Sector in South Asia" developed jointly by IUCN and ADPC as part of the CARE for South Asia project has the scope of replication and scaling up to address the capacity gaps of the water sector government stakeholders discussed above. The training covers the fundamentals, global standards of NbS, planning cycle of NbS interventions, green-gray infrastructure solutions and cost-benefit analysis, sustainable finance and mainstreaming NbS in policies and governance.

7.4 Develop national financing strategies for NbS

The government survey identified a diversity of financing schemes for NbS that have been adopted by countries but also highlighted an overall gap in financing NbS. The current funding for NbS is largely project-based. Some countries are making an effort to develop national financial mechanisms for ecosystem restoration and NbS, such as Pakistan's Ecosystem Restoration Fund. As recommended in Section 7.2, catalyzing of NbS is an important catalyzing factor for developing a national NbS strategy and needs to be developed at the country level. Several options for financing NbS exist.

Grants and public funding such as the GCF, Adaptation Fund, International Climate Fund (IKI), and GEF remain key sources of funding for medium to large NbS initiatives. Accredited Agencies including the World Bank and IUCN can help countries to develop large-scale NbS programs, such as the GCF- funded Improving Climate Resilience of Vulnerable Communities and Ecosystems in the Gandaki River Basin, Nepal. These donor-funded projects increasingly serve as a basis for developing long-term financial strategies, involving government financial commitments and engagement with private sectors and banks.

Blended finance is also increasingly considered in NbS projects involving the private sector. GCF has developed specific guidance on this (Green Climate Fund 2023). Blended finance is the strategic use of public funds or concessional finance to mobilize private sector investment in sustainable development projects. This financial approach is aimed at de-risking investments for the private sector. Public or concessional finance can absorb some of the risks, making projects more attractive to private investors.

Blended finance can be implemented through several mechanisms:

- Concessional loans: Loans provided with terms more generous than market loans, such as lower interest rates or longer grace periods from the GCF, GEF, and other bilateral and multilateral financiers.
- Guarantees: Financial guarantees can be used to protect investors against specific risks, thereby making investments more attractive.
- Equity investments: Direct ownership stakes in projects or companies, which can be attractive if a project's risk profile is reduced through other blended finance mechanisms.

For water-related NbS, blended finance can be particularly effective for larger national level projects or those with the potential for revenue generation. For instance, the BFL Secretariat since since its inception has mobilized 7,270,034 USD and 9,996,940 USD of private donor funding (GCF 2022. FP050). The project is also providing support for the development of PES mechanisms, one example of which is the agreement between the Yakpugang community and the Mongar municipality, which aims to incentivize the members of Yakpugang Community Forest to protect the watershed that acts as the water source for Mongar city.

Other mechanisms to fund NbS include:

- Impact investment: Investors looking for both financial returns and positive environmental or social impacts might invest in NbS projects.
- Green bonds: These are bonds issued specifically for environmentally-friendly projects. They could be used to fund NbS water projects.
- Public-private partnerships (PPPs): Collaborative arrangements between government agencies and private-sector companies can help fund NbS projects.
- Grants and philanthropy: Non-profits, international organizations, and philanthropic entities might provide funding, especially in the initial stages or for pilot projects.
- Payments for ecosystem services (PES): This approach involves direct payments to those who manage and maintain environmental services. For instance, downstream water users might pay upstream communities to protect and restore watersheds.

Some international organizations, like IUCN, are also offering specific NbS Finance instruments:

- Nature+ Accelerator Fund: A platform fostering innovation and support for NbS and start-ups.
- This is a first-of-its-kind private sector-focused nature conservation fund and aims to demonstrate measurable conservation and social benefits while ensuring financial returns for investors. The GEF is an anchor investor for the fund. The fund has secured 8 million USD in risk-tolerant financing (IUCN, 2020). The fund is managed by asset management company Mirova and targets the following areas: marine conservation and coastal resilience; smallholder production systems and sustainable agriculture; ecosystem conservation and restoration; and innovation in services, finance, and technology. The fund is offering investment money to early- stage pilots and project ideas, impacting enterprises with high potential for scalability. The fund also provides technical assistance and capacity building in three complementary investment windows: Seed, Early Venture (together defined as the incubation period), and Venture windows.
- Subnational Climate Fund: A funding mechanism targeting local and regional climate action projects (SCF 2024).
- The SCF is a global blended finance initiative that aims to invest in and scale mid-sized (5–75 million USD) sub-national infrastructure projects in the fields of sustainable energy, waste and sanitation, regenerative agriculture, and NbS in developing countries. The SCF Fund is managed by Pegasus Capital Advisors. To popularize and increase the expertise and awareness of stakeholders, the Fund has established a dedicated funding window of 28 million USD in grants for Technical Assistance (TA). This TA provides technical support, as well as grant funding to stakeholders for identifying and strengthening investment proposals for the Fund. In the mitigation of Urban climate and water challenges, the SCF is working as a part of Leadership for Urban Climate Investment (LUCI), which includes 20+ initiatives helping deliver urban climate projects worldwide.

These funds provide an opportunity for mobilizing resources for public-private-partnership projects (PPP) on NbS for water resilience, which are often perceived by the private sector to be risky investments, due to a variety of reasons, including misconceptions that NbS are slow acting and a lack of awareness and studies on cost-benefit analysis of NbS for water resilience.

7.5 Enhance regional exchange and cooperation on NbS

The survey identified limited interactions between South Asian countries on topics linked to NbS. The South Asian Association for Regional Cooperation (SAARC)-supported dialogues currently do not include NbS-related topics. It is important that governments and development partners continue advocating that SAARC includes NbS and consider it as a key strategy for regional development.

Existing bilateral agreements, such as the Joint Consultative Commission (JCC) between Bangladesh and India, can provide a platform for capacity building and development of joint NbS projects. The JCC is chaired by the external affairs ministers of Bangladesh and India and includes officials from the Ministries of Water, Trade, and Finance.

The implementation of regional, transboundary NbS is often required in the water sector as several river basins are transboundary, therefore requiring joint solutions. This was identified as a gap in the survey and government partners should continue promoting regional projects focusing on joint management of ecosystems for shared benefits including DRR. This is sometimes difficult to implement as countries tend to prefer national-level initiatives, but could be piloted in small areas, or basins to evidence the need for cooperation.

In the short term, project-supported regional exchanges of experience between the countries seem to be the most effective way to ensure cross-learning. There is an appetite for site visits and knowledge forums, which can later initiate more formal regional cooperation on NbS.

These exchanges could be extended to other parts of Asia, where significant cooperation on water management is in place, such as the Mekong basin and the Mekong River Commission. Twinning between sites facing the same issues and planning similar NbS approaches in different countries should be encouraged to move the NbS dialogues to a technical level of cooperation and shared sustainable development.

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Annex I: Government Survey Participants

Bangladesh

Name	Designation and institution
Md. Aminul Haque	Principal Scientific Officer, Water Resources Planning Organisation, Water Resources Planning Organization (WARPO)
Atika Sultana	Extension Officer (LR), Administration & Finance Wing, Department of Agricultural Extension
A.K.M. Saifuddin	Superintending Engineer (Civil), Bangladesh Water Development Board (BWDB)
Md. Shamsul Islam	Deputy Chief, Agriculture, Water Resources and Rural Institutions Division, Bangladesh Planning Commission
Farah Anzum	Strategic Communications Associate Global Strategic Communications Council (GSCC)
Chowdhury Sarwar Jahan PhD	Professor, Dept. of Geology and Mining University of Rajshahi, Bangladesh

India

Name	Designation and institution
Anil Kumar Gupta	Professor, National Institute of Disaster Management
Uday Bhonde	Senior Program Specialist, Water & Environment Vertical, Climate Centre for Cities, National Institute of Urban Affairs
Shri Ashok Goel	Commissioner SPR, Ministry of Jal Shakti, Department of Water Resources
Shri Anuj Kanwal	Commissioner, Command Area Development & Water Management Information System, Ministry of Jal Shakti, Department of Water Resources, River Development, GR

Nepal

Name	Designation and institution
Saroj Adhikari	Senior Agricultural Engineer, Department of Agriculture (DoA), Agricultural Engineering and Post-Harvest Section
Shanta Karki	Deputy Director General (DDG), Department of Agriculture (DoA)
Hari Prasad Sharma	Climate Change Management Division (CCMD)
Dinesh Bhatta	Senior Divisional Engineer, Department of Irrigation (DoI)
Mukesh Pathak	Senior Divisional Engineer, Department of Irrigation (DoI)
Kapil Gyawali	Senior Divisional Hydrologist/Engineer, Water and Energy Commission Secretariat (WECS)
Dinesh Parajuli	Senior Divisional Engineer, Water Resources Research and Development Centre (WRRDC)

Name	Designation and institution
Ananta Man Pradhan	Senior Divisional Engineering Geologist, Water Resources Research and Development Centre
Krishna Rijal	Chief, Bagmati River Basin Improvement Project (BRBIP), Project Implementation Irrigation Unit (PIIU)
Gayatri Joshi	Engineer, Water and Energy Commission Secretariat (WECS)
Jaya Laxmi Singh	Department of Water Resources and Irrigation (DoWRI)

Pakistan

Name	Designation and institution
Ahmed Kamal	Chairman, Federal Flood Commission (FFC), Ministry of Water Resources
Muhammad Ashraf	Chairman, Pakistan Centre of Research on Water Resources (PCRWR), Ministry of Water Resources
Ashfaq Ahmed Sheikh	Additional, Registrar, Pakistan Engineering Council
Sarfraz Ali Memon	Senior Agriculture Engineer, Federal Water Management Cell
Shahid Habib	Deputy Director Environment and Social Safeguards, Punjab Irrigation Department Lahore
Ejaz Tanveer	Superintending Engineer (Floods-I), Ministry of Water Resources
Idrees Mehsud	Member Disaster Risk Reduction, National Disaster Management Authority
Noor Fatima	In charge Political Science and International Relationship Department, at Islamic International University
Zain Ijaz	Section Officer Water, Ministry of Water Resources, Islamabad
Azeem Khoso	Director of Urban Affairs, Ministry of Climate Change and Environmental Coordination
Yasir Gull	Assistant Chief, Ministry of Planning, Development and Special Initiatives, Islamabad
Muhammad Hamza	Assistant Director Hydrology, Pakistan Centre of Research on Water Resources (PCRWR) Islamabad
Mufeeza Ahsan	Research Officer Hydrology, Pakistan Centre of Research on Water Resources (PCRWR), Islamabad

Annex II: Summary from Regional NbS Survey

The text below presents a summary and key message from the regional government survey in four South Asian countries (Bangladesh, India, Nepal, and Pakistan) on climate change impacts and mainstreaming NbS in the Water Sector for long-term climate resilience. The survey was conducted by IUCN staff in Bangladesh, India, Nepal, and Pakistan and engaged more than thirty government agencies from these countries. The survey was designed to gather information and insights on the impacts of climate change-induced water stress and the current government strategies (policies, plans, and initiatives) to deal with the situation. The survey also asked participants about the current recognition of NbS in national policies and agencies that need to work together to mainstream/upscale NbS in the water sector.

The summary of the region survey results is presented below by country.

(I) Bangladesh

1. Survey participants

Six individuals were interviewed, including four from the government, one from academia, and one from a civil society organization (CSO). The government agencies surveyed include the Water Resources Planning Organisation, Bangladesh Water Development Board, Department of Agricultural Extension, and Bangladesh Planning Commission.

2. Climate change-induced water stress and strategies

Bangladesh faces significant climate risks, including increasing temperatures, changing rainfall patterns, and frequent natural hazards such as floods and cyclones. Bangladesh is highly vulnerable to the impacts of sea level rise due to its low-lying coastal geography. These challenges impact water availability and quality, leading to issues such as water scarcity, salinity intrusion, and reduced agricultural productivity. From the interviewees' perspective, drought is important because it negatively affects the country's food security, due to its impact on agriculture and inland fisheries.

To address these challenges Bangladesh is implementing several strategies to address water stresses and enhance climate resilience. The Bangladesh Delta Plan 2100 (BDP 2100) provides a framework for achieving this, by providing a long-term vision for the development and sustainable management of the country's delta and river systems. Complementing this is the National Adaptation Plan (NAP) 2022, designed to promote coordinated action among various ministries and sectors. The Bangladesh Climate Change Strategy and Action Plan (BCCSAP), updated in 2022, also provides a comprehensive framework for building climate resilience and promotes the ecosystem approach.

To finance these initiatives, Ministries receive allocations from the Annual Development Programme (ADP) for climate resilience projects. The Bangladesh Climate Change Trust Fund (BCCTF) is also established by the government to provide funding for projects promoting climate resilience, such as embankment construction, water reservoir development, and afforestation. Additionally, multilateral funding agencies like the World Bank, ADB, GCF, and others are supporting several initiatives promoting ecosystem-based adaptation in the Haor region and coastal areas in Bangladesh.

The government recognize the value of research and knowledge in development of innovative solutions for climate resilience. The country promotes Climate Smart Agriculture (CSA), which includes the development and use of climate-resilient crop varieties, efficient water management techniques and conservation agriculture practices. The Flood Forecasting and Warning Center has been established by the national government for flood prediction and disaster preparedness.

Partnerships are also key to Bangladesh's approach to reduce the vulnerability to climate-induced water stress. The Ministry of Water Resources (MoWR) has collaborated with various other ministries and international organizations such as FAO, the World Bank, and ADB. These partnerships facilitate the sharing of technical expertise and the implementation of standardized frameworks for climate resilience projects.

3. Government agencies working on water sector resilience

Below is the list of institutions, mentioned by the interviewees during the regional government survey, that are responsible for building water sector resilience to climate change.

National:

- **Ministry of Water Resource (MoWR):** Overall planning, development, and management of water resources.
- **Bangladesh Water Development Board (BWDB):** Implements and manages water resource projects, including flood control and irrigation.
- **Water Resources Planning Organisation (WRPO):** Conducts studies and research on water resources and assists in policy formulation.
- **Ministry of Agriculture (MoA):** Develops agricultural policies and programs, ensuring food security and sustainable practices.
- **Ministry of Environment, Forest and Climate Change (MoEFCC):** Manages natural resources, environmental conservation, and climate change issues.
- **Ministry of Disaster Management and Relief (MoDMR):** Disaster risk reduction, preparedness, and response.
- **Ministry of Finance (MoF):** Manages economic and financial affairs, supporting development projects.
- **Ministry of Local Government, Rural Development & Co-operatives (MoLGRDC):** Develops local government bodies and rural areas.
- **Ministry of Planning:** Formulates national development plans and coordinates development programs.

Province/Division:

- **Department of Agricultural Extension (DAE):** Disseminates agricultural knowledge and technology to farmers.
- **Bangladesh Agricultural Development Corporation (BADC):** Promotes agricultural development and food security.
- **Local Government Engineering Department (LGED):** Plans and maintains local infrastructure projects.
- **Department of Public Health Engineering (DPHE):** Ensures safe drinking water supply, sanitation, and environmental hygiene.

Local level:

- **Water Resources Coordination Committee:** Coordinates water resources management at the district level.
- **Community-Based Organizations:** Involved in water management and climate adaptation initiatives at the grassroots level.
- **Upazila Parishad:** Local government body responsible for water security and sanitation at the sub-district level.

4. Challenges hindering NbS mainstreaming for water resilience

Funding Constraints: Securing adequate funding for water sector projects is a significant challenge. The government currently spends about 0.8% of GDP on BDP 2100 programme, but the estimated need is around 2% of GDP. Mobilizing an additional 1.2% of GDP for public financing is difficult. Projections suggest that Bangladesh should aim to secure at least 0.5% of GDP per year from the private sector for water resource management projects.

Policy and Institutional Challenges: Effective climate adaptation requires coordination among various government agencies, stakeholders, and NGOs. Addressing institutional barriers and streamlining policies are essential for successful outcomes.

Data and Information Gaps: Accurate and up-to-date data on rainfall patterns, river flows, and groundwater levels are crucial for planning and implementing climate change adaptation strategies. Inadequate data hinders effective decision-making.

Community Participation and Awareness: Involving local communities in project planning and execution is vital for sustainability. Raising awareness about climate change impacts and water conservation can enhance community engagement.

Long-Term Monitoring and Evaluation: Continuous monitoring and evaluation of water sector projects are necessary to assess the impact of adaptation measures and identify areas for improvement. The Implementation Monitoring and Evaluation Division (IMED) oversees the implementation of public sector development projects.

Research and Knowledge Gap: Establishing a dedicated research department within the Ministry of Water Resources (MoWR) is essential for evidence-based project implementation. Regular publication of research findings should be prioritized. Despite its importance, research is not widely emphasized in the water sector. Research is crucial before initiating projects, such as the bamboo bundling project. Projects exceeding 50 crores require an environmental impact assessment.

(II) India:

1. Survey participants

The survey included five government officials from India from different agencies. The participants included a Professor from the National Institute of Disaster Management, a Senior Program Specialist from the Water & Environment vertical at the National Institute of Urban Affairs, and a Senior Programme Lead in Climate Resilience at the Council on Energy, Environment and Water (CEEW). Additionally, two commissioners from the Ministry of Jal Shakti, from the Department of Water Resources in India, were interviewed from the Command Area Development & Water Management Information System, as well as River Development and Ganga Rejuvenation.

2. Climate change-induced water stress and strategies

Increasing temperatures, changing rainfall patterns, and natural hazards impact water availability. For example, urban floods and heat stress are becoming more frequent, leading to drying water bodies and reduced quality of freshwater. The increasing frequency of urban floods is a major concern for planners and implementers. Previously localized, the heat stress is now more widespread due to climate change. Rising temperatures and heat stress are causing water bodies to dry up, reducing availability for agriculture and household use.

To deal with climate change risks in the water sector, the country is working on the integration of ecosystem approaches in the National Water Policy and National and State Action Plans for Climate Change, and various national missions (e.g., Clean Ganga, Green India, Sustainable Habitat). India is also encouraging large-scale research initiatives such as hydrological modeling to support the development of Urban River Management Plans (URMP) and River Sensitive Master Plans (RSMP).

For synergizing interventions with national and state missions, engaging the private sector, and leveraging multilateral and bilateral funding schemes (e.g., GEF, GCF). Utilizing CSR funds and partnerships under the National Action Plan on Climate Change (NAPCC).

For improving data consolidation and management, India has established the Water Resources Information System (WRIS) under NHP for standardized data and information management. To deal the water stress, the water policy emphasizes the need for groundwater development and recharge mechanisms, as well as comprehensive flood management programs under the National Hydrology Project (NHP). Government agencies like NIUA are working on the *River Cities Alliance (RCA)* and *NbS Forum of Cities* to promote its mainstreaming in planning and cross-sectoral collaboration.

3. Government agencies working on water sector resilience

National level:

- **Ministry of Jal Shakti:** Responsible for preparing policy guidelines and coordinating the development and regulation of the country's water resources.
- **Ministry of Environment, Forest, and Climate Change:** Focuses on conservation and management of natural resources, biodiversity, wildlife, and pollution.
- **Ministry of Housing and Urban Affairs:** Formulates policies and supports programs related to housing and urban affairs.
- **Ministry of Earth Sciences:** Provides information services related to weather, climate, ocean and coastal state, hydrology, seismology, and natural hazards to different stakeholders and agencies.
- **National Disaster Management Authority**: Coordinates disaster management efforts and mobilization of funds for mitigation.
- **Central Ground Water Board:** Manages groundwater resources through exploration, monitoring, and assessment.
- **National Mission for Clean Ganga:** Focuses on rejuvenating, protecting, and managing the River Ganga.

State and local level:

- **Department of Urban Development:** Responsible for policy and planning for the development of integrated and sustainable human settlements.
- **Public Health Engineering Department (PHED):** Responsible for water supply and sanitation services.
- **State Disaster Management Authorities:** Monitors flood and drought situations and is responsible for taking preventive measures and post-disaster recovery.
- **Municipal Corporations**: Regulate land use, urban planning, and water supply.
- **Block Development Offices:** Implement development schemes at the sub-district level and in the rural areas.

4. Challenges hindering NbS mainstreaming for water resilience

Conflicting policies: Climate change, disaster risk, and water management are often treated as separate issues rather than interconnected challenges. There is a lack of cross-sectional policies that synergize various environmental agendas. Multiple programs exist, but a cohesive policy framework is missing.

NbS recognition: There is a limited understanding of the NbS approach as a comprehensive mitigation strategy, leading to inconsistent policies and implementation. NbS is relatively new to the government, which currently favors the Ecosystem-based Adaptation (EbA) approach.

Innovation in green-grey infrastructure: Green-grey infrastructure solutions are crucial for sustaining urban growth meeting the demand for water from different sectors and mitigating the impacts of climate change. However, there is a poor understanding of how green-grey infrastructure can be applied in specific contexts. More research and sectoral guidance are needed.

Financing: Implementing NbS requires substantial funding. However, existing funding mechanisms are issue-specific, focusing on climate change mitigation, disaster risk reduction, biodiversity conservation, and sustainable livelihoods. Also, NbS implementation is perceived as costly, which affects its adoption and limits its application at scale reducing the overall effectiveness.

(III) Nepal

1. Survey participants

The survey included a diverse group of eleven participants from different government departments working on water management and climate resilience. From the Department of Agriculture (DoA), the Senior Agricultural Engineer from the Agricultural Engineering and Post-Harvest Section and the Deputy Director General (DDG) were interviewed. The Department of Irrigation (DoI) was represented by two Senior Divisional Engineers, emphasizing the importance of irrigation in water management. The interview also included the officials from the Climate Change Management Division (CCMD), Water and Energy Commission Secretariat (WECS), and Water Resources Research and Development Centre (WRRDC). Additionally, the Chief of the Bagmati River Basin Improvement Project (BRBIP). The Department of Water Resources and Irrigation (DoWRI) also participated, highlighting its overarching role in managing water resources and irrigation projects. These participants provided comprehensive insights into various aspects of water management, climate resilience, and agricultural engineering, and acknowledged the need for a holistic approach to addressing water-related challenges.

2. Climate change-induced water stress and strategies

Water is a crucial sector for Nepal's economy, playing a key role in livelihood, economic, political, and social development. Nepal is home to approximately 5,358 lakes and 6,000 rivers and rivulets. Climate change is significantly impacting these water resources by increasing the frequency and intensity of droughts, floods, landslides, and storms, and rising sea levels.

Key impacts include changes in temperature, precipitation patterns, glacier melt, and increased variability in river flows, which affect water availability for drinking, irrigation, hydropower generation, and aquatic ecosystems. Rising temperatures directly impact water resources and hydropower, affecting resource availability and energy supply. Climate variations have accelerated glacier melting and increased the risk of Glacial Lake Outburst Floods (GLOFs), as well as increased sediment loading and evaporation, leading to cascading disasters. For instance, the Melamchi flood disaster in 2021 was triggered by both anthropogenic and climatic factors.

The government survey participants mentioned the following policies and strategies are applied by the national governments to enhance climate resilience:

Integrated Water Resources Management (IWRM) principles and the Water-Energy-Food-Ecology Nexus are being adopted in policies to enhance water management efficiency. In this regard, the interview mentioned several policies that are being updated, including the Water Resource Act 1992 (currently being updated), National Water Resources Policy 2020, Water Resources Strategy 2002, and National Water Resources Plan 2005. To address the water scarcity issue, the government is working on increasing groundwater extraction through the Department of Irrigation (DoI) but lacks a strategy for restoring groundwater tables. Other policies mentioned by interviewees include the National Climate Change Policy 2019, National Adaptation Plan 2021, and various sector-specific policies such as the Irrigation Policy (draft) 2022, National Drinking Water, Sanitation and Hygiene Policy (draft) 2022, and the Environment Protection Act 2019.

For funding climate resilience initiatives, the government relies primarily on national budget allocations, with additional support from development partners and international funding sources such as GCF, GEF, ADB, World Bank, JICA, and others. The Environment Protection Act 2019 established an Environment Protection Fund (EPF) to finance climate resilience activities. However, financial resources are insufficient to address all climate change challenges in the water sector. Bilateral and multilateral funds are mainly focused on research and capacity building, with less emphasis on infrastructural interventions. The government is exploring new funding sources, such as the digitalization of water data with support from the Korean government. The government is also implementing and testing the Payment for Ecosystem Services (PES) as a part of the Dhulikhel Drinking Water Supply Scheme, which is managed and financed by the community contributing to both sustainable water supply and ecosystem conservation.

There is a growing recognition of the importance of research-based development for sustainable water management. The Water Research and Development Center (WRDC) was established under the Ministry of Water Resources, Energy, and Irrigation (MoWERI) to focus on research and development, including landslide early warning and management. The government collaborates with development partners like the World Bank and the ADB for action-oriented research.

3. Government agencies working on water sector resilience

National Level:

- **Ministry of Finance (MoF)**: Manages the country's financial resources, serves as the focal point for GCF and GEF, and oversees financial policy and program implementation.
- **Ministry of Forests and Environment (MoFE)**: Develops and implements policies related to national parks, wildlife reserves, wetlands, environment, and climate change. The **Climate Change Management Division (CCMD) under MoFE** is responsible for the development and implementation of climate change policies.
- **Ministry of Agriculture and Livestock Development (MoALD)**: Develops policies related to food security and agricultural business.
- **Ministry of Energy, Water Resources and Irrigation (MoEWRI)**: Develops policies for water resource and energy development, including river basin planning.

- **Department of Hydrology and Meteorology (DHM)**: Manages hydro-meteorological data and provides forecasting information.
- **National Planning Commission (NPC)**: Formulates national development policies and plans and advises the government on project approvals.
- **National Disaster Risk Reduction and Management Authority (NDRRMA)**: Develops national policies for disaster risk management and guides to provincial and local levels.
- **Nepal Academy of Science and Technology (NAST)**: Manages climate change knowledge through the Climate Change Knowledge Management Centre.
- **National Agriculture Research Council (NARC)**: Conducts research on agricultural varieties and climate-resilient breeds of crops and cattle.
- **Alternative Energy Promotion Center (AEPC)**: Focuses on research and innovation in alternative energy development, including renewables.
- **Ministry of Water Supply (MoWS)**: Develops policies for drinking water and sanitation.
- **Water and Energy Commission Secretariat (WECS)**: Formulates plans and policies related to water resources and regulates water allocation to different sectors.
- **Department of Water Resources and Irrigation (DoWRI)**: Plans and implements irrigation and water-induced disaster management projects.
- **Department of Water Supply and Sewerage Management (DoWSSM)**: Leads the development of water supply and sanitation systems.
- **Department of Forest and Soil Conservation (DoFSC)**: Studies and monitors forestry and soil conservation strategies.
- **Department of Agriculture (DoA)**: Provides research, extension, and regulatory services related to agriculture.

Province level:

- **Ministry of Physical Infrastructure Development (MoPID)**: Develops policies and projects related to energy, irrigation, and water-induced disasters.
- **Ministry of Social Development (MoSD)**: Supports education, health, sanitation, and drinking water.
- **Ministry of Industry, Tourism, Forest and Environment (MoITFE)**: Manages watershed conservation and water use.
- **Ministry of Land Management, Agriculture and Cooperatives (MoLMAC)**: Develops policies related to agriculture and livestock development.
- **Ministry of Internal Affairs and Law (MoIAL)**: Manages disaster response and supports local disaster funds.
- **Provincial Policy and Planning Commission (PPPC)**: Develops provincial development policies and advises on project funding.
- **Ministry of Water Resources and Irrigation (MoWRI)**: Develops policies for water resource and energy development at the provincial level.

Local Level:

- **Municipalities and Rural Municipalities Executives**: Develops and implements local development policies, including water supply and disaster management.
- **Ward Offices**: Implements and monitor locals plans based on community needs.
- **District Coordination Committee (DCC)**: Coordinates and monitors development activities within districts.
- **Water Users Committees/ Associations**: Manages and maintains water infrastructure and promotes community ownership of water schemes.

- **Agriculture and Livestock Enterprise Promotion Section of Municipalities**: Supports agricultural and livestock activities at the municipal level.
- **Environment and Disaster Management Division of Municipalities**: Manages environmental and disaster-related activities at the municipal level.
- **Disaster Management Section of Municipalities**: Develops and implements disaster management activities at the municipal level.

4. Challenges hindering NbS main-streaming for water resilience

Policy and institutional barriers: There is a lack of synergy between different policies and sectors, leading to fragmented efforts. NbS is relatively new and not fully integrated into government strategies and sectoral guidance and monitoring frameworks are underdeveloped. Additionally, there is no intergovernmental mechanism with a clear mandate for promoting NbS.

Awareness: A limited understanding of NbS and its benefits among policymakers and stakeholders is a significant challenge. NbS is often seen as a new mitigation approach rather than a comprehensive strategy, leading to poor integration into existing policies.

Financial constraints: There is a perception that NbS has a high implementation cost and limited funding mechanisms. While some international funding is available, there is no dedicated funding for NbS mainstreaming. The government relies heavily on national budgets, which are insufficient to cover the costs of comprehensive NbS projects.

Data and research gaps: Inadequate data on critical factors such as rainfall patterns, river flows, and groundwater levels hamper effective decision-making. Although research institutions like the Water Research and Development Center (WRDC) and the Water and Energy Commission Secretariat (WECS) are involved in research, the findings often remain in the demonstration phase and are not fully implemented.

Implementation challenges: NbS approaches are time-consuming and require long-term investment and multi-stakeholder engagement. The lack of coordinated development and mass implementation, coupled with the need for long-term investment, makes it difficult to adopt NbS at a large scale. Additionally, there is a need for better monitoring and evaluation to assess the effectiveness of NbS projects.

(IV) Pakistan

1. Survey participants

The survey included a diverse group of 14 participants from various institutions involved in water management, disaster risk reduction, and climate change in Pakistan. Among the participants were the Chairman of both the Federal Flood Commission (FFC) and the Pakistan Centre of Research on Water Resources (PCRWR), both under the Ministry of Water Resources. The survey also included the Superintending Engineer (Floods-I) from the Ministry of Water Resources, a Registrar from the Pakistan Engineering Council, a Senior Agriculture Engineer from the Federal Water Management Cell, a Deputy Director from the Punjab Irrigation Department (Lahore) also participated in the survey.

Additionally, from the National Disaster Management Authority, the Member for Disaster Risk Reduction provided insights into policies and strategies aimed at reducing the impact of natural hazards. The survey also included the Charge of the Political Science and International Relations Department at the Islamic International University, Director of Urban Affairs from the Ministry of Climate Change and

Environmental Coordination, Assistant Chief from the Ministry of Planning, Development, and Special Initiatives, Assistant Director and Research Officer from the Pakistan Centre of Research on Water Resources (PCRWR) in Islamabad.

2. Climate change-induced water stress and strategies

Pakistan faces significant water stress due to climate change, including reduced water availability, increased frequency of floods and droughts, and declining water quality. These challenges are exacerbated by rapid population growth, urbanization, and industrialisation.

Strategies for addressing the climate risks in the water sector include:

A comprehensive framework of strategies, policies, and planning initiatives to address water stresses and build resilience in the water sector has been developed. Key strategies include the Federal Flood Protection Plan IV (NFPPIV), which focuses on developing protection structures like spurs, dykes, and embankment walls. The National Water Policy 2018 and the National Climate Change Policy (NCCP) 2012 provide guidelines for water resource management and climate change adaptation. The Implementation Framework of the National Climate Change Policy, National Disaster Risk Reduction (DRR) Policy, and Climate Change Adaptation Policy, with its Implementation Framework, further support these efforts. Furthermore, the National Water Policy 2018 advocates for Integrated Water Resource Management (IWRM) approaches to optimize water allocation and management. Nature-based solutions (NbS) are also being integrated into policy and planning, promoting sustainable development and enhancing resilience.

Other significant policies include the National Food Security Policy, the National Drinking Water Policy 2009, the National Forest Policy, the National Wildlife Policy, and the Draft National Water Conservation Strategy. The National Disaster Management Plan, National Adaptation Plan, and Vision 2025 outline long-term strategies for disaster management and sustainable development of the country. The Resilient, Recovery, Rehabilitation, and Reconstruction Framework (4RF), was developed in response to the 2022 mega floods in Pakistan. Furthermore, the Nationally Determined Contributions (NDCs) reflect Pakistan's commitment to international climate agreements and explicitly include NbS projects, such as Recharge Pakistan which focuses on building on ecosystem approaches to minimize the impacts of drought and floods.

The government has approved the solarization of agricultural, industrial, and residential tube wells, allocating PKR 377.23 billion to the 2023 water and power sector development budget. This initiative aims to reduce groundwater abstraction. Water conservation strategies include the reuse of wastewater after proper treatment, rainwater harvesting, and groundwater recharge. Large, medium, and small dams are being constructed to store water during flood seasons, acting as buffers against drought.

3. Government agencies working on water sector resilience

National Level:

- **Ministry of Climate Change:** Leads the formulation and implementation of climate policies and strategies.
- **Pakistan Council of Research in Water Resources (PCRWR):** Conducts research on water resources and provides policy recommendations.
- **National Disaster Management Authority (NDMA):** Manages disaster risk reduction and response.

- **Ministry of Water Resources:** Oversees water resource management and related infrastructure.
- **Ministry of Agriculture:** Focuses on sustainable agricultural practices and water use.
- **Ministry of Environment:** Responsible for environmental conservation and climate change adaptation.

Provincial Level:

- **Provincial Irrigation Departments:** Manage irrigation systems and water distribution at the provincial level.
- **Provincial Environment/Climate Change Departments:** Implement environmental and climate change policies at the provincial level.
- **Provincial Environment Protection Departments (EPDs):** Enforce environmental regulations and standards.
- **Public Health Engineering Department (PHED):** Manages water supply and sanitation services.
- **Provincial Agriculture Departments:** Promote sustainable agricultural practices and manage water use in agriculture.
- **Provincial Forest Departments:** Oversee forest conservation and management.
- **Urban Planning Units:** Plan and implement urban development projects.
- **City Development Authorities:** Manage urban infrastructure and development.
- **City District Governments:** Oversee local governance and development projects.
- **Water and Sanitation Authorities (WASAs):** Provide water supply and sanitation services in urban areas.
- **Provincial Disaster Management Authorities (PDMAs):** Coordinate disaster management efforts at the provincial level.
- **Local Government Municipalities:** Implement local development projects and services.
- **Provincial Fisheries Departments:** Manage fisheries resources and promote sustainable practices.
- **Agriculture Extension Departments:** Provide support and services to farmers.
- **Provincial Planning and Development Departments:** Plan and coordinate development projects at the provincial level.

Local Level:

- **Tehsil Municipal Authorities (TMAs):** Manage municipal services and infrastructure at the tehsil level.
- **District Councils:** Manage local governance and development projects at the district level.
- **Masjid Committees:** Engage in community development and welfare activities.
- **Farmers Organizations:** Represent farmers' interests and promote sustainable agricultural practices.
- **Community Organizations:** Engage in local development and community welfare activities.
- **Water and Sanitation Agencies:** Provide water supply and sanitation services at the local level.
- **Irrigation Departments:** Manage irrigation systems and water distribution at the local level.
- **City Development Authorities:** Oversee urban development and infrastructure at the local level.

- **Local Government & Community Development Departments:** Implement local development projects and services.
- **Ministry of Climate Change:** Leads the formulation and implementation of climate policies and strategies.

4. Challenges hindering NbS mainstreaming for water resilience

Despite the recognition of NbS, there are challenges related to funding, awareness, and capacity for NbS implementation. There is a need for better coordination among stakeholders and integration of NbS into existing frameworks. The main challenges highlighted by survey participants in Pakistan are listed below.

Conflicting policies: Lack of alignment between different policies and sectors, leading to fragmented efforts and inefficiencies. The fragmentation makes it difficult to implement NbS effectively, as policies may contradict or fail to support each other.

Limited awareness and understanding: Limited awareness and understanding of NbS among policymakers, stakeholders, and the public. This lack of awareness hinders the adoption and implementation of NbS, as stakeholders may not fully appreciate the benefits or know how to integrate these solutions into existing frameworks.

Capacity gaps: Limited technical and institutional capacity for planning, implementing, and monitoring NbS initiatives.

Funding constraints: Insufficient funding and financial mechanisms to support the implementation and scaling up of NbS projects. Without adequate financial resources, it is challenging to initiate and sustain NbS projects, limiting their potential impact.

Data and information gaps: There is inadequate data on climate impacts, water resources, and ecosystem services, which hinders effective decision-making and planning. Without reliable data, it is difficult to design and implement NbS projects that are tailored to local conditions and needs.

Photo credit: Mostafijur Rahman Nasim

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Asian Disaster Preparedness Center

SM Tower, 24th Floor, 979/67-69 Paholyothin Road,
Phayathai, Bangkok 10400 Thailand
Tel: +66 2 298 0681-84
Fax: +66 2 298 0012
Email: adpc@adpc.net

www.adpc.net
@ADPCnet
Asian Disaster Preparedness Center - ADPC
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